ONEMIND OS · RESEARCH RUN F-10

OneMind OS Foundation Report

What OneMind OS should stand on for the next ten years, from a Raspberry Pi greenhouse to a robotics-simulation cluster, and the path to build it.

Prepared for	Zeus Delacruz, founder of OneMind OS
Date	2026-11-11 (research run of 2026-11-10/11)
Status	Draft for founder review. Synthesises `research-output/` files 00–30 at foundation decision revision 4.1.
Evidence	Every factual sentence carries a claim ID that resolves in the research ledger; Appendix C lists each source.

The answer in one sentence. Own a read-only image with two OS slots (RAUC) and a fresh-signal health gate, on an Ubuntu 26.04 LTS base if Canonical allows it (the locked scorecard's narrow, conditional result, with a Debian 13 version built alongside until the base is locked), k3s on every real box, NATS as the bus, one capability gateway in front of every app, device and AI agent, and ten signed plugin sockets; what each pending test result means for the base was written down in advance.

Not legal, tax or financial advice. Vendor prices are unpublished where stated. Lab results come from disposable virtual machines unless marked otherwise.

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Chapter 1
The executive answer

Explain it like I'm the founder. You asked one question: *what should OneMind OS stand on for the next ten years, and how do we get there?* The short answer is: **build your own read-only operating-system image, keep two copies of it on every box so a bad update rolls itself back, run small Kubernetes (k3s) on every real box, wire everything together with NATS, and put one "capability gateway" in front of every app, device and AI agent.** That is what ADR-001 proposed, and after four rounds of attack that shape still stands. **One thing under it changed in revision 4: the base.** Under the contest rules we locked before scoring, **Ubuntu 26.04 LTS** narrowly beats Red Hat's bootc and Debian 13 (65, 64 and 63 points), but only if Canonical allows OneMind to ship a modified Ubuntu, or OneMind strips the brand and rebuilds it (Z-01). Revision 3 had kept Debian with a tie-break rule written after the scores were known; that rule is withdrawn [C-PROTO4-018]. Until the base is locked, one recipe builds both an Ubuntu and a Debian version. What else changed is detail: the update engine (RAUC instead of Rugix), how "healthy" is checked, how backups work on a single box, which boxes run full Kubernetes, and when Secure Boot arrives. The base and the update engine (RAUC over Mender) both won narrowly, so each has a written, testable way to the runner-up, and for the base the result of each pending answer was written down in advance. Revision 4.1 changed no score: three new attackers who saw only the one-line decision all called it *weakened*, none *wrong*, and the industrial-IoT gate passes with conditions. The one pending test that needs no hardware, Ubuntu's own Secure Boot test, could not run here (Ubuntu's servers are blocked from the research computer), so it is yours to run on day 1 (LT-VM-20). Nearly everything was proven in virtual machines on a Mac; the first 180 days prove it on your real hardware.

How to read this report. Every factual sentence carries a claim ID in square brackets, for example [C-W2-001]. Each ID is a row in the research ledger (`research-output/evidence/claims/*.csv`) with its source URL, access date and re-check date; Appendix C lists the source of every ID cited here. Sentences without an ID are recommendations or reasoning, and they name the research file that argues them. Requirement IDs (for example F-14, R-1) come from `docs/REQUIREMENTS.md`. Nothing here is legal advice.

1.1 The answer, one layer at a time

Layer	Decision	vs ADR-001	Confidence
Base OS	**Ubuntu 26.04 LTS** core image + RAUC for x86, generic arm64 and Pi 5 (the frozen-protocol result), **conditional on Canonical's permission or a debranded rebuild (Z-01)**; a **Debian 13** version is built from the same recipe until the base is locked before the first box ships; **NVIDIA's L4T Ubuntu 24.04** for Jetson; Ubuntu LTS for GPU servers; **AlmaLinux 10 or RHEL 10 image mode** as a year-2 Industrial profile, CentOS Stream and Fedora IoT as CI canaries. Written in advance: bootc fails on a real Pi 5 → Debian; Canonical refuses → bootc; Ubuntu	**Changed (conditional)**	**Low** (a tie in all 14 weight runs; Ubuntu wins 11, Debian 2, bootc 1)

Layer	Decision	vs ADR-001	Confidence
	fails its own Secure Boot test (U-SB, attempted and blocked here; you run it as LT-VM-20) → bootc		
Image	OneMind owns the image: one container-style build definition with an Ubuntu 26.04 leg and a Debian 13 leg, turned into A/B disk images; a distro base (not built from source) because of NVIDIA's CUDA path and one-person upkeep	Confirmed	High
Updates	**RAUC** A/B with verity bundles on x86 and Pi 5, built by OneMind from a pinned upstream tag; NVIDIA's own A/B on Jetson; one release contract; **Mender** is the named runner-up	Changed (engine)	Medium-low
Health gate	Commit an update only on **fresh** signals from *this* boot; apps excluded; 3 boot tries where the boot path allows; never auto-retry a bad release; shown safe for **one box on SQLite**; cluster rollback still untested	Confirmed, made concrete	**High** (single box)
Kubernetes	**k3s** on every box with ≥ 4 GB RAM and an SSD; robots and field kits run the k3s *agent* only; smaller devices are **NATS leaf** nodes; RKE2 only when a buyer needs FIPS/STIG	Changed	Medium
Backups	**restic** with three keys; online SQLite backup of k3s on a single box; automatic verified restores	Changed (method)	High
Security	Disk layout for Secure Boot and encryption fixed on day 1; encrypted data from first boot; signed releases; human approval in the gateway for anything physical	Changed (timing)	Medium
Plugins	Ten extension points (E1–E10) in release 1, signed-only	New	Medium

Source: `01-FOUNDATION-DECISION.md` §1, §3.1.6 and §7 (revision 4.1; no score or ranking change from revision 4). Key evidence: the rev-4 scorecard and its 14 weight runs [C-DEC4-001, C-DEC4-006], Canonical's redistribution rule [C-W2-050], the Ubuntu 26.04 lab leg [C-LAB-ubuntu-2604-leg-011], the health-gate lab [C-LAB-rauc-k3s-gate-008], the k3s footprint [C-LAB-k3s-footprint-restore-007] and the single-box restore [C-LAB-k3s-footprint-restore-011].

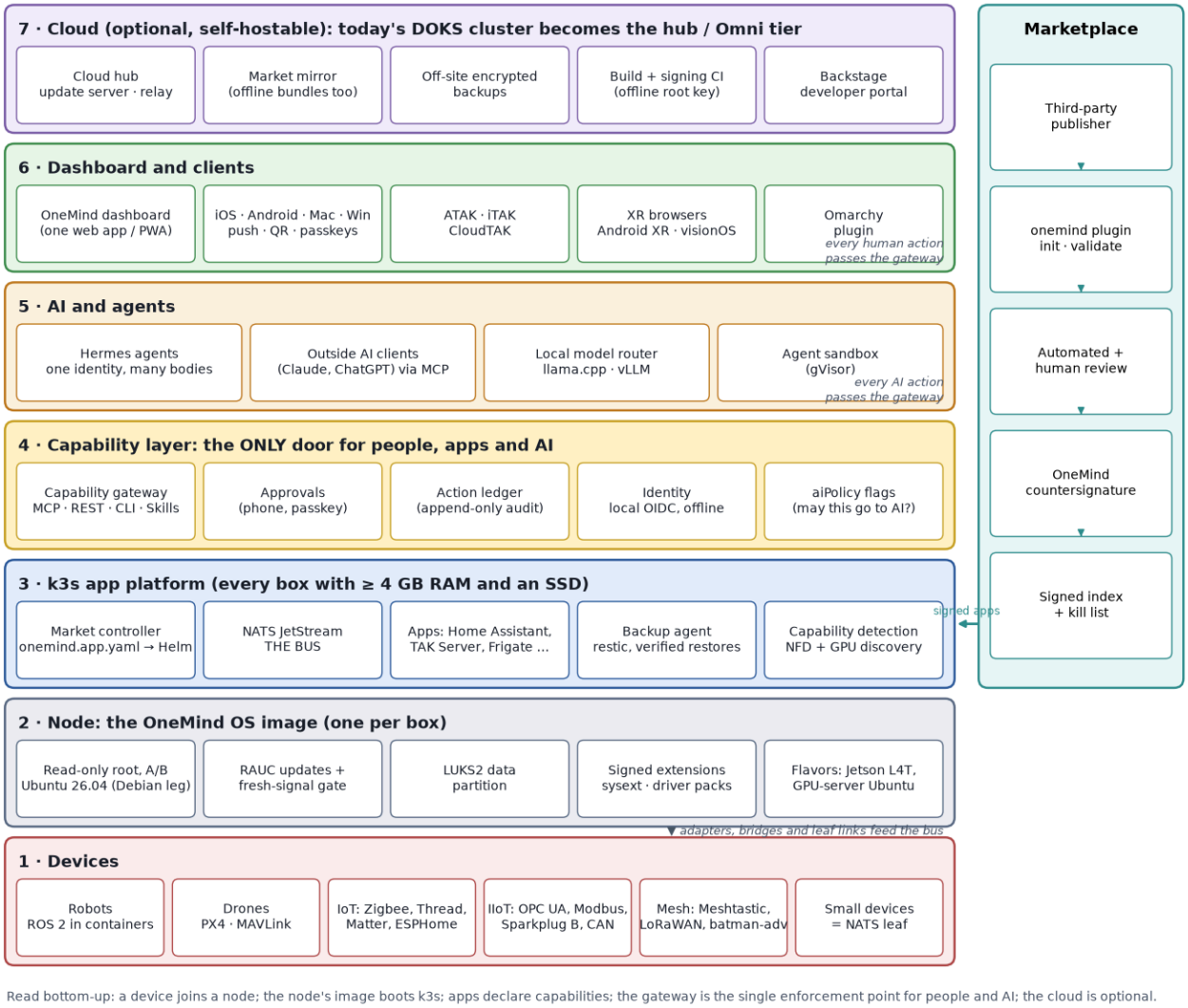


Figure 1: ADR-001's shape survived four rounds of attack; the base under it is now Ubuntu 26.04, conditionally. The diagram shows the whole system that results; Chapter 6 explains each layer.

1.2 Ten findings that matter most

1. **The three bases are within 2 points, and the locked rules pick Ubuntu.** The rev-4 card is Ubuntu 65, bootc 64, Debian 63 [C-DEC4-001]. The frozen rule calls a gap under 3 points between the **top two** a tie, so only Ubuntu and bootc go to the tie-break; Debian, third, is not a finalist even though it is within 3 points of the leader. On the decision-critical evidence Ubuntu beats bootc **6–4** [C-DEC4-006]. Revision 3 had brought Debian in as a third finalist after the scores were known; that rule is withdrawn [C-PROTO4-018]. The result is fragile: Ubuntu wins 11 of 14 weight runs, Debian 2 and bootc 1 (`01` §3.1.6).
2. **Ubuntu's win needs Canonical's say-so.** A modified Ubuntu may be shipped only with Canonical's approval, or with the trademarks removed and the binaries recompiled [C-W2-050]. That includes the Jetson flavor on every base, because NVIDIA's Jetson root filesystem is Ubuntu [C-REF-DC-cost-007 (SECONDARY: NVIDIA staff on forum)]. Debian's core has no such condition [C-W2-043]. So we wrote down in advance what each answer means: Canonical refuses and you reject the debranded rebuild → bootc; bootc fails on a real Pi 5 (W2-HW-3) → Debian; both → your decision; Canonical agrees with a price → Ubuntu outright; Ubuntu fails its own Secure Boot test (U-SB) → bootc (`01` §3.1.6 h). U-SB was attempted on 2026-11-11 and is **blocked**, not failed: Ubuntu's servers do not resolve from the research environment [C-FIX5-001], so it is a founder-run test, LT-VM-20 (Chapter 4).

3. **"Healthy" must mean healthy now.** After an update, Kubernetes shows the *previous* boot's "all good" status. A naive check committed a broken update with Rugix [C-LAB-debian-rugix-023] and again with RAUC and real k3s [C-LAB-rauc-k3s-gate-008]; the fresh-signal gate caught every one.
4. **Nobody ships exactly our combination.** A targeted search found no field-proven product running Debian or Ubuntu with RAUC [C-DEC-013 (UNVERIFIED: negative search)]; the closest, IronFlockOS (Debian 13 + RAUC + Docker), is pre-1.0 [C-CIT4-001]. Each piece is proven somewhere (Debian in umbrelOS, RAUC in Home Assistant OS, k3s in Olares), so our labs and your hardware tests are the proof of the whole.
5. **Debian's own RAUC is too old for newer kernels.** RAUC 1.13 cannot install any update on Linux 6.19 or later [C-LAB-ubuntu-2604-leg-010], while upstream 1.15.2 passed every test [C-LAB-rauc-1152-011]. So OneMind builds its own RAUC, as Home Assistant does.
6. **Secure Boot works with our layout, but it does not stop downgrades.** Signed per-slot kernels, TPM-unlocked data and rollback all passed in emulation [C-LAB-secureboot-rauc-013], and again on RAUC 1.15.2 [C-LAB-secureboot-1152-005], but an *old* signed kernel still boots against a newer root [C-LAB-secureboot-1152-009]. OneMind adds an anti-downgrade rule.
7. **The plugin sockets must be locked.** systemd's extension feature accepts unsigned add-ons by default [C-REF-W19-security-004]; OneMind switches that off before any third party plugs in.
8. **NVIDIA's Jetson software ends before its hardware does.** JetPack 4 ended 26–32 months before Nano and TX2 NX stopped shipping [C-REF-W7-T-001], and JetPack 5 ended on 2026-10-30 [C-REF-W7L-001].
9. **Robotics simulation lives on Ubuntu.** Isaac Sim lists Ubuntu and Windows hosts only [C-REF-DC-robotics-013], and NVIDIA's DKMS driver broke on routine Debian kernels [C-REF-DC-cost-009]. This is part of why Ubuntu scores highest on ecosystem.
10. **The EU Cyber Resilience Act applies from 2027-12-11** [C-W6-001] and expects security updates to install automatically by default [C-W6-008]. That collides with "never auto-install" and is your call (Z-02).

1.3 "Pi greenhouse → enterprise robotics simulation": one OS at every scale (F-14)

You asked for an OS *"as simple as using a Raspberry Pi, up to full enterprise"*. Here is the same OneMind OS growing from one Pi in a greenhouse to a robotics-simulation cluster, step by step, with what stays the same and what changes.

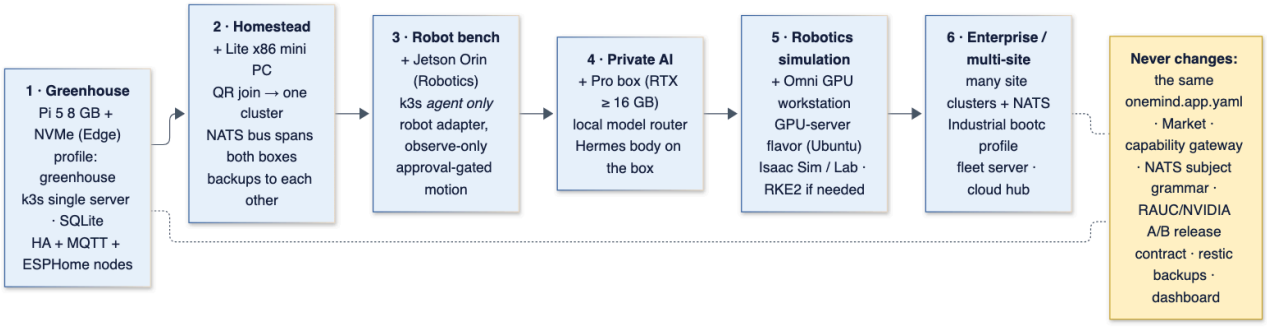


Figure 2: The six steps. The yellow box is the promise: these parts never change as you grow.

Step 1: the greenhouse Pi. Buy a Pi 5 with 8 GB and an NVMe drive (the Edge size; production of the Pi 5 and CM5 is committed to at least January 2036 [C-W10-035]). Flash the OneMind image, scan the QR code on first boot and pick the `greenhouse` profile. You get Home Assistant, a small MQTT broker and ESPHome sensor boards, on a single-box k3s server (k3s adds about 936 MiB at idle on a 4 GiB arm64 box [C-LAB-k3s-footprint-restore-007]). The important design rule: **the climate loop runs on the ESP32 board, not on the Pi**, so a crashed Pi never cooks the plants. ESPHome's defaults are unsafe for this (a node reboots after

15 minutes without a client, and relays default to OFF at boot), so the profile overrides both [C-W8-029, C-W8-030]. The AI may suggest set-points; opening a valve needs your approval on the phone. Updates use RAUC's Pi 5 "tryboot" path, and backups go to restic.

Step 2: the homestead. Add a Lite x86 mini PC. Scan its QR code and it joins the same cluster. The NATS bus now spans both boxes; each box backs the other up. A single box keeps k3s's simple SQLite database, and when a third server joins, one restart with `--cluster-init` moved all records to etcd in our lab [C-W5-005].

Step 3: the robot bench. Plug in a Jetson Orin robot. It runs the **k3s agent only**, never the part that holds every password, because a robot can be stolen [C-REF-W4-security-008]. The box recognises it, offers a signed adapter, starts it in *observe-only*, and lets AI move it only with your approval (Chapter 9).

Step 4: private AI. Add a Pro box with an NVIDIA RTX card (≥ 16 GB VRAM). The capability detector labels it "Private AI"; the Market places language and vision models there; the Hermes agent gets a body on the box. The same app manifest says what each app needs, and the dashboard answers "runs here / runs on another of your machines".

Step 5: robotics simulation. Add an Omni GPU workstation. This is the one place the OS family changes: Isaac Sim supports Ubuntu and Windows hosts only [C-REF-DC-robotics-013], so simulation boxes run the **Ubuntu GPU-server flavor** (gated on Canonical's answer, Z-01) with k3s or RKE2. It still uses the same Market, the same manifest format, the same gateway and the same NATS subjects.

Step 6: enterprise and many sites. Ten thousand devices are many site clusters joined by NATS, not one giant cluster: k3s documents 16-CPU/32 GB servers for about 1,800 agents [C-W8-049]. Industrial customers can pick the bootc profile on the Red Hat lineage, which holds IEC 62443-4-2 at security level 2 on RHEL 9.6 [C-W2-042]. The regulated tier can run RKE2 for FIPS.

The honest caveat. "One OS" means **one app platform, one release contract and one user experience**, not literally one disk image. There are three image families (the core image on Ubuntu 26.04 or Debian 13, NVIDIA's Jetson Ubuntu, the Ubuntu GPU-server flavor), because NVIDIA rules its hardware. The research scored "one OS from Pi to cluster" as *conditional* for every candidate base, a tie (`01` §3.1.6, DC-17). What makes the promise real is that the parts you touch never change.

What stays the same at every scale	What changes as you grow
`onemind.app.yaml` app format and the Market	Hardware size, and the image flavor on Jetson and GPU boxes
The capability gateway, approvals and audit	k3s role (single server → cluster → agent-only robots)
NATS subject grammar and leaf links	Datastore (SQLite → etcd at three servers)
Signed A/B release contract and health gate	Optional RKE2 for FIPS; optional bootc Industrial profile
restic backups with a recovery key	Storage (local disk → Longhorn on 3+ nodes)
Dashboard, alerts, identity	Number of sites and the optional cloud hub

1.4 What competitors do at both ends of the scale

Product	Smallest box	Largest deployment	Same OS both ends?	Constraint fit with OneMind
Home Assistant OS	Pi / HA Green, Buildroot image [C-TD-home-assistant-os-003]	One home server	Yes, but no cluster, no Kubernetes	High for the appliance; none for scale-out
umbrelOS 2.0	Pi 5 + x86, Debian image [C-TD-umbrelos-003]	One box	Yes, single box only	High for the base; no cluster

Product	Smallest box	Largest deployment	Same OS both ends?	Constraint fit with OneMind
Olares	x86 + NVIDIA, 8 GB floor [C-W4-042]	Multi-node k3s	No image of its own; installs on a host distro [C-TD-olares-007]	High for the app platform; none for OS safety
Red Hat Device Edge	Orin and x86, no Pi 5 [C-W2-041]	Fleets with MicroShift	Yes, paid	High for Industrial; none for consumer
OneMind (target)	Pi 5 Edge profile	Many site clusters + GPU simulation	One app platform; three image flavors	The only one aiming at both ends with robots

1.5 Recommendations and the requirements they serve

Recommendation (source)	Requirement IDs
Own the image; Ubuntu 26.04 core (conditional on Z-01) with a Debian 13 leg until the base lock; pre-registered decision map; Jetson flavor (`01` §3.1.6, §4.1, REC-DEC-17, REC-DEC-42)	RS-1, R-1, M-6, F-7, F-12, F-14, D-002
RAUC A/B with the fresh-signal gate; one release contract (`01` §4.2)	R-1, R-3, M-2, P-7, U-4
k3s on platform nodes, NATS leaf below, agent-only robots (`01` §4.5)	N-9, N-2, N-6, F-14, M-7
restic three-key backups and verified restores (`01` §4.6)	R-2, R-3, UX-6, U-4
Day-1 Secure Boot layout, gateway-enforced approvals (`01` §4.7)	M-2, M-4, AI-5, AI-7
Ten plugin extension points, signed-only (`01` §4.8)	A-5, A-6, F-3, F-4, F-5, F-6
Profiles for greenhouse and TAK node (`09` §8)	P-8, F-13, F-14

Chapter 2
Everything from the founding session

> **Explain it like I'm the founder.** On 10 November 2026 you spent one long session talking through phones, earlier plans, strategy, competitors, Linux, Kubernetes, robots, IoT, VillageOS, kernels and plugins. This chapter walks that conversation in order (from `context/00-CONVERSATION-RECORD.md`) and, for each topic, says **what the research found** and **where the full answer lives**. Appendix A is the checklist proving every topic is covered.

2.1 Phones and Android sovereignty (where it started)

What you said. You use a Samsung S24 Ultra, love Android's freedom and run ATAK. GrapheneOS on a Pixel was suggested as a hardened ATAK field device, and you remembered an ATAK crash report on a GrapheneOS Pixel Tablet.

What the research found (`15-phones-field-devices.md`):

- **The phone is a client, not the OS we build.** OneMind OS runs on boxes; phones connect through the OneMind app and ATAK/iTAK. Nothing about phones changes the foundation.
- **Your field phone: a Pixel 8 or newer with GrapheneOS.** GrapheneOS recommends only Pixel 8 and later, which carry a 7-year support guarantee and hardware memory tagging [C-W14-002].
- **The Pixel Tablet crash could not be found anywhere public** (GrapheneOS tracker, ATAK tracker, the web) [C-W14-018]. More important: **the Pixel Tablet has no GPS and no cellular modem** [C-W14-011], so it can never be a standalone ATAK field device. ATAK's own location code warns, rather than crashes, when GPS is missing [C-W14-015]. A one-hour test in `15-…` Appendix A tells the likely causes apart (memory-protection toggle, a tablet OS bug, or a plugin).
- **The real GrapheneOS + ATAK gotcha is networking, not crashes:** multicast over a VPN is blocked by GrapheneOS's leak protection (open issue) [C-W14-010].
- **Your S24 Ultra stays a great stock ATAK phone.** A US S24 Ultra cannot take a custom OS; forcing one trips Samsung's one-time Knox e-fuse for good [C-W14-021]. Use no-root tools (Shizuku, UAD-ng, Obtainium, Shelter, Tasker, DeX).
- **Fairphone and friends:** CalyxOS is back and supports Fairphone 4–6 and Pixels [C-W14-027]; iodéOS, /e/OS and Volla are covered in `15-…` as client options.
- **Linux phones are not ATAK phones in 2026.** postmarketOS has renamed itself Nura [C-W14-029]; FuriLabs FLX1s runs FuriOS on 8 GB [C-W14-030]; but Waydroid has no GPS pass-through, open since 2021 [C-W14-031]. On a Linux phone, the TAK answer is **CloudTAK in the browser**.
- **Termux and Android's built-in Linux Terminal (AVF):** the Terminal runs an experimental Debian VM behind developer options [C-W14-033]. It is a developer convenience, not a OneMind node runtime: do not plan a "phone as a k3s node" (REC-W14-09).

2.1.1 The mobile OS landscape, HarmonyOS and the dead OSes

- **HarmonyOS NEXT** dropped Android compatibility entirely and runs only native apps (SECONDARY, unverified) [C-W14-037]. Its open sibling's discovery layer, DSoftBus, is Apache-2.0 [C-W14-036]. **Lesson**

written into OneMind's architecture: no single store, push service or cloud in the client path; OneMind gets Huawei's "super device" outcome with NATS, QR onboarding and the capability layer (REC-W14-10).

- **The microkernel lesson:** even Huawei kept Linux API/ABI compatibility to reuse drivers [C-W18-036] (Chapter 5).
- **Android forks and the rest:** GrapheneOS, LineageOS, /e/OS, CalyxOS and iodéOS are client options compared in `15–…`; KaiOS serves feature phones and is out of scope. A custom Android ROM stays deferred, as the earlier planning documents also concluded.
- **Dead OSes** (BlackBerry 10/QNX, Windows Phone, Symbian, webOS, Firefox OS, Tizen) and Fuchsia teach the same thing: an OS without an app and driver ecosystem dies. Fuchsia's Zircon shipped commercially on one device class (SECONDARY) [C-W18-039].

2.1.2 "Build your own OS" levels and the OneMind EUD

Level	What it is	Verdict for OneMind
1. Launcher + device-owner app on stock Android or GrapheneOS	OneMind app + open-source device management (Headwind MDM, Apache-2.0)	**Build first** (REC-W14-08)
2. AOSP/GrapheneOS fork signed with your own keys	GrapheneOS ships about 80 releases a year [C-W14-040]; a solo founder cannot keep a fork secure	Not before year 2, only for a paying customer
3. Linux-phone distro	No ATAK with GPS today [C-W14-031]	No
4. From scratch	Huawei's path [C-W14-037]	Never

The "OneMind EUD" becomes a kit, not an OS: a Pixel with GrapheneOS, the OneMind app, ATAK and plugins, enrolled by QR into the customer's box. A small vendor already sells exactly that shape (Pixel 9 + GrapheneOS + ATAK) for about USD 1,500 (`15–…` §11).

2.2 The earlier planning documents (the Blueprint)

What you decided: the Nov 3–4 **Blueprint is canonical**; the Build Handoff is outdated (old pillars, "no k8s", NATS "later", Letta primary, ClickUp). The critique was too much hedging and a 90-day plan too big for one engineer.

What the research did with it (`20-PLAN-180-DAYS.md`): the 180-day plan keeps the Blueprint's milestone order (M0 reconcile → M1 secure vertical slice → M2 extensions → M3 disconnected client → M4 independent owner → M5 managed appliance) and maps all 24 acceptance tests A01–A24, but moves dates, because the Blueprint's plan did not include the OS image, A/B updates, Secure Boot, Pi 5, Jetson or GPU work. **M6 (multi-site federation) is outside the 180 days**, following the Blueprint's own rule "if M1 or recovery fails, cut breadth". The four questions repeated across the old documents (is Google/Pixel hardware acceptable, the Dell's state, the first vertical slice, which task system) are answered or scheduled in Chapter 14.

2.3 The strategy conversation

- **OneMind is a platform first and an image second.** The research confirms the order for development (platform built first in a VM, image from the same layers from about week 3) but makes the **image the product**: every competitor that ran both a script install and an image later dropped the script. Home Assistant ended its Core and Supervised installs because they "generate more issues" [C-REF-DC-longevity-004], and Umbrel dropped script installs at 1.0 [C-REF-DC-longevity-005].
- **The moat: homestead + AI agents + TAK/field ops + self-hosting, and robots.** Across 11 competitor code bases, zero files mention ROS 2 or MAVLink [C-MATRIX-002]; robotics is open ground (Chapter 10).

- **AI is the turning point** (“who owns my AI’s memory”): OneMind’s answer is that memory, models and data live on the owner’s box by default, and a model router keeps health and family data local (Chapter 3).
- **Market view.** Most people won’t leave Apple and Google; the first customers are people with a strong reason to (homesteaders, privacy-minded families, TAK teams, small businesses, sovereignty-minded governments). 17–… recommends homesteaders and self-hosters first, then TAK teams, robotics through partners in years 2–3, and public safety later, because enterprise sales sank Sandstorm [C-W16-050].
- **Business model:** open core + boxes + support + education, the Home Assistant playbook. Its foundation took in CHF 8.84 million in 2025 from about 2.28 million homes [C-W16-013] (Chapter 3).
- **You want an Umbrel/Home Assistant/CasaOS-like product** with NATS and your services plug-and-play, access from any device, simple → powerful hardware tiers, robot and agent simulation, and XR. All of these have homes in Chapters 3, 8, 9 and 12.
- **NVIDIA at KubeCon NA 2026 (Salt Lake City, 9–12 Nov).** NVIDIA released NVKE/NVKR as a public preview, “not yet recommended for production” [C-W4-039], and Agent Substrate entered the CNCF Sandbox [C-W4-040]; the larger 2026 moves came at KubeCon Europe, where the DRA GPU driver was donated to Kubernetes and KAI Scheduler entered the CNCF Sandbox [C-W4-032]. **The warning stands:** these tools validate Ubuntu, RHEL, SLES, Talos and RKE2, not Debian or k3s (`05` §7), and NVIDIA lock-in is the opposite of sovereignty, so OneMind’s app manifests must run on AMD, Intel, Hailo and Rockchip too (Chapter 12).
- **Fedora CoreOS / Fedora IoT as the core.** Answered in full in Chapter 4: a profile and a canary, not the base, today.

2.4 Your existing cloud cluster: the proven concept, and why an OS now

Your DigitalOcean cluster already runs NATS JetStream, Argo CD, Backstage, Hermes, an MCP server, TAK Server with CloudTAK, Authentik and Tailscale (`context/01`, reported state). It proves the *software*. What it cannot give is what only the machine owner can: atomic OS updates with rollback, hardware detection, first-boot onboarding, working through internet loss, physical device access and per-device keys (`05` §12). The two products that won households, Home Assistant OS and umbrelOS, ship their own image and keep the app runtime simple [C-W4-043, C-W4-044]; Olares, which only installs on a stock distro, needs 8 GB and has no OS rollback [C-W4-042].

- **Converting the cluster is not “three changes”.** Your own read-only inventory lists 13 dependencies, four of them real work (control plane, 95 block volumes, load balancers, snapshots) [C-W4-063]. The plan is six phases, app by app, with the NATS bus authority flipping last (Chapter 6).
- **Fix this first:** the public ingress-nginx controller has been unmaintained since March 2026 [C-W4-016]. It is risk RR-04, critical, and live today.
- **Backstage stays** as the developer and enterprise portal in the cloud tier; it needs at least 6 GB RAM just to run [C-W4-059], too much for a home box. The OneMind dashboard becomes the front door.
- **Hermes moves as-is.** The upstream agent image already builds for amd64 and arm64 on Debian trixie [C-W4-058]; its NATS contracts become the OS’s internal bus contract.

2.5 The big brief (82, now 97, requirement IDs)

Every item you listed has an owner in the research and a place in this report:

You asked for	Where it is answered
Reverse-engineer competitors; research all Linux distros	Chapter 10; Chapter 4

You asked for	Where it is answered
Sell hardware and become a distributor	Chapter 8 (no own boxes in year 1; certified list first)
OS profiles for a box or a Pi (situational awareness, greenhouse)	Walkthrough; Chapter 3 (TAK node, greenhouse)
Robotics, drones, physical AI as the moat	Chapter 12; Chapter 9 (robot plug-and-play)
Every device is a node; mesh native	Chapter 6; Chapter 3 (mesh)
Every app readable by AI (MCP, CLI, API, SSH, MHS) with permissions	Chapter 3 (AI capability layer)
Native alert system	Chapter 3 (dashboard and alerts)
Cloud apps (Strava) with API keys shown and given to AI	Chapter 3 (“may this go to AI?” flag)
Internal apps: finance, wellness, robotics, infrastructure, PM, mesh, IoT, family, OSINT	Chapter 3
Multi-user family, parental controls, kids’ education	Chapter 3
Hardware capability auto-detection; QR onboarding	Chapter 8; Chapter 9
Multiple machines in one UI, mirror, infrastructure map	Chapter 3
Lite/Pro/Omni editions; NVIDIA as a partner	Chapter 8; Chapter 12
Website (blog, Circle.so, docs, downloads, store, affiliates, Zoho Forms, support, CMS)	Chapter 3
Omarchy-style plugin ecosystem; WebXR and Android XR	Chapter 9; Chapter 3
Updates users control; strong support	Chapter 7; Chapter 14 (Z-02)
Become the industry standard for sovereign operations	Chapter 11 (roadmap exit criteria)
No guessing. Do research.	Every claim ID; Appendix C

The Olares images you shared (Market catalog, Olares + Home Assistant, Olares One): the Market research read both catalogs line by line, 399 Umbrel apps and 396 Olares apps with only 56 in common [C-W12-001, C-W12-008]. Olares’s app repository is Apache-2.0 and reusable with credit [C-W12-004]; Umbrel’s has no licence at all, so its files cannot be copied [C-W12-002]. Olares One is benchmarked in Chapter 8.

2.6 What phase-1 research reported, re-checked

Phase-1 fact	Status after this run
umbrelOS is PolyForm-Noncommercial; Olares AGPL-3.0 with a leftover non-commercial file	Confirmed [C-TD-umbrelos-045, C-TD-olares-047]; Olares moved its main licence to AGPL in 2025 [C-W16-006]
Kairos is Apache-2.0	Confirmed [C-TD-kairos-002]; its Pi 5 support is still an open issue since 2023 [C-REF-DC-longevity-008], so it was dropped as a lead
Project N.O.M.A.D. is Apache-2.0	Confirmed [C-TD-project-nomad-002]
MHS = Anthropic’s Model Hardware Standard, research preview 2026-08-27	Confirmed; no public spec yet, Hugging Face (LeRobot) and Raspberry Pi are early adopters [C-W7-047]
Strava bars AI use of its data (secondary sources then)	Now **primary**: the Strava API Policy in force since 2026-06-01 bans its data in any AI context window, even with consent [C-W12-044]
MCP spec 2026-07-28; MCP donated to the Linux Foundation’s Agentic AI Foundation	Confirmed [C-W9-001, C-W9-009]

2.7 Founder decisions D-001 to D-006

ID	Decision (`docs/DECISIONS.md`)	What the research did with it
D-001	Open core, Apache-2.0	Kept. Your protection is the **trademark**, not the licence: Apache-2.0 grants no right to your name [C-W16-001]. Use a DCO, not a CLA, as Home Assistant effectively does

ID	Decision (`docs/DECISIONS.md`)	What the research did with it
		[C-W16-003]. The CRA makes the open/paid boundary a compliance question too (Z-05).
D-002	Design all tiers together	Kept. One image pipeline, many profiles; x86, Pi 5 and Jetson are all in the week-4 to week-8 bake-off (Chapter 11).
D-003	One app shell with modules; scaffold all	Kept. Each module is also a Market app; build the thin parts, integrate proven engines for the heavy ones (Chapter 3).
D-004	Hardware on hand: Mac, Android, Beelink U55, Dell 5540, cloud	Kept. The Dell is your best Secure Boot test bed (custom keys allowed) [C-W10-043]; the Beelink's i3-5005U is a "worst-case old box" (no listed TPM firmware) [C-W10-041]. Buy a Pi 5 and a Jetson Orin Nano Super now.
D-005	iOS push via APNs with an opaque payload	Kept. Catch: Apple accepts pushes only signed with OneMind's developer key, so a tiny content-free relay is needed (Chapter 3).
D-006	Apply to NVIDIA Inception and the MHS preview	Kept. Inception is free but requires incorporation and a working website [C-W16-031].

2.8 The learning track

You are learning OS engineering as you go (Lesson 01: the big four bases; Lesson 02: Omarchy). `27-LEARNING-PATH.md` turns this into **12 modules in build order**, each with why-now, reading (official material first), one hands-on exercise in the Linux VM on your Mac, vocabulary and self-check questions. The fastest single resource is Bootlin's free embedded-Linux training [C-W17-001]; the single most important module is **L5, A/B updates with RAUC**, because it is what keeps boxes in homes from bricking. Omarchy is a desktop *client* edition (Chapter 3), not the box base.

2.9 ADR-001 and your questions about it

ADR-001 was **proposed, not approved**; you said "I have questions". Here are the questions and the answers in one line each:

Your question	Short answer	Chapter
Why Ubuntu vs Debian?	Under the locked rules **Ubuntu 26.04 wins narrowly** (65 vs bootc 64 and Debian 63; 6–4 over bootc on the tie-break) on NVIDIA, ROS 2 and ecosystem, but a modified Ubuntu needs Canonical's permission or a debranded rebuild [C-W2-050]. Debian is built alongside until the base is locked, and becomes the base if bootc fails on a real Pi 5	4
Why do competitors ship their own image?	Because only an image gives atomic updates with rollback; script installs were dropped [C-REF-DC-longevity-004]	4, 7
Why k3s when I run full Kubernetes?	Same API and manifests, one binary, about 1 GB of RAM [C-LAB-k3s-footprint-restore-007]; RKE2 is the bigger sibling from the same vendor	6
How does the dashboard work?	One web app, role-based "Pro" screens, rescue screen outside Kubernetes	3
Backups, rollback, security, updates (your top concerns)	Fresh-signal health gate, restic with a recovery key and verified restores, day-1 Secure Boot layout	7
"Use the proven competitors"	Done for 30 products, each with a constraint-fit verdict	10

Final verdict on ADR-001 (`01` §7): **confirmed** in architecture (own image, A/B with a health gate, k3s, restic); **changed, conditionally,** in the base (Debian 13 → Ubuntu 26.04 by the frozen scorecard, gated on

Canonical, Z-01), and **changed** in the update engine (Rugix → RAUC, OneMind-built), the k3s datastore backup method, the RKE2/FIPS text, CRA timing and Secure Boot timing; **overturned** on Kairos for Jetson, the system-upgrade-controller on image nodes, "DOKS converts in three changes" and "no Fedora/RHEL role".

2.10 The second to sixth briefs

- **Second brief (F-1…F-10): the future, NVIDIA, ease of use, "I want to connect a robot → use OneMind OS", the Marketplace, your own kernel or image, the Mac toolchain, a second build session and this PDF.** Physical AI and NVIDIA: Chapter 12. Kernel and image ownership, Mac toolchain: Chapter 5. Robot plug-and-play and the Marketplace: Chapter 9. The build brief for the second session is `30-NEXT-ULTRACODE-BUILD-BRIEF.md` (summarised in Chapter 11).
- **Third brief (F-11…F-13): IoT and IIoT without limits, Fedora IoT "super important", VillageOS.** IoT/IIoT: Chapter 3; Fedora IoT: Chapter 4; VillageOS: Chapter 10.
- **Fourth brief (F-14): as simple as a Pi, up to full enterprise.** The walkthrough in Chapter 1.
- **Fifth brief (F-15): kernel families, Darwin/XNU vs BSD vs Linux.** Chapter 5.
- **Sixth brief: the plugin ecosystem.** Chapter 9.

Chapter 3
The product people touch

> **Explain it like I'm the founder.** The foundation (Chapters 4–7) is invisible when it works. This chapter covers what people actually see and use: the dashboard and alerts, family accounts, XR, the IoT and mesh features, TAK, the AI layer that lets agents use everything safely, your own apps (finance, wellness, OSINT and the rest), the Market, the website, education and the business. Each section gives the research's recommendation, what competitors do, and the claim IDs behind it.

3.1 Dashboard, clients and alerts

What competitors do (from their code).

Product	How the dashboard works	Phone alerts	Constraint fit
umbrelOS	One React page; each app opens in a new browser tab on its own port [C-W11-003]	None: notifications are text lines on the box [C-W11-004]	High (appliance, small team)
Olares	A desktop in the browser; apps open in windows (iframes); its phone app is one web codebase wrapped for each platform [C-W11-009, C-W11-011]	Via LarePass	High for the app platform
Home Assistant	Separate native iOS (Swift) and Android (Kotlin) apps [C-W11-015]	Google's push service, not encrypted [C-W11-016]	High for UX lessons

Recommendations (`12-dashboard-clients-xr.md`):

1. **One web app is the dashboard everywhere.** It installs as a PWA and is wrapped for iPhone, Android, Mac and Windows only to add what the web cannot: push alerts, QR scanning, Face ID, a Mac menu-bar icon (UX-5, UX-8).
2. **Simple vs Pro, and the Control Hub.** Simple mode shows Home, Apps, Files, Alerts and Family; owners and operators also see Nodes, the **Control Hub** (Kubernetes-level, as in Olares), AI & Agents audit and System (12-… §5). **No global "Pro mode" switch.** Home Assistant is removing its "Advanced mode" toggle because one switch gated unrelated features [C-W11-019]. Pro screens appear by *role* (owner, operator), and detail sits one click deeper (UX-1, UX-2).
3. **A small rescue screen lives outside Kubernetes**, inside the update agent, so you can still roll back if k3s breaks (R-1).
4. **Alerts travel on NATS.** Android, desktop and web use self-hosted push (ntfy/UnifiedPush); on iPhone, Apple only learns *that* something happened (D-005). Apple accepts pushes only signed with OneMind's own key, so OneMind runs a tiny content-free relay, which brushes against "no OneMind cloud yet" (M-1); your call (UX-4).
5. **Multi-machine view, mirror and infrastructure map.** A *Nodes* app shows every box and device; "mirror" turns out to mean four different things (screen mirror, data replica, failover twin, config clone), so the word needs your definition before it is built (Z-13); the infrastructure map draws from the same capability data as the Market (UX-6, UX-7).
6. **QR onboarding** for boxes, nodes and robots uses the same flow as first boot (N-4, Chapter 9).

3.2 The Omarchy desktop edition

Omarchy (Arch + Hyprland, MIT) calls itself an "agentic Linux distribution" and already lists Hermes in its agents panel [C-W11-047]. It shipped 72 releases since July 2025, 32 in the last year [C-W11-048]. **Recommendation:** ship a **OneMind client and plugin pack for stock Omarchy** (a top-bar widget with box health and alerts, an agent skill that calls the box's gateway with scoped tokens, and `onemind pair`), **not** a second OneMind OS on rolling Arch, which would double the security workload (UX-10, F-4). A "desktop that is also a node" can come later on bootc if customers ask.

3.3 XR (WebXR, Android XR, visionOS)

Platform	WebXR today	What OneMind does
W3C WebXR	Candidate Recommendation Draft, 2026-11-09 [C-W11-043]	Stable core to build on
Android XR (Chrome)	AR, hand input, anchors, depth [C-W11-044]	First target for immersive "spatial ops"
visionOS (Safari)	Immersive VR only; no AR session [C-W11-045]	2D dashboard panel; VR views

Plan: day one, the dashboard works as a 2D window in every headset browser (it only needs HTTPS on the LAN). Year 2, a WebXR route with a 3D infrastructure map, TAK layer and robot cameras. Year 3+, native apps only for low-latency teleoperation, still behind the approval gateway (UX-9).

3.4 Family accounts, parental controls and a child's device

Every person gets a role: Owner, Adult, Teen, Child or Guest. Jellyfin already models per-user ratings, tag blocks and schedules [C-W13-001], and OneMind copies that model for every module; Umbrel and Olares have no parental controls at all (14–…). The child's own phone is the hard part, because Apple and Google decide what outside apps may lock; OneMind offers a ladder: (1) the child signs in and sees only their apps; (2) the box filters the child's internet through its DNS filter, even away from home over the private network; (3) a walkthrough of Screen Time or Family Link; (4) for a dedicated kids' tablet, a QR reset into full management with an Apache-2.0 open-source MDM (U-1, U-2, U-3).

New laws target operating systems directly. California's AB 1043 makes an "operating system provider" ask for a birth date and pass an age range to apps from 2027 (SECONDARY) [C-W13-012]; a 2026 amendment may exempt openly licensed OSes, but whether that covers a company selling boxes is unclear (SECONDARY) [C-W13-014]. systemd already added a birth-date field for this [C-W13-015]. **OneMind collects a birth date for child profiles anyway**, so the age signal is built in from the start.

3.5 Education and offline knowledge

Children's learning stays on the box and offline: **Kiwix** (offline Wikipedia, 13 GB to 119 GB per edition [C-W13-022]) and **Kolibri** (courses; about 1 GB RAM recommended, runs on a Pi [C-W13-020]). Project N.O.M.A.D. seeds a similar offline library [C-TD-project-nomad-022]. **Circle.so** is for adults learning to build and run boxes (W-2); keep the course source files in your own Git repository, because Circle's export gives members, posts and comments but not lessons or media [C-W13-025].

3.6 Website, store and community

Copy what competitors converged on: a static site for blog and docs, **Shopify** for the cart, **Discourse** for the forum. Six of seven competitors fingerprinted run Discourse [C-W13-051]. Plugins and apps get website pages with an **"Open in OneMind"** button; the website never sees your box, because your browser remembers the box address, the trick Home Assistant's "My" links use [C-W13-043] (W-1…W-9). Zoho Forms, a support desk, affiliate links and a headless CMS fit around this core (14–… §7–§9).

3.7 IoT, Industrial IoT, mesh, off-grid and TAK

> The question you asked: does anything in the foundation **block** IoT or IIoT for ten years? **No.** Four things need work, and one tool is off-limits (`09-iot-mesh-offgrid-tak.md`).

- **Industrial vendors already ship Debian.** Siemens' Industrial Edge device image is Debian 12 [C-W8-037]; Moxa's Debian 13-based industrial Linux holds an IEC 62443-4-2 security-level-2 declaration [C-W8-033]; the Debian-based CIP project announced 62443-4-1 and 4-2 compliance [C-W8-034]. **Certification attaches to a product (device + OS + configuration), never to "an OS".**
- **Protocols run as apps:** OPC UA (open62541), Modbus, CAN (SocketCAN in the kernel), BACnet and MQTT are all available on Debian [C-W8-004, C-W8-005, C-W8-002]. **NATS can be the site's MQTT 3.1.1 + Sparkplug B broker, but it rejects MQTT 5 clients**, so an MQTT 5 broker app (Mosquitto or NanoMQ) is bridged in [C-W8-007, C-W8-008]. EMQX ≥ 5.9 is excluded by its licence.
- **Zero-touch onboarding (FDO) is no longer a Fedora advantage:** Fedora IoT's Rust FDO was deprecated in February 2026, and the maintained code is distro-agnostic Go under Apache-2.0 [C-W8-042, C-W8-044]. The Pi 5 has no TPM, so industrial Pi SKUs need a secure element [C-W8-041].
- **Store-and-forward is proven:** when a site's link to the hub dropped, a NATS leaf delivered 600 of 600 readings, with no gaps or duplicates, once the link returned [C-W8-050].
- **The four things to build:** a Pi 5 real-time kernel variant [C-W8-040], a packaged FDO client, the MQTT 5 broker app, and a 10,000-device fleet test.
- **Off-limits: Reticulum.** Its licence forbids use in systems that can harm people and in AI training data [C-W8-014], so it may only be a user-installed, non-default app.
- **OneMind's native mesh** is batman-adv + WireGuard/Headscale + NATS leaf, all in Debian's kernel [C-W8-003], with **Meshtastic and LoRaWAN as radio gateways**. The Meshtastic ATAK plugin sends CoT through the Meshtastic app, with a 3-hop default and a 56 KB file cap [C-W8-015]. ChirpStack (MIT) runs LoRaWAN for battery sensors over kilometres [C-W8-032]. Satellite direct-to-cell is covered in `09-…` §6.
- **Home Assistant is the IoT engine** (Zigbee, Z-Wave, Matter, Thread, ESPHome). Every competitor's HA app needs host networking and device passthrough: umbrelOS runs it privileged with all of `/dev` [C-W8-026]. So the OS exposes those as a **permissioned capability**, not a blanket privilege (E2, Chapter 9).

Profiles you asked for (P-8):

Profile	Hardware	Runs	Key rule
Pi TAK node	Pi 5 with ≥ 8 GB (16 GB recommended)	TAK Server in a container; OpenTAKServer for 4 GB boards	TAK recommends 8 GB for a minimal server [C-W8-018]; Debian 13 lacks Java 17 for the host package [C-W8-019]
Greenhouse	Pi 5 4–8 GB + NVMe + ESPHome boards	Home Assistant, MQTT, NATS leaf for sensors	Control loop on the ESP32, AI writes set-points only [C-W8-029, C-W8-030]

TAK after the cloud move is Z-12: keep TAK Server in the cloud as a relay, move it home, or both. ATAK, iTAK and CloudTAK are the clients either way.

3.8 The AI capability layer: everything readable and actionable by AI

Design (`10-ai-capability-layer.md`): every app ships one manifest, `onemind.app.yaml`, listing what it can *do* and *show*. The OS publishes that list four ways (**MCP tools, a REST API, CLI commands and Agent Skills**), all through **one capability gateway**, the single place that decides "may this agent do this, for this person,

now?" and writes it to an append-only ledger (AI-1, AI-2, AI-3, AI-4). SSH stays available to the owner for the box itself; agents get shell access only as a declared, approval-gated capability.

- **MCP rewrote itself in July 2026** (version 2026-07-28: no sessions, no handshake) [C-W9-001, C-W9-002]. Apps never speak MCP directly, so protocol churn changes adapters, not apps.
- **Authentik cannot yet be the MCP front door:** MCP clients must use RFC 8707 resource indicators [C-W9-004], which Authentik does not implement; the gateway plays that role and still uses Authentik or Kanidm to log people in.
- **Agents are their own principals,** acting as *a named user intersected with a grant*, never as the owner. Umbrel's MCP server lets an agent install an OS update and reboot [C-W6-052]; OneMind requires human approval for physical actuation, OS updates, installs and data export, because prompt injection has no reliable prevention [C-W6-068] (AI-5, AI-7).
- **"May this data go to AI?"** Every integration declares an `aiPolicy`. **Strava** is the textbook "display yes, AI no": its policy bans its data in any AI context window, even with consent [C-W12-044], and bans running an MCP server that exposes it [C-W9-032] (UX-3).
- **Local inference per tier:** llama.cpp/Ollama on Lite, Edge and Pro; vLLM on GPU servers; one model router with data-class rules such as "health and family data never leave the box" (AI-8).
- **Biometrics:** year 1 is passkeys only (the fingerprint or face check happens on the phone [C-W9-043]); on-box face or voice ID waits until year 2+, opt-in, never on by default for children. Illinois lets each person sue for $1,000–$5,000 [C-W9-035], and the GDPR household exemption does not cover the company supplying the software [C-W9-037] (AI-6).
- **MHS (Anthropic's Model Hardware Standard):** a research preview with standardized drivers and declared safety limits, but **no public spec** yet [C-W7-047]. OneMind's device adapters are designed in the same shape, so an MHS driver host can be added the day the spec is published (N-10, D-006).
- **Hermes (AI-9)** becomes an agent principal of the gateway; NATS `svc.*` stays the internal agent bus.

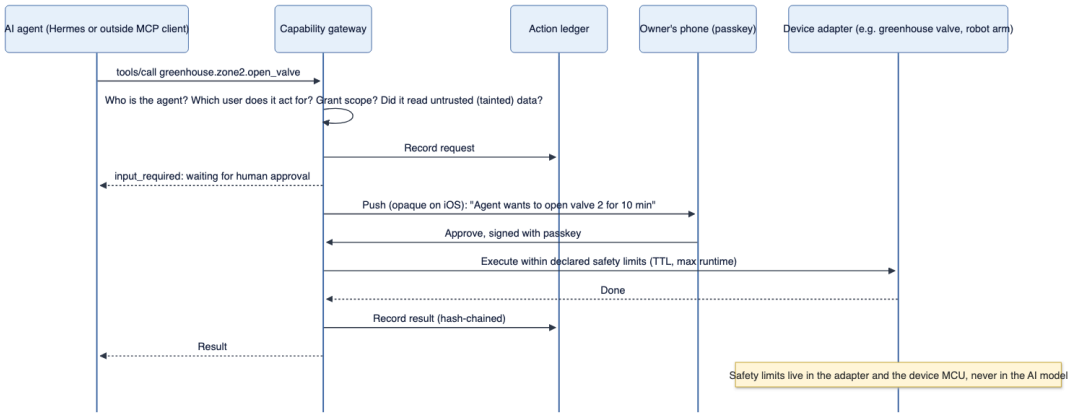


Figure 3: Every AI action that touches the physical world waits for a human. Safety limits live in the adapter and the device, not in the model.

3.9 OneMind's own apps: build the thin parts, integrate the heavy ones

D-003's app shell hosts modules; each module is also a Market app. The research recommends **building** what makes OneMind special (the shell, tasks and knowledge over your Markdown vault, the robot registry, the AI capability layer) and **integrating** proven engines for the rest (`13-apps-market-plugins.md` §8):

Module	Integrate	Key constraint
Finance	Actual Budget (MIT) [C-W12-147]; SimpleFIN for North-American bank data, read-only by protocol [C-W12-061]	Plaid's own policy lets Plaid train AI on end-user data [C-W12-065], so it is not a default. AI may read; write only with approval; **never move money**
Wellness	wger, Endurain and similar self-hosted apps	Strava bans AI use [C-W12-044]; Oura allows AI only through its own MCP server and forbids caching [C-W12-043]; HealthKit data may not be mined for marketing [C-W12-041]; Health Connect bans transfer to brokers [C-W12-042]. The `aiPolicy` switch is **off by default** for these
OSINT & intel	SpiderFoot (MIT) [C-W12-157], OpenCTI CE, IntelOwl/MISP unmodified	Scope: world events, infrastructure and your own assets; **people-search is excluded**. Public pages only: the US CFAA bars access "without authorization" [C-W12-070]. OpenSky data is non-commercial without a licence [C-W12-075]; Maltego is not bundled [C-W12-076]. OSINT text is tagged *tainted*, so an agent that read it cannot trigger commands without approval
Family, food	Mealie, Grocy	Roles from the family section
Education	Kiwix, Kolibri	Offline first
Robotics, infrastructure, mesh/IoT, PM	Own thin modules on the capability layer	Robot registry and map are OneMind's core value

3.10 The Market

- **Format:** `onemind.app.yaml` plus a Helm chart, with an importer for Compose files; it imports from upstream projects and Apache-licensed stores (Olares, CasaOS [C-W12-009]), **never** from Umbrel's unlicensed store [C-W12-002].
- **Licence classes:** several "open source" apps carry business restrictions that block preinstalling them on boxes you sell, among them Open WebUI's branding clause [C-W12-015]; at least one Olares listing calls an app "MIT" when its server part is under an enterprise licence [C-W12-007]. Every app gets a licence class, and only green classes ship preinstalled.
- **Quality tiers** and the plugin types are in Chapter 9.

3.11 Business, licensing and go-to-market

> **Give the whole OS away and sell what people can't easily copy: boxes, the name, the update service, support and teaching** (`17-business-licensing-gtm.md`).

- Home Assistant pays more than 50 staff with no investors through a foundation and commercial partners [C-W16-012]; 2.28 million homes [C-W16-013].
- Licence switches go badly: HashiCorp, Elastic and Redis all paid for theirs [C-W16-007, C-W16-008, C-W16-009]. Stay Apache-2.0 and protect the **trademark**.
- **Segments:** homesteaders and self-hosters first, then TAK teams; robotics through partners in years 2–3; public safety later.

- **Revenue:** support plans, pre-orders and education in year 1; boxes from year 2; OEM per-device fees, certification and a Marketplace share later (Chapter 11). Market sizes are third-party estimates given as ranges in 17–… §11 (SECONDARY).
- **Strong support (R-2):** Community (forum, docs, and security updates for the declared support period at no charge), Personal (human help, about $100–200 a year; Start9 charges $200 [C-W16-021]), Business and Fleet tiers. Paid support sells help and response times, **never access to security fixes**, which the CRA requires for the support period [C-W6-002].
- **Before spending money:** incorporate (NVIDIA Inception requires it [C-W16-031]), clear the name with a trademark attorney, and get prices from Canonical and NVIDIA. Outreach drafts are ready but not sent.

Chapter 4
Which Linux, and why Ubuntu wins narrowly

Explain it like I'm the founder. The base OS is the ground floor every OneMind box stands on: the programs, the kernel, how security fixes arrive and which chips it runs on. We put 12 candidates through the same contest. First, **gates**: must-pass rules (licence, x86 + Pi 5 + Jetson, a support path to 2036, no IoT/IIoT blocker, one OS from Pi to cluster). Then a **score out of 100** with weights and rules frozen *before* scoring, one source per score. The rules say: if the **top two** are less than 3 points apart, it is a tie, settled by the decision-critical evidence between those two. Revision 4 applies that rule exactly as locked. **Ubuntu 26.04 (65) and bootc (64) tie, and the evidence gives it to Ubuntu, 6 rows to 4.** Debian (63) is third, so it is not in the tie-break. The win is narrow and conditional: Canonical must allow OneMind to ship a modified Ubuntu, or OneMind must strip the brand and rebuild (Z-01). What each pending answer will mean was written down before it arrives, and one of those answers gives Debian.

4.1 The gates

Candidate	Licence to ship it	x86 + Pi 5 + Jetson	Support to 2036	Verdict
Debian 13 (+ L4T Jetson flavor)	Conditional: the core is free to redistribute modified [C-W2-043]; the Jetson flavor is modified Ubuntu [C-W2-050]	Pi 5 via Raspberry Pi's kernel, as umbrelOS ships [C-W2-044]	LTS to 2030-06-30; paid ELTS to 2035, quote-only and not shareable with customers [C-W2-001, C-W2-004]	Ranked
Ubuntu 26.04	Modified Ubuntu needs Canonical's approval or trademark removal and recompiled binaries [C-W2-050]	Pi 5 certified [C-W2-057]; Jetson is Ubuntu	Pro to May 2036, Pro for Devices unpriced [C-W2-005, C-W2-006]	Ranked
RHEL-family bootc (CentOS Stream / AlmaLinux / RHEL image mode)	RHEL needs an agreement; AlmaLinux free [C-W2-041]	No Pi 5 on RHEL; experimental AlmaLinux images [C-SCORE-025]; RHDE on Orin [C-W2-041]	AlmaLinux 10 to 2035-05-31 [C-W2-013]	Ranked
Fedora IoT, Yocto + RAUC, Buildroot + RAUC, Ubuntu Core	Pass or conditional	Conditional	Fedora: about 13 months per release [C-W2-015]	Ranked
Talos, Kairos, SUSE Linux Micro, balenaOS, NixOS, Alpine, CIP Core	—	**Fail**: no merged Orin support in Talos [C-W4-064]; Kairos has no Pi 5 [C-REF-DC-longevity-008]	**Fail** for short-lived ones	Out

The IoT/IIoT gate (G4), checked for Ubuntu in revision 4.1. Revision 4 had scored Ubuntu's IIoT gate from a ROS 2 source only, so revision 4.1 checked its five sub-checks one by one. Result: **conditional, and no sub-check fails**. Industrial protocols pass (an MQTT 5 broker ships in 26.04 [C-G4U-003]). Real time is conditional: Canonical documents a Raspberry Pi real-time kernel for 26.04, but no one has measured it on a Pi 5 [C-G4U-001, C-G4U-002]. Zero-touch onboarding is conditional: the open-source FDO client and server are released, but only as RPM packages [C-G4U-004, C-G4U-005]. Fleet scale is conditional: Canonical's Landscape manages packages but not A/B updates, so the update server is hawkBit [C-G4U-008, C-G4U-010].

Because nothing fails, the gate does not change the result (`01` §3.1.1, `09` §1). The “one OS from Pi to cluster” gate (G5) is conditional for every base, because the Jetson flavor is NVIDIA’s Ubuntu 24.04 on all of them.

4.2 The scores

Weights (frozen in `evidence/scorecard-weights.md` before scoring): longevity/support 20, hardware and robotics fit 20, security and update safety 15, proven in production 15, solo-developer operability 10, cost and licensing 10, ecosystem 10.

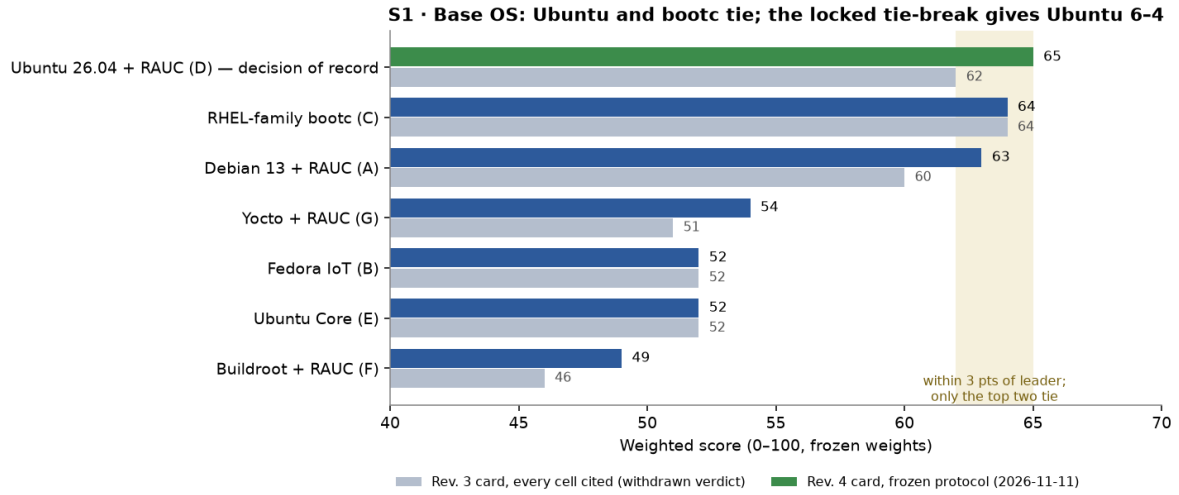


Figure 4: Base OS scores. Grey: the revision-3 card, where every cell first carried a citation. Blue/green: the revision-4 card, after one evidence correction applied to every RAUC stack. The gold band marks 3 points from the leader, but under the frozen rule only the top two tie.

Candidate	Longevity	HW + robotics	Update safety	Proven	Solo ops	Cost	Ecosystem	Total
Ubuntu 26.04 + RAUC	3 [C-W2-005]	3 [C-W2-026]	**4** doc-only [C-LAB-ubuntu-2604-leg-011, C-LAB-secure-boot-1152-005]	3 [C-W7-039]	3 [C-W2-043]	2 [C-W2-006]	5 [C-W2-028]	**65**
RHEL-family bootc	3 [C-W2-013]	3 doc-only [C-SCORE-025]	3 lab [C-W2-054]	3 [C-W2-041]	4 [C-W2-056]	3 [C-W2-013]	4 [C-W2-026]	**64**
Debian 13 + RAUC	3 [C-W3-039]	3 [C-W2-044]	**4** lab [C-LAB-secure-boot-rauc-012…015, C-LAB-secure-boot-1152-005]	3 [C-W2-043]	3 [C-W2-045]	3 [C-W2-001]	3 [C-W2-026]	**63**
Yocto + RAUC	2 [C-W2-034]	3 [C-W2-049]	**4** doc-only [C-DEC4-001]	2 [C-W2-040]	2 [C-W18-050]	4 [C-W2-049]	2 [C-W3-052]	54
Fedora IoT	3 [C-W2-015]	2 [C-SCORE-027]	3 [C-W2-054]	1 [C-W2-019]	4 [C-W2-056]	4 [C-W2-024]	2 [C-W2-028]	52
Ubuntu Core	3 [C-W2-008]	2 [C-W2-009]	4 [C-W2-006]	2 [C-SCORE-023]	2 [C-W2-006]	2 [C-W2-006]	3 [C-SCORE-023]	52

Candidate	Longevity	HW + robotics	Update safety	Proven	Solo ops	Cost	Ecosystem	Total
Buildroot + RAUC (the HAOS way)	1 [C-W2-035]	2 [C-W2-047]	**4** doc-only [C-DEC4-001]	3 [C-W2-047]	2 [C-W2-047]	4 [C-W2-049]	2 [C-W2-047]	49

Each cell is 0–5; total = Σ(weight × score)/5 [C-DEC4-001]. **What changed from revision 3:** the update-safety cell (C3) of every RAUC stack rose from 3 to 4, because our Secure Boot lab passed on RAUC, also on the 1.15.2 OneMind ships [C-LAB-secureboot-1152-005], and the updater contest (S2) already scored RAUC 4 on that same lab [C-DEC4-002]. A proposed cut of bootc's cell to 2, for its power-cut downgrade bug, was **rejected**: S2 scores the same bootc stack 3 on the same lab row, so cutting it here alone would break the "same standard for everyone" rule. It is shown as a variant below. Full cell citations are in `evidence/scorecard-rev3.md` and `01` §3.1.6 (d).

Shaking the weights. Each of the seven weights was moved up and down by 5 points (14 runs). Every run is a tie, so the tie-break decides every run: **Ubuntu wins 11, Debian 2** (more weight on security, or less on solo operability) **and bootc 1** (less weight on ecosystem) [C-DEC4-006]. Under the rules that makes the result **fragile**, not settled.

4.3 The tie-break between the top two: why Ubuntu

The frozen rule settles a tie between the **top two only**, row by row on the decision-critical evidence (`01` §3.1.6 f). Revision 3 instead brought Debian in as a third finalist and compared all three head-to-head, which kept Debian. That step was a rule written after the scores were known and never logged, so revision 4 **withdraws** it [C-PROTO4-018].

Question (decision-critical row)	Ubuntu 26.04	RHEL-family bootc	Favours
Can OneMind legally ship the image? (G1)	Needs Canonical's approval, or the brand removed and binaries recompiled [C-W2-050]	AlmaLinux free to redistribute [C-STEEL4-bootc-014]	bootc
Cost per device at 10,000 units (DC-03)	Pro for Devices has no published price [C-W2-006]	AlmaLinux bootc costs about the same as Debian (`lifecycle-cost-2036.md`)	bootc
Upgrade to the next major release without a reflash (DC-05)	No production precedent found	By design, shown in our lab [C-LAB-fedora-bootc-015]	bootc
CRA free-software route (DC-04)	Modified-Ubuntu terms add a vendor dependency	AlmaLinux clean [C-W6-013]	bootc
One update framework (DC-07)	RAUC on x86 and Pi 5, NVIDIA's on Jetson [C-W3-049]	greenboot is GRUB-only; Pi experimental [C-W3-058]	Ubuntu
Pi 5 pipeline (DC-19)	Certified [C-W2-057], with an open boot bug on older Pi 5 firmware [C-DEC-006]	Experimental, manual steps [C-SCORE-025]	Ubuntu
NVIDIA GPUs on the base (DC-13)	Validated with k3s [C-W2-026]; Canonical builds signed NVIDIA modules per kernel [C-STEEL-ubu-016]	Validated for RKE2 [C-W2-026]	Ubuntu
ROS 2 support tier (DC-15)	Tier 1 on x86 and Pi [C-W7-027]	Tier 2 on amd64	Ubuntu
Who carries NVIDIA forward?	Canonical ships CUDA in 26.04 [C-STEEL-ubu-003]	Red Hat Device Edge on Orin, paid [C-W2-041]; the free Jetson route is frozen [C-STEEL4-bootc-013]	Ubuntu
Lab: the same RAUC + real-k3s exam	Pass [C-LAB-ubuntu-2604-leg-011]	A power cut during finalize **silently downgraded a healthy box** [C-LAB-fedora-bootc-011]	Ubuntu

Question (decision-critical row)	Ubuntu 26.04	RHEL-family bootc	Favours
One OS Pi → cluster (G5), IIoT vendor process, governance, proven competitor, signed GPU modules under Secure Boot	—	—	Equal

Result: Ubuntu beats bootc 6–4. Frozen-protocol S1 result: Ubuntu 26.04 LTS + RAUC A/B, with NVIDIA's L4T Ubuntu flavor on Jetson. Fragile (11 of 14 runs). Conditional on Canonical's permission (gate G1, Z-01) and on G5. The same rows also give the other pairs, which matter when a run flips or an answer arrives: Debian beats Ubuntu 6–5 and bootc beats Debian 6–5 [C-DEC4-006]. **Every pair is decided by one or two rows.**

Card	Frozen rules (top-two tie, row-count tie-break)	Revision-3 three-way rule (withdrawn)
Revision 3 (bootc 64 · Ubuntu 62 · Debian 60)	**Ubuntu** > bootc > Debian	Debian > Ubuntu > bootc
Revision 4 (Ubuntu 65 · bootc 64 · Debian 63)	**Ubuntu** > bootc > Debian	No winner (a cycle)

The case for Debian is real, but in the rules it is only tie-break rows. No one's permission is needed to ship its core image; the closest constraint-matched competitor (umbrelOS) runs it; and every lab ran on it, now including native Arm and Secure Boot on RAUC 1.15.2 [C-W2-043, C-LAB-aarch64-gate-004, C-LAB-secureboot-1152-005]. You can still choose Debian yourself: Z-38 sets out the options, and the choice is then logged as yours, not as the scorecard's (Chapter 14).

4.3.1 Revision 4.1: three blind attacks, no change

Earlier attackers (refuters) had read the research's own reasoning. In revision 4.1 three new ones were given **only the one-line decision** and asked to disprove it with their own sources. All three returned **weakened**; none returned **refuted**, and none found a score that disagrees with its source, so no score moves (`01` §5.1.3).

Lens	Verdict	Strongest point	What would turn it into "refuted"
Technical	Weakened	"One OS" is really a family: a 26.04 core, a 24.04 Jetson flavor whose standard support ends in 2029, and a separate Pi kernel [C-REF5-S1-technical-003, C-REF5-S1-technical-004]. RAUC has no merged Pi firmware fail-safe backend [C-REF5-S1-technical-006]	Refuted if Canonical refuses and the rebuild costs more than Debian with paid extended support, or if RAUC cannot replace Ubuntu's Pi boot fallback while a Debian A/B image passes the same power-cut test
Longevity / vendor	Weakened	Canonical's permission "can be revoked at any time" [C-REF5-S1L-002]; JetPack 7 stays on Ubuntu 24.04 [C-REF5-S1L-009]	Refuted by a refusal, or a revocable-only offer with an unaffordable rebuild, or a Pro for Devices price that breaks the bill of materials
Security	Weakened	If OneMind must strip the brand and rebuild, it loses Canonical's signed boot chain and must get its own shim signed [C-REF5S1SEC-001, C-REF5S1SEC-002]; "supported to 2036" holds only with Ubuntu Pro [C-REF5S1SEC-006]	Refuted if the rebuild route is forced with no shim plan, no hardware key store and no promise on how fast security fixes are rebuilt

What this changes: no score and no ranking. It sharpens two conditions: Canonical's answer (Z-01) must cover revocation and say which route ships, and "one OS" means one app platform over a 26.04 core and a 24.04 Jetson flavor. **The decision of record stays Ubuntu 26.04 + RAUC: fragile (11 of 14 runs), conditional on Z-01 and G5.**

4.4 What changes the result: written in advance

Three results are pending: **Canonical's reply (Z-01)**, **the bootc test on a real Pi 5 (W2-HW-3)**, and **Ubuntu's own Secure Boot test (U-SB)**. U-SB was added after three refuters attacked this result: Ubuntu's one-point lead rests on a Secure Boot score earned by a lab that ran **Debian** (01 §5.1.2) [C-FIX4-010, C-FIX4-014]. On 2026-11-11, before any of the three was known, the research applied the frozen rules to each outcome and recorded the result (01 §3.1.6 h) [C-DEC4-007]:

Event	Card	Base-OS result
None of the three results yet (today)	Ubuntu 65 · bootc 64 · Debian 63	**Ubuntu** (fragile)
U-SB fails, or passes only after a partition or bootloader change	bootc 64 · Debian 63 · Ubuntu 62	**RHEL-family bootc**, 6–5 over Debian (bootc 11, Ubuntu 3 of 14 runs)
U-SB passes unchanged in two runs	Ubuntu 65 · bootc 64 · Debian 63	**Ubuntu** (as today; its Secure Boot score is then lab-tested)
U-SB fails and W2-HW-3 fails	Debian 63 · Ubuntu 62 · bootc 60	**Debian 13**, 6–5 over Ubuntu (14 of 14 runs)
U-SB fails and Canonical refuses	bootc 64 · Debian 63	**RHEL-family bootc**, 6–5 (14 of 14 runs)
W2-HW-3 fails (bootc on a real Pi 5)	Ubuntu 65 · Debian 63 · bootc 60	**Debian 13**, 6–5 over Ubuntu
Canonical refuses, and you reject the de-branded rebuild	bootc 64 · Debian 63	**RHEL-family bootc**, 6–5 over Debian
Canonical refuses and W2-HW-3 fails	Debian 63 · bootc 60	Debian leads, but only in 7 of 14 runs: **unstable, your decision**
Canonical agrees and publishes a price	Ubuntu 67 · bootc 64 · Debian 63	**Ubuntu outright**

U-SB was tried, and is blocked, not failed. The test's pass/fail rule was frozen at 15:51 UTC on 2026-11-11, before any build, and the run started at once in a disposable virtual machine. It stopped because Ubuntu's download servers do not resolve from the research environment, while Debian, Fedora, kernel.org and GitHub do [C-FIX5-001, C-FIX5-002]. The rules forbid working around a block, so it was not run. **"Blocked" is not an event in the table:** no score moves, and Ubuntu's lead still rests on a Secure Boot score with no Ubuntu lab behind it. The kit is ready, with only the pre-registered Ubuntu changes [C-FIX5-003]: **you can run it as lab test LT-VM-20 on your Mac on day 1 (2026-11-16)**, about 1.5–2 hours unattended, twice (22 LT-VM-20, 20 D1). That is decision Z-41 (Chapter 14).

What this means in practice. One image recipe builds an **Ubuntu 26.04 version and a Debian 13 version** side by side (switching between them costs about 40 lines of bootstrap change and no gate change [C-LAB-ubuntu-2604-leg-011]). Nothing Ubuntu-based ships before Canonical answers. The base is **locked at gate G6** (April 2027), before anyone else installs OneMind. bootc is a different build, and a box shipped on Debian or Ubuntu + RAUC cannot move to bootc without a reflash [C-STEEL4-bootc-025], so the **bootc Pi 5 test runs before week 4**.

Variants a reviewer might prefer. At least seven other readings change the winner (01 §3.1.6 g, V1–V6 and V8), and none was applied, because each would treat candidates unequally or change a rule after scoring [C-DEC4-007]. Scoring "experimental" Pi 5 support as 2 for bootc gives **Debian** (V1); scoring Ubuntu's longevity cell 2 gives **bootc** (V2); keeping Ubuntu's Secure Boot cell at 3 until U-SB runs gives **bootc** (V3); cutting bootc's update-safety cell to 2 for the power-cut bug gives **Debian** (V4); the security refuter's own card gives **Debian** (V5); scoring "proven in production" on the whole stack including the update engine gives **bootc** outright (V6); and reading Canonical's condition as a fail for Ubuntu alone gives **bootc** (V8). Z-40 asks you to write these readings down **before** W2-HW-3 runs, so no rule is chosen after its result is known.

4.5 Lifecycle and cost to 2036

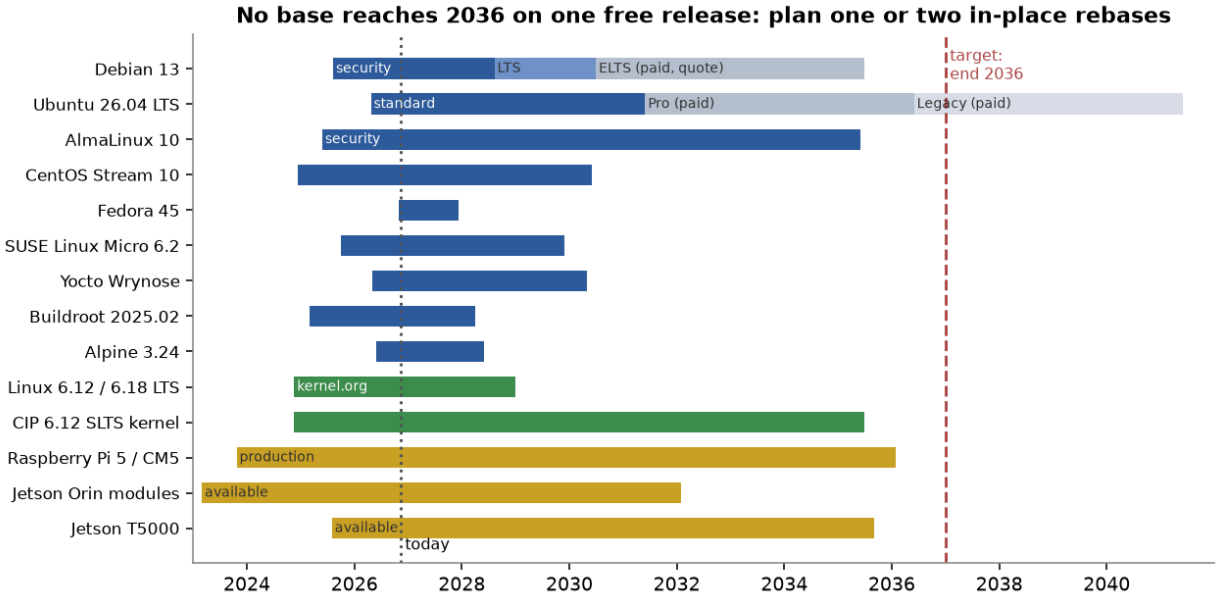


Figure 5: Support windows (end dates cited in 03 §6.1; bar starts approximate). Every base needs a rebase or a paid tail to reach 2036.

A box sold in late 2027 with a 7–10-year promise lives to about 2035–2037. **Every** base needs either one in-place rebase (Debian 13 → 14/15, AlmaLinux 10 → 11, Ubuntu 26.04 → 28.04) or a paid tail [C-W2-001, C-W2-005, C-W2-013]. Debian releases about every two years (11 in 2021, 12 in 2023, 13 in 2025) [C-COST-007]. The Jetson Orin module itself ends in January 2032 [C-W2-030], before any OS end.

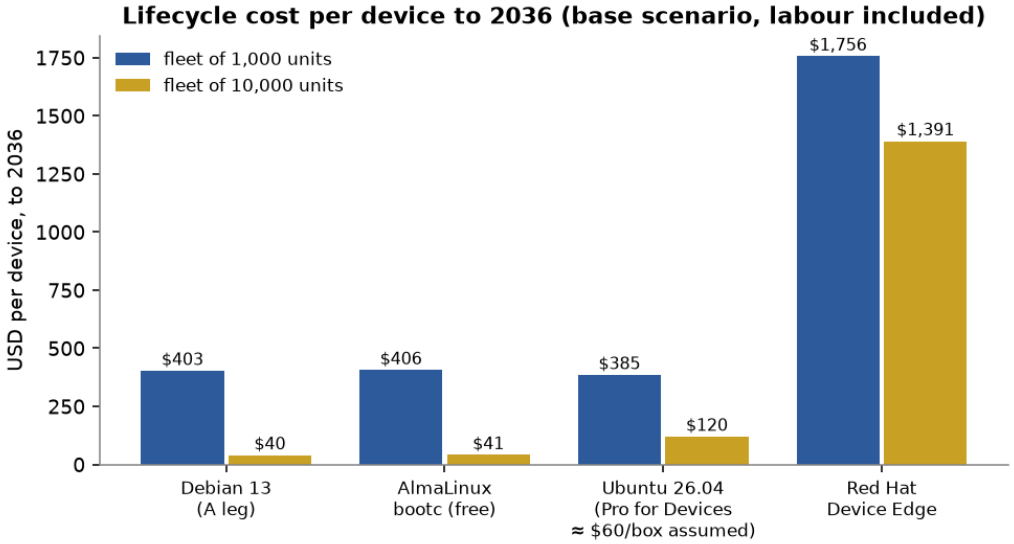


Figure 6: Cost per device to 2036 in the base scenario (01 §3.1.5, evidence/lifecycle-cost-2036.md). Ubuntu's figure assumes about $60 per box for Pro for Devices, which Canonical does not publish.

Ubuntu sells Pro as an annual or one-time fee for up to 10 years, with Legacy beyond [C-COST-002]; Red Hat Device Edge is "RHEL at a reduced cost" [C-COST-009]. **Labour, not licences, dominates every base**: about 1,680 base hours plus 1,480 CRA hours to 2036 on Debian at the middle assumption (a computed estimate, `evidence/lifecycle-cost-2036.md`). That is the main argument for hiring a release engineer first (Chapter 11).

4.6 Your Fedora IoT question (F-12)

> **Short answer: Fedora IoT is a profile and a canary, not the base, today. The Red Hat bootc lineage is genuinely excellent and becomes OneMind's Industrial profile in year 2.**

What we checked (`03-base-os.md` §9, at Tier A depth):

- **It works.** In our VM lab, Fedora IoT bootc with greenboot booted with composefs and SELinux, accepted an update only after k3s was healthy, and rolled back a broken update with zero manual commands [C-W2-054].
- **But not fast enough or safe enough yet.** That automatic rollback took 32 minutes against a 15-minute target [C-W2-054]. A power cut during the update's "finalize" step **silently downgraded a healthy box**, twice out of two [C-LAB-fedora-bootc-011], and the default update timer re-staged a known-bad image [C-LAB-fedora-bootc-009].
- **Each Fedora release lives about 13 months** with no "stay one behind" option [C-W2-015]; Fedora 45 ends 2027-12-08.
- **Pi 5 is experimental.** Fedora 44/45 run on Pi 5 with gaps (no sound, no USB boot; CM5 still in progress, SECONDARY) [C-SCORE-027]; bootc images need a bootloader change that is still an open pull request [C-PLAN180-003].
- **FDO zero-touch onboarding is no longer a Fedora advantage:** Fedora IoT's own FDO was deprecated, and the maintained code is distro-agnostic [C-W8-042, C-W8-044].
- **The Red Hat lineage has real industrial credentials:** RHEL 9.6 and MicroShift hold IEC 62443-4-2 security level 2 [C-W2-042], and Red Hat Device Edge runs on Jetson Orin (paid) [C-W2-041].

So: one Containerfile builds on **AlmaLinux 10** (free) or **RHEL 10 image mode** (paid, IEC 62443 path), gated by greenboot on k3s, as the **Industrial profile** in year 2 (Z-10). CentOS Stream 10, which ends 2030-05-31, and Fedora IoT bootc build in CI only as **early-warning canaries** [C-REF4-DC-longevity-001]. Two industrial tests (FDO on both stacks, a Flight Control fleet simulation) were pulled into the first 180 days, so if the core base blocks industrial onboarding, the switch happens in year 1, not year 2 (Z-27). If bootc instead becomes the core base through the decision map above, this profile becomes the main line. This overturns ADR-001's implicit "no Fedora/RHEL role".

4.7 One base or two? And why not build from source?

- **Two image families plus a flavor is unavoidable** while NVIDIA owns Jetson's software: CUDA 13.5 lists Debian 13 for x86 and Arm servers but **not** for Jetson, where only Ubuntu 24.04 is listed [C-CIT4-003]. Even with an Ubuntu 26.04 core, Jetson stays on NVIDIA's own Ubuntu 24.04, a second release. OneMind keeps *one release contract* over them (Chapter 7).
- **A Jetson image that does not depend on Ubuntu** (Yocto meta-tegra with the 6.18 LTS kernel) is tested inside 180 days: it is packaged upstream for Orin and Thor, and balenaOS ships Orin on it [C-STEEL-jb-002, C-STEEL-jb-010]. It becomes the Jetson default only if CUDA, cameras and A/B all pass (Z-16).
- **Building from source the Home Assistant way is not a security shortcut.** We built the same tiny OS both ways: Buildroot is 4–6× smaller, but has only 7–18% fewer known vulnerabilities, because most of them live in the kernel either way [C-LAB-minimal-image-dc20-011]. The refuters showed that CVE count alone cannot settle this, so the distro base now stands on two other reasons: the NVIDIA/CUDA path, and what one person can maintain (`01` R4-9).

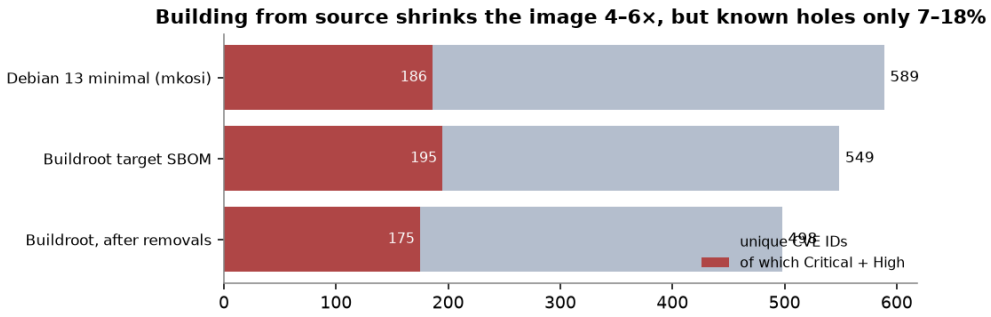


Figure 7: Distro vs build-from-source, measured. Size drops a lot; known holes barely move.

4.8 What competitors build on (constraint fit)

Product	Base	Constraint fit with OneMind	What OneMind takes
umbrelOS 2.0	Debian 13, snapshot-pinned, built as a container [C-TD-umbrelos-003]	**High** (Pi 5 + x86, small team)	Image-as-container-build, over-the-air major rebase; the Debian leg's precedent
Home Assistant OS	Buildroot, one image per board [C-TD-home-assistant-os-003]	High for the appliance; base differs	RAUC, three boot tries, CA rotation
StartOS	Debian 13 via live-build [C-TD-startos-005]	Medium	Second Debian appliance precedent
Proxmox VE	Debian 13 + own packages [C-TD-proxmox-ve-003]	Medium	Debian holds up as an enterprise base
TrueNAS SCALE	Debian 12 from a frozen mirror [C-TD-truenas-scale-003]	Medium	Moved apps off Kubernetes: keep apps portable
Olares	Host Ubuntu or Debian, no image [C-TD-olares-007]	None for OS safety	Nothing at this layer
balenaOS	Own Yocto distribution [C-TD-balenaos-010]	Medium (fleet product, no k3s)	meta-tegra is shippable on Orin
Red Hat Device Edge	RHEL bootc on Orin [C-W2-041]	High for Industrial only	The Industrial profile

Requirements served: RS-1, RS-2, R-1, R-3, M-6, M-7, N-1, N-2, N-7, N-8, H-3, P-5, F-1, F-2, F-7, F-11, F-12, F-14, D-001 (`03` §13, `01` §11).

Chapter 5
Owning the kernel and the image, and building from a Mac

Explain it like I'm the founder. You said: *"I technically want my own kernel, and everything on top of it, or our own image if necessary."* The research answer is: **own the image, own the kernel's *contract*, own the update and signing chain, own the SDK. Do not write a kernel, and do not compile one at launch.** That is exactly what Umbrel and Home Assistant own, and a small team has kept it running in production for years. Writing or forking a kernel is what Google and Huawei did with thousands of engineers, and it would not help any of your goals. And your Mac is a fine build machine, as long as every build runs inside a Linux virtual machine.

5.1 Kernel families: Darwin, BSD, Linux and microkernels (F-15)

- **Linux** is *monolithic*: drivers, filesystems and networking run inside the kernel. It is GPLv2, so anyone shipping it must offer the source [C-W18-069]. Containers are built from Linux features (namespaces and cgroups), which is why Kubernetes is Linux-native.
- **Darwin/XNU** (macOS, iOS) is a *hybrid*: Apple describes XNU as combining the Mach kernel from Carnegie Mellon with FreeBSD components and the IOKit driver API, under the Apple Public Source License 2.0 [C-W18-030].
- **The BSDs** are monolithic Unix kernels under permissive licences; FreeBSD's container primitive is the jail [C-W18-032].
- **Microkernels** (seL4, Zircon, HongMeng, Redox) keep only the minimum in the kernel and run drivers as separate processes: safer in theory, but every driver must be written for that kernel.

Family	Pi 5 / Jetson	NVIDIA / CUDA	Kubernetes	ROS 2	Verdict
Linux	Mainline device trees and vendor kernels for both [C-W18-018, C-W18-019]	CUDA 13.5 on Debian, Ubuntu, RHEL and Fedora; on Jetson, Ubuntu 24.04 only [C-CIT4-003]	Control plane runs only on Linux [C-W18-029]	Tier 1 on Ubuntu [C-W18-034]	**Choose**
FreeBSD	No Pi 5 or Jetson evidence	Driver for x64 only, no CUDA [C-W18-028]	Not a node OS [C-W18-029]	Not listed [C-W18-034]	Reject. TrueNAS moved new work to Linux because innovation lands "on Linux first" [C-W18-033]
Darwin/XNU	Apple hardware only	No CUDA: absent from the supported-OS tables of the CUDA 13.5 guide, which list only Linux distributions [C-CIT4-003]	No Linux containers natively; Apple runs each in a small VM [C-W18-031]	macOS Tier 3 [C-W18-034]	Reject as a product kernel; it is your **development host**

Family	Pi 5 / Jetson	NVIDIA / CUDA	Kubernetes	ROS 2	Verdict
seL4	Pi 4B/5B configs verified, but boot and IOMMU outside the proof [C-W18-035]	None	None	None	Reject for the OS; maybe a future verified safety island
Zircon (Fuchsia)	One consumer device class (SECONDARY) [C-W18-039]	None	None	None	Reject
HongMeng (HarmonyOS)	Huawei devices; kept Linux ABI compatibility to reuse drivers [C-W18-036]	None	None	None	Reject; learn the lesson
Redox	VMs and a few PCs; version 0.9.0 [C-W18-040]	None	None	None	Reject (research OS)

What Darwin means for developing on the Mac: macOS cannot run a Linux container or build a Linux kernel by itself; every Linux container on a Mac runs inside a Linux VM (Docker Desktop, OrbStack, Lima, Colima, Apple's `container`) [C-W18-031]. The Mac is the editor, the terminal and the VM host; Linux does the building.

5.2 What "our own OS" realistically means

Product	Kernel	Owns	Team	Constraint fit
umbrelOS	Debian's stock kernel on PCs [C-W18-012]; Raspberry Pi's on Pi [C-W18-013]	Image, update chain, apps, UI; **no kernel build**	Small company	**High**
Home Assistant OS	Upstream LTS 6.18 with its own config [C-W18-008]; Pi fork on Pi 5 [C-W18-009]	Image, **kernel config**, update chain, apps	Funded team; about one kernel bump a week, mostly one maintainer [C-W18-010]	**High**
Talos Linux	Builds its own kernel from latest stable [C-W18-045]	Everything	Vendor	Medium
ChromeOS	Own branches, rebases every 3–6 months [C-W18-043]	Everything	Google	Low: copy the config discipline, not the effort
SteamOS 3	Valve's (SECONDARY); Arch + RAUC + casync A/B [C-W18-044]	Image, kernel, updates	Valve + Collabora	Low-medium
Android (GKI)	Vendor kernels fragmented until GKI moved board code into modules [C-W18-041]	Everything	Google + OEMs	Low; the lesson is central
HarmonyOS NEXT	Own microkernel, Linux-ABI compatible [C-W18-036]	Everything	Huawei	None

OneMind owns: the **image** (layers, partition layout, profiles), the **kernel contract** (required features checked in CI) and later the kernel config, the **update and signing chain**, and the **SDK and app platform**.

5.3 The kernel route (decision S5)

Route	What it is	Solo-founder cost	Score
K1 · board maintainer's kernel	The base's own on PCs (Canonical's, or Debian's on the Debian leg), Canonical's `linux-raspi` or Raspberry Pi's on Pi 5, NVIDIA's on Jetson	Lowest: refresh and rebuild, no kernel compile	**71**
K2 · OneMind-configured upstream LTS	Home Assistant's way	About one bump a week per kernel line (HAOS: 51 bumps a year [C-W18-010])	66
K3 · CIP super-long-term kernel	Industrial 10-year line, to mid-2035 [C-W18-003]	Like K2 on a slower branch	61
K4 · patched OneMind kernel	K2 plus own patches	Highest; not needed for real time	46
K5 · microkernel or new kernel	seL4, Zircon, own	Not feasible	Not ranked

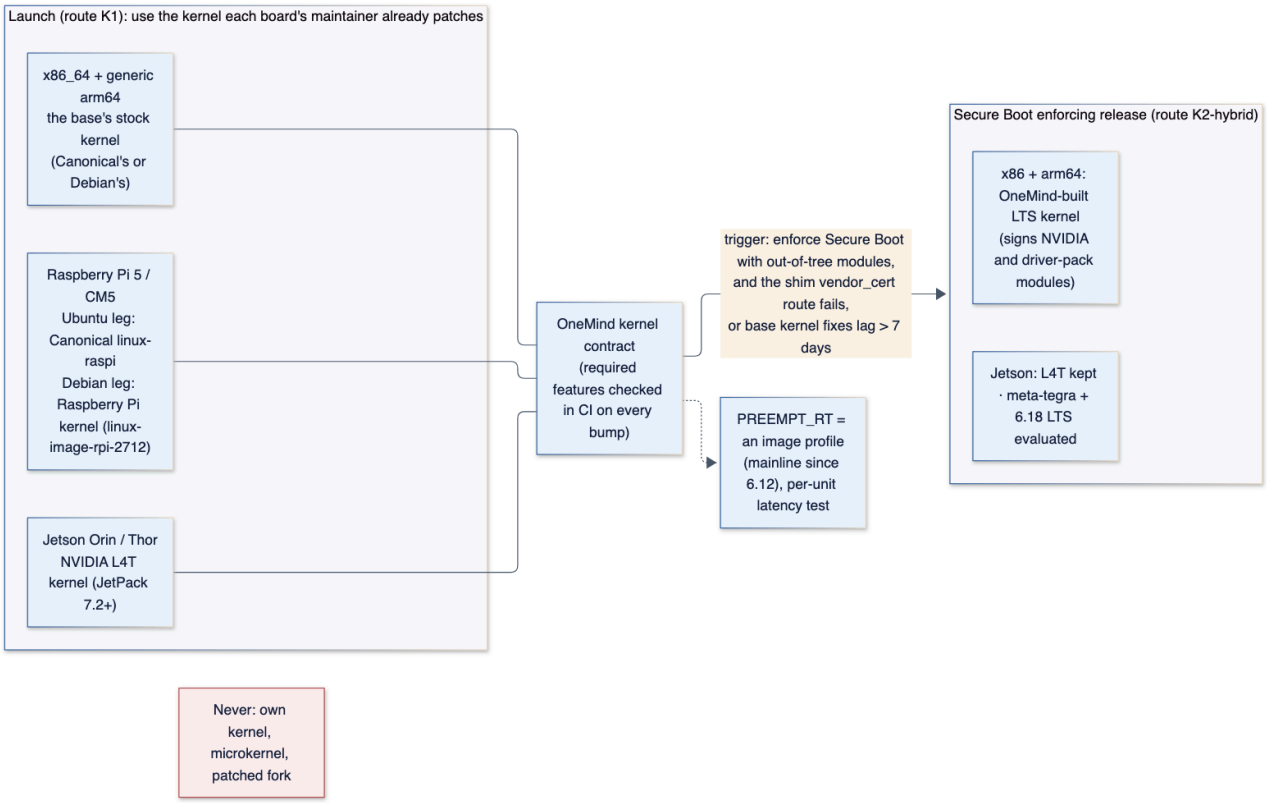


Figure 8: The kernel route: K1 at launch, a OneMind-built LTS kernel (K2-hybrid) only when Secure Boot is enforced with out-of-tree drivers and the shim route fails.

Why K1 now and K2 later. A Secure Boot key placed in the PC firmware cannot sign kernel modules for a stock kernel: the kernel trusts only built-in and "machine owner" keys for modules [C-DEC-008, C-DEC-010]. **Revision 4 found a route that may avoid that:** a OneMind *shim* (the small first-stage loader) carrying OneMind's own certificate, which the stock kernel then trusts for modules through its "machine" keyring, with no console step. It is read from source only, not yet run, so it becomes the **first arm** of the Secure Boot test T-W19-03 (`01` R4-10). Only if it fails does the release that *enforces* Secure Boot with NVIDIA or driver-pack modules move x86 and generic arm64 to a OneMind-built LTS kernel (K2-hybrid), keeping NVIDIA's L4T on Jetson. A second trigger: if the base's kernel security fixes lag upstream by more than 7 days on median, K2 comes early (`20-…` §4.1).

Real time is no longer a custom kernel. PREEMPT_RT has been a normal option in mainline Linux since 6.12 on x86, arm64 and RISC-V [C-W18-005]; Debian ships an RT kernel package [C-W18-006]; and our lab switched it on for both architectures without a single patch [C-W18-067]. It becomes an image profile with a per-unit latency test.

Kernel lifetimes that matter: kernel.org 6.12 and 6.18 LTS to December 2028 [C-W18-001]; CIP 6.12 to mid-2035 [C-W18-003]; NVIDIA's JetPack 7.2 kernel is 6.8, not an upstream LTS [C-W18-019].

5.4 Image build tooling

The image is defined like a container build on a snapshot-pinned base, Ubuntu 26.04 or Debian 13 by leg (the umbrelOS pattern [C-W2-043]), then assembled into A/B disk images and RAUC bundles per board family. umbrelOS uses a Dockerfile root filesystem plus Rugix Bakery [C-W18-015]; Home Assistant uses Buildroot; Kairos uses one Dockerfile and its own tool to produce ISO, raw and UKI images [C-TD-kairos-007]; Jetson uses NVIDIA's flash tools. 28–… §7 compares debos, mkosi, Rugix Bakery, Buildroot, Yocto, bootc/osbuild and Kairos; the lab used **mkosi** for the minimal Debian image [C-LAB-minimal-image-dc20-011].

5.5 Platform first or image first?

Both, in that order: develop the platform first as a script on plain Debian and in a VM, and produce the image from the same layers from about week 3 (28–… §8). Precedent is mixed: Home Assistant shipped its platform years before its OS [C-W18-062], while Umbrel shipped a Pi image first [C-W18-063]. The script path is a developer loop with a written sunset at image GA (Z-24), because Home Assistant and Umbrel both dropped theirs [C-REF-DC-longevity-004, C-REF-DC-longevity-005]. A self-contained container or VM developer preview may stay, as Home Assistant kept its Container install, used by about 17% of opted-in installs [C-REF3-DC-technical-004], while dropping Core [C-REF3-DC-technical-005].

5.6 The Mac development setup and CI/CD (F-8)

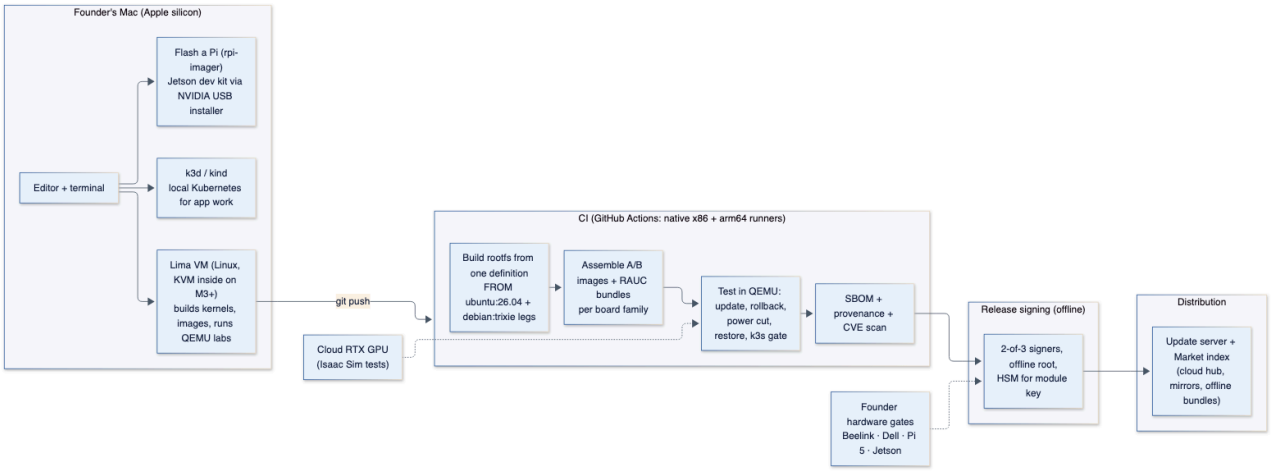


Figure 9: From the Mac to a signed release. Every build runs in Linux; signing is offline.

- **On an M3-or-newer Mac, a free Lima VM gives full Linux with KVM inside it** [C-W18-054, C-W18-065]. In our lab it compiled a full arm64 kernel in 5.5 minutes and cross-compiled an x86 kernel in under 2 [C-W18-066].
- **k3d or kind** for local Kubernetes app work; **Docker buildx** for multi-arch images.
- **A Jetson dev kit can now be set up from a Mac** with NVIDIA's new USB installer [C-W18-023]. Production Jetson flashing and fusing still need a Linux host [C-W18-021].
- **Isaac Sim needs an RTX GPU** [C-W18-058], so it runs on a rented cloud GPU, not the Mac.

- **CI on native x86 and arm64 runners** builds the root filesystem from one definition in two legs (`FROM ubuntu:26.04` and `FROM debian:trixie`, at parity until the base lock), assembles images and bundles, runs the QEMU update/rollback/power-cut/restore battery, and produces an SBOM and provenance. Signing happens offline, 2-of-3, with the module key in an HSM.

5.7 What competitors do (constraint fit)

Product	Kernel approach	Constraint fit	OneMind takes
umbrelOS	Distro and vendor kernels, no compile [C-W18-012]	High	K1 at launch
Home Assistant OS	Upstream LTS + own config, ~51 bumps/year [C-W18-010]	High	K2 when Secure Boot is enforced; the kernel-contract idea
Talos	Own kernel from latest stable [C-W18-045]	Medium	Not needed
ChromeOS, Android, HarmonyOS	Own branches or own kernel	Low / none	Config discipline and the “stable module interface” lesson only

Requirements served: F-7, F-8, F-15, D-002, M-6, R-3, N-7 (28–… §14).

Chapter 6
How everything connects

Explain it like I'm the founder. Read the picture from the bottom. A **device** (a sensor, a robot, a drone) joins a **node** (a box running the OneMind image). The image boots **k3s**, which runs the apps and the **NATS bus**. Every app declares what it can do, and the **capability gateway** turns that into tools that people and **AI agents** can use, only through the gateway, with approvals for anything risky. People reach it through the **dashboard and clients**; the **cloud** is optional and self-hostable. The **Marketplace** feeds signed apps and plugins in from the side.

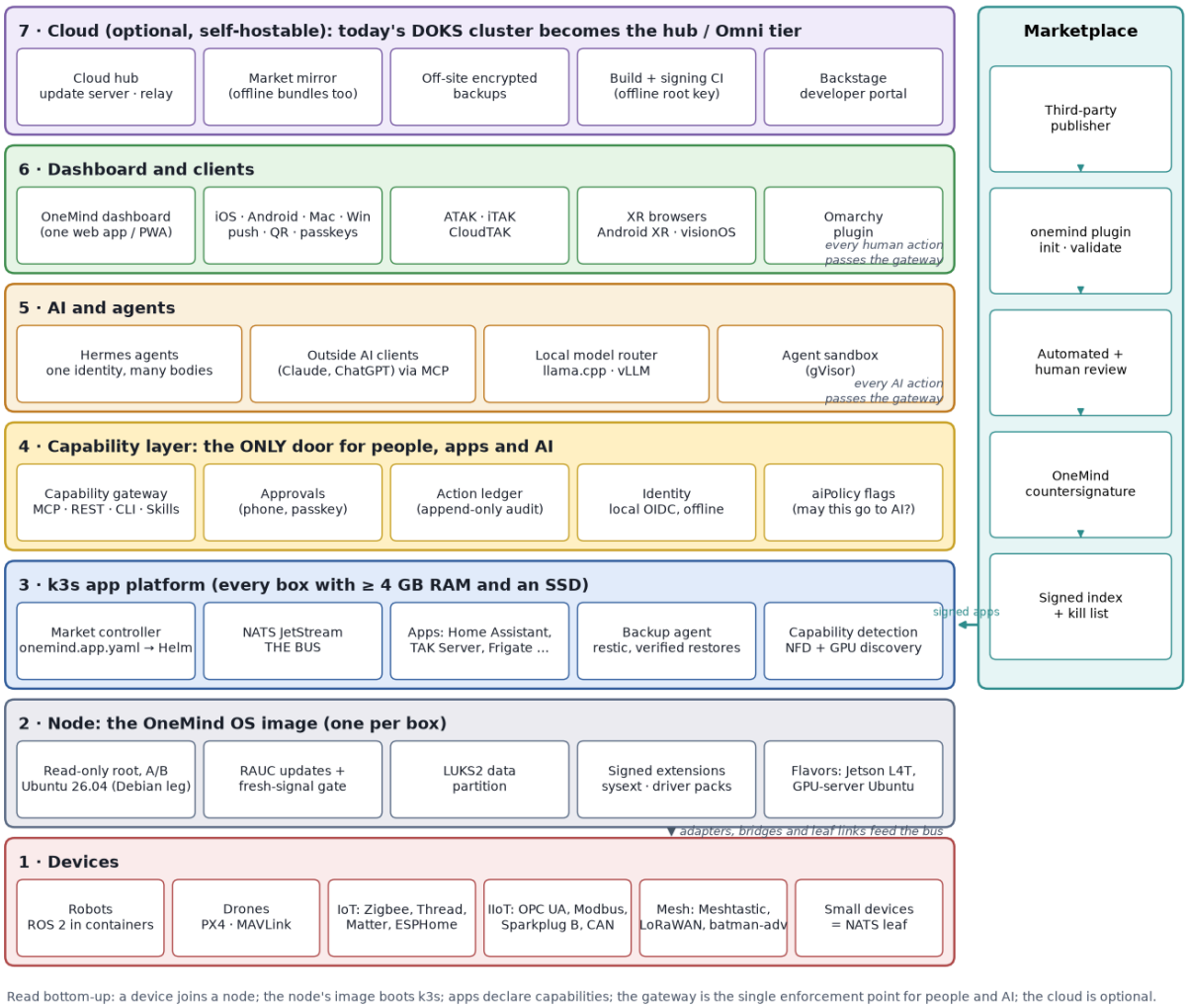


Figure 10: The OneMind layer cake (repeated from Chapter 1 for reference).

6.1 The layers, mapped onto Olares's model (P-3)

Olares organises itself in five layers (infrastructure, platform, framework, built-in apps, community apps). OneMind keeps that shape and adds two layers Olares lacks: an owned OS underneath and a device-and-robot layer beside the platform (`01` §8).

OneMind layer	What it is	Different from Olares because
L0 · OS image	Read-only A/B (RAUC) core image on Ubuntu 26.04 (decision of record, conditional on Z-01), with a Debian 13 leg until the base lock; signed releases, health gate, LUKS2 data; Jetson flavor on L4T	Olares has no OS rollback and only same-origin checksums [C-TD-olares-003, C-TD-olares-004]
L1 · Cluster	k3s (SQLite on one box, etcd at three servers), Flannel (Cilium on GPU/enterprise), Retain storage, Longhorn on 3+ nodes	Olares ships a Kubernetes minor past its end of life [C-TD-olares-022]
L2 · Platform services	NATS JetStream as the authority bus, object storage, restic backup agent, identity store	Each box has its own JetStream domain and leaf links [C-W4-054]
L2b · Devices and robots	MCP/MHS device adapters, ROS 2 bridge, MAVLink, MQTT 5 site bus bridged to NATS, TAK	Olares has none
L3 · Framework	`onemindd` update agent, Market controller, capability gateway, offline identity, Headscale/Tailscale	Default-deny egress (Olares has ingress-only policies [C-TD-olares-034])
L4 · App shell + modules	Dashboard, files, alerts, settings, map, family, finance, wellness, robotics, mesh/IoT, OSINT, education	Each module is also a Market app (D-003)
L5 · Market	Helm + `onemind.app.yaml`, public linter, signed publishers, Compose importer	Trust by signature, never by image name: Olares's linter exempts any image path containing "beclab/" [C-TD-olares-030]

6.2 Why Kubernetes at all, and where it stops

The two most-used self-hosting appliances run Docker, not Kubernetes [C-W4-043, C-W4-044], and on a relaxed scorecard Compose ties k3s (`05` §1). Kubernetes still wins for OneMind because the same app contract must run from a Pi to a GPU cluster with GitOps, multi-node placement and GPU scheduling, which is your existing cluster's whole stack. **The cost is measured:**

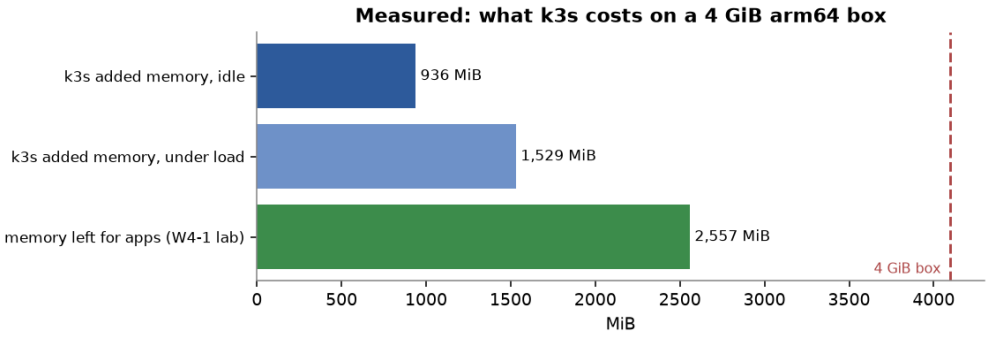


Figure 11: k3s on a 4 GiB arm64 box: about 0.9 GiB idle, 1.5 GiB under load, 2.5 GiB left for apps.

k3s added 936 MiB at idle and 1,529 MiB under load on a 4 GiB arm64 VM [C-LAB-k3s-footprint-restore-007, C-LAB-k3s-footprint-restore-008]; with an app running, 2.50 GiB stayed free and writes were about 2.75 GB/day [C-W4-078]. So:

Node class	Rule	Why
Platform node (≥ 4 GB RAM **and** SSD/NVMe; 8 GB recommended)	Runs k3s; 4 GB boxes run a slim profile sold as a profile (TAK node, greenhouse)	Measured footprint above
Capturable field node (robots, field kits)	**k3s agent only**: no API server, no datastore, no admin credentials on disk	A k3s server holds every Secret; Kubernetes assumes physically secure nodes [C-REF-W4-security-008]

Node class	Rule	Why
Pi as a cluster server	Never the server of a cluster that holds GPU-tier or fleet secrets	A Pi has no TPM, and k3s keeps its encryption key in a plain file [C-REF3-DC-security-008]
Below 4 GB or SD-only	**NATS leaf** with store-and-forward, plus an MCP/MHS adapter	600/600 messages survived an outage [C-W8-050]
Regulated / AI-conformant GPU cluster	RKE2, only when a buyer needs it	RKE2 is on the AI-conformance list; k3s is not [C-W4-038]

What competitors run (constraint fit):

Product	App runtime	Smallest box	Kubernetes?	Constraint fit with OneMind
Home Assistant OS	Docker via the Supervisor [C-W4-044]	Pi-class (HA Green)	No	Partial: appliance UX and Pi target match; no multi-node, GPU scheduling or robots
umbrelOS 2.x	Docker Compose read and written by `umbreld` [C-W4-043]	Pi 5 class	No	Partial: single box only
Olares	k3s with external etcd and add-ons disabled [C-W4-041]	8 GB RAM; Pi 4B/5 8 GB only [C-W4-042]	Yes	Closest on platform, but an 8 GB floor and no A/B image
Your DOKS cluster	Managed Kubernetes + Argo CD (`context/01`, reported)	Cloud nodes	Yes	Exact on software, wrong on hardware and hosting

Other platform choices: ingress through Gateway API with Traefik (ingress-nginx is unmaintained since March 2026 [C-W4-016]); about 8–10 Kubernetes minor upgrades to 2036, never skipping one, each preceded by a datastore snapshot [C-W4-012]; apps never depend on k3s-only features, because TrueNAS left Kubernetes for Docker in 24.10 [C-DEC-007].

6.3 The NATS bus: one nervous system from sensor to cloud

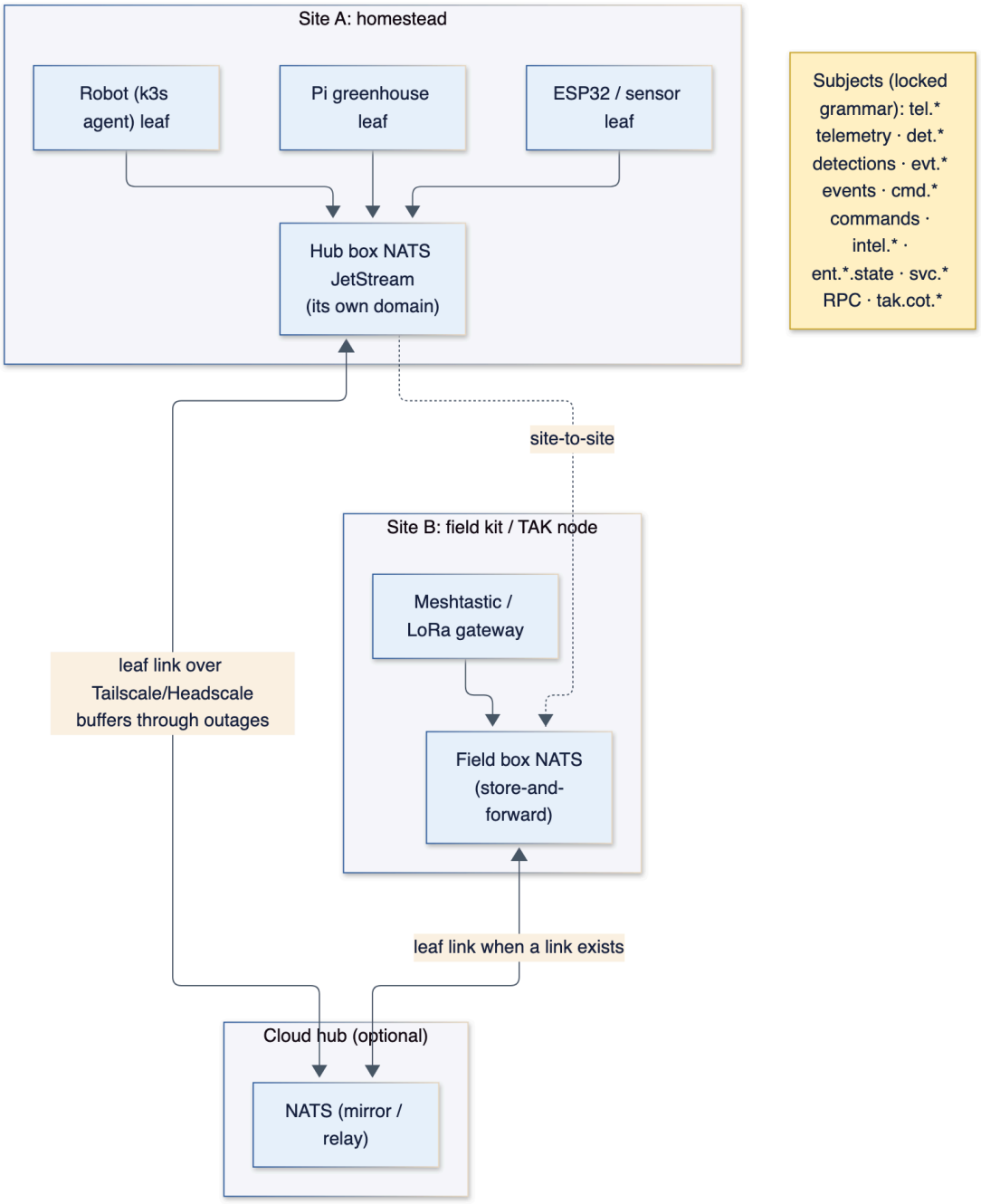


Figure 12: Every site has its own hub; small devices and robots are leaves; the cloud is a leaf or mirror. The subject grammar is locked.

NATS JetStream is the authority bus, as on your cluster today. Each box runs its own JetStream domain; devices, robots and the cloud connect as leaf nodes that buffer through outages, but only for subjects captured by a stream on the leaf [C-W8-050]. The CNCF keeps NATS under Apache-2.0 after the 2025 licence dispute [C-W4-056]. The locked subject grammar (`tel.*`, `det.*`, `evt.*`, `cmd.*`, `intel.*`, `ent.*.state`, `svc.*`, `tak.cot.*`) becomes the OS's published **bus contract**, and plugins get NATS permissions only inside it (E7).

6.4 Robots, drones, IoT and TAK on the bus

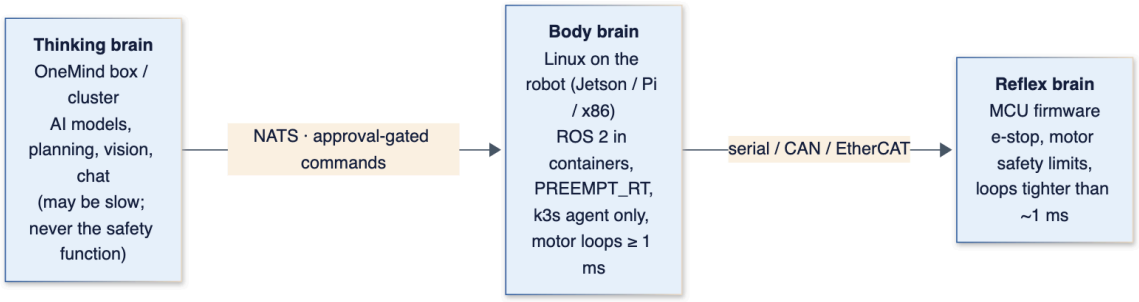


Figure 13: A real robot has three brains. OneMind owns the thinking brain and hosts the body brain; it never replaces the reflex brain.

- **Robots:** ROS 2 runs **in containers** on every flavor, because ROS 2 does not officially support Debian and its newest release does not support Jetson's Ubuntu 24.04 [C-W7-026, C-W7-027]; NVIDIA itself recommends containers for Isaac ROS. ROS 2 stays local to each robot; NATS links robots to the box (`08` §11.3). Fast DDS shared memory needs host IPC and network, so signed robot apps get a narrow, allowlisted exception [C-W7-033].
- **Drones:** PX4 (BSD-3) or ArduPilot (GPL-3.0) on the flight controller [C-LIC-047]; the OneMind Jetson or Pi is the companion computer with a MAVLink-to-NATS gateway. PX4's DDS link is unauthenticated, so it never crosses Wi-Fi or a VPN [C-W7-044].
- **IoT and IIoT:** Home Assistant for consumer radios; protocol adapters (OPC UA, Modbus, Sparkplug B) publish into a unified namespace on NATS, with an MQTT 5 broker bridged in (Chapter 3).
- **TAK:** TAK Server runs as an app; CoT flows in and out through `tak.cot.inject` and `tak.cot.>`; field devices reach it over the tailnet.

6.5 The NVIDIA toolbox on k3s, and the lock-in guard

Component	Supports Jetson?	Supports Debian?	Supports k3s?
Driver + Container Toolkit	In JetPack	Yes [C-W4-026]	n/a
k8s device plugin	Yes	Not OS-specific	Yes [C-W4-029]
GPU Operator	**No** (discrete GPUs)	**Not listed**	Listed [C-W4-022]
DRA driver, KAI Scheduler	Not stated	n/a	Needs DRA [C-W4-030, C-W4-034]
AI Cluster Runtime (AICR)	No	**No**	**No** (RKE2, k0s, Talos yes) [C-W4-037]
HAMi GPU sharing	No	n/a	Yes [C-W4-036]

Lock-in guard (F-2, F-6): apps declare accelerator *capabilities*, not vendor resources; models ship as ONNX plus per-backend engines. The same manifest must run on a Rockchip RK3588 (mainline NPU driver since Linux 6.18 [C-W7-049]), a Pi with Hailo [C-W7-053], or an x86 GPU. Google Coral is avoided for new SKUs: its repository has been archived since 2021 [C-W7-050].

6.6 Converting your DigitalOcean cluster

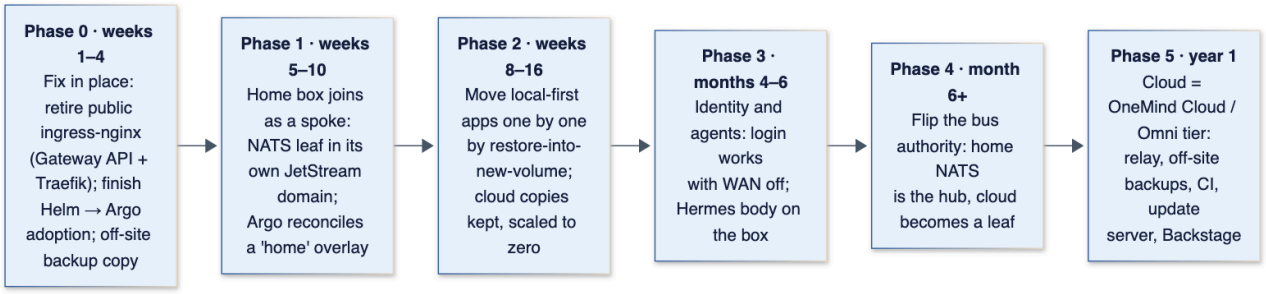


Figure 14: Six phases. Backups are the moving van; the bus authority flips last.

Principles: never move the authority twice; one app at a time; test with the internet unplugged before each cut-over. The DOKS cluster ends as the **OneMind Cloud / Omni tier**: relay, off-site backups, image build and signing CI, the update server and Backstage (`05` §9). Hermes gets a body on the box only after the bus flips (AI-9).

Requirements served: N-2, N-3, N-5, N-6, N-8, N-9, P-2, P-3, P-9, AI-8, AI-9, M-7, F-2, F-6, F-14.

Chapter 7
Updates, rollback, backups and security, as flows

Explain it like I'm the founder. These were your top four worries. Here is how each one works, drawn as a picture. **Updates:** the box keeps two copies of the OS, writes the new one into the spare copy, tries it, and only keeps it if fresh checks pass; otherwise it quietly goes back. **Backups:** everything is encrypted with restic, there is a printed recovery key, and the box proves every week that it can actually restore. **Security:** each stage of boot checks the next, the data disk is encrypted from day one, and nothing physical happens on an AI's say-so without your approval.

7.1 The update flow

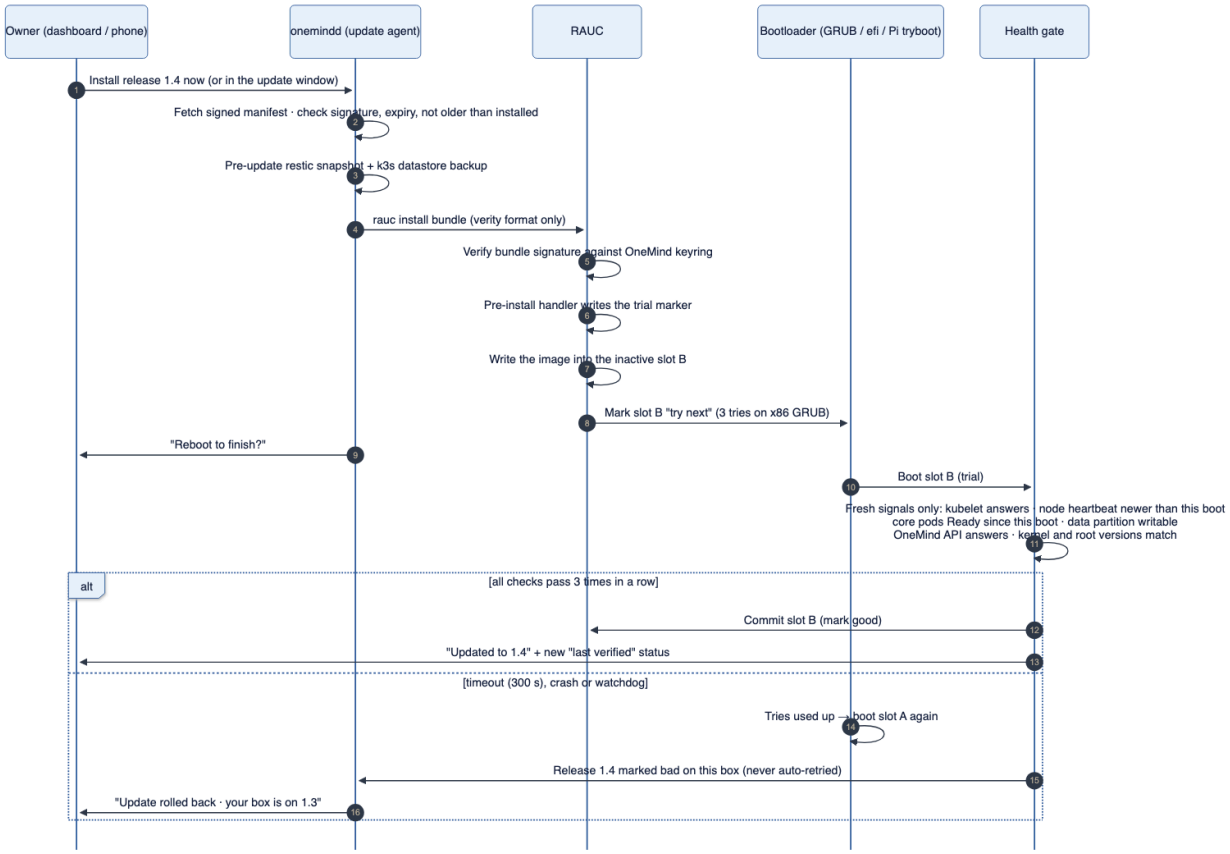


Figure 15: One update, end to end. The gate uses only signals produced after this boot.

Why the gate matters. After an update, Kubernetes shows the *previous* boot's "all good" status. A gate that only read that status committed an update whose Kubernetes never started, first with Rugix [C-LAB-debian-rugix-023] and again with RAUC and real k3s, in about a minute, in both runs [C-LAB-rauc-k3s-gate-008]. The fresh gate rejected it every time. Competitors commit earlier: Umbrel commits the moment its daemon starts [C-W3-036], and Home Assistant commits even after logging "unhealthy" [C-W3-025].

Rules that came out of the labs (`01` §4.2):

- Write the trial marker in RAUC's **pre-install** handler, because RAUC marks the slot active before post-install runs [C-LAB-rauc-k3s-gate-001].

- k3s needs bind-mounts to run on a read-only root [C-LAB-rauc-k3s-gate-004].
- Accept only **verity** bundles; a validly signed “plain” bundle was refused [C-DEC-005].
- Ship **OneMind’s own RAUC** from a pinned upstream tag: Debian’s 1.13 cannot install on kernels ≥ 6.19 [C-LAB-ubuntu-2604-leg-010].
- **Never auto-retry** a release that failed on this box: bootc’s default timer re-staged a known-bad image in our lab [C-LAB-fedora-bootc-009].
- **The same gate passed on native Arm** (RAUC 1.15.2 + k3s for arm64, gate unchanged) [C-LAB-aarch64-gate-004].
- **Where “safe” is proven, and where not yet (revision 4).** The design is shown safe for **one box on SQLite**. A multi-server cluster rolls back through k3s’s snapshot restore, which is not yet tested. Three more rules came from the refuters: app database migrations wait until the update commits; the gate gives two verdicts, **“image bad”** (final) and **“environment bad”** (for example, a core pod starved by an app: rolled back, then re-offered a limited number of times with an alert, so a hostile app cannot block a security update for ever); and after a commit, RAUC’s `prevent-late-fallback` retires the old slot and security releases raise a version floor [C-LAB-secureboot-1152-010, C-REF4-DC-security-001] (`01` R4-8).
- **Sign from release 1:** offline root, 2-of-3 signers, manifest expiry with an offline grace period, revocation, and a CA rotation drill using Home Assistant’s cross-signing method [C-W3-024]. Umbrel’s *stable* release still runs a script supplied by its server [C-W3-034].

7.2 The rollback states

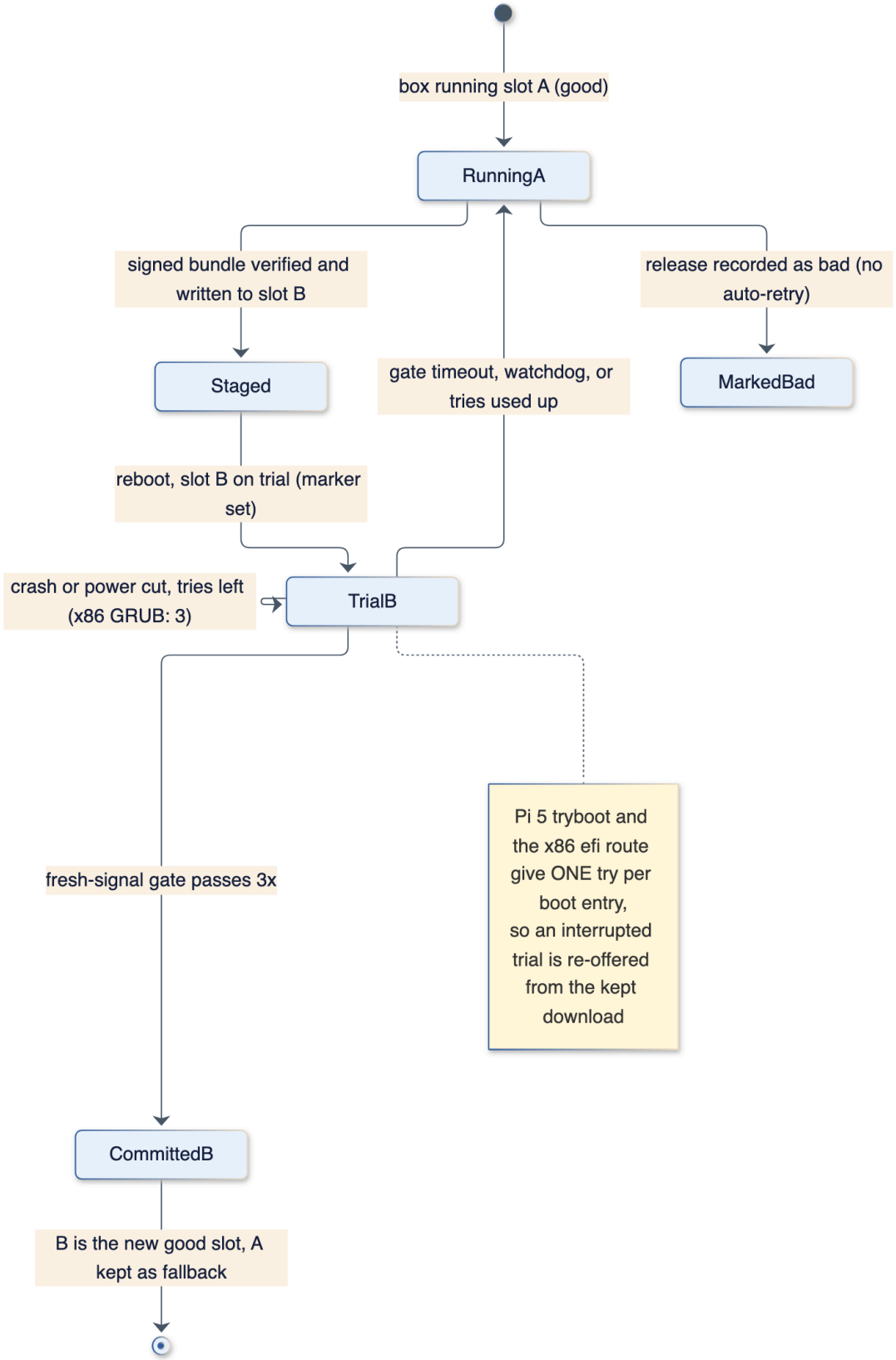


Figure 16: Every path out of a trial boot ends on a good slot.

Boot tries differ by board. The x86 GRUB route allows three tries, as Home Assistant does [C-LAB-rauc-010]. The Pi 5 firmware falls back after one missed tryboot [C-LAB-rauc-1152-012], and the x86 `efi` route gives one try per boot entry [C-LAB-secureboot-rauc-003]. So an interrupted trial is **re-offered automatically** from

the kept download, never for a release the gate marked bad. Stock Mender also allows only one try, so one power cut during that boot discards a good update [C-LAB-mender-013]. **Jetson** uses NVIDIA's own rootfs A/B and image-based OTA with OneMind's gate in NVIDIA's validation hook [C-W3-049, C-W3-050].

7.3 The update engine decision (S2)

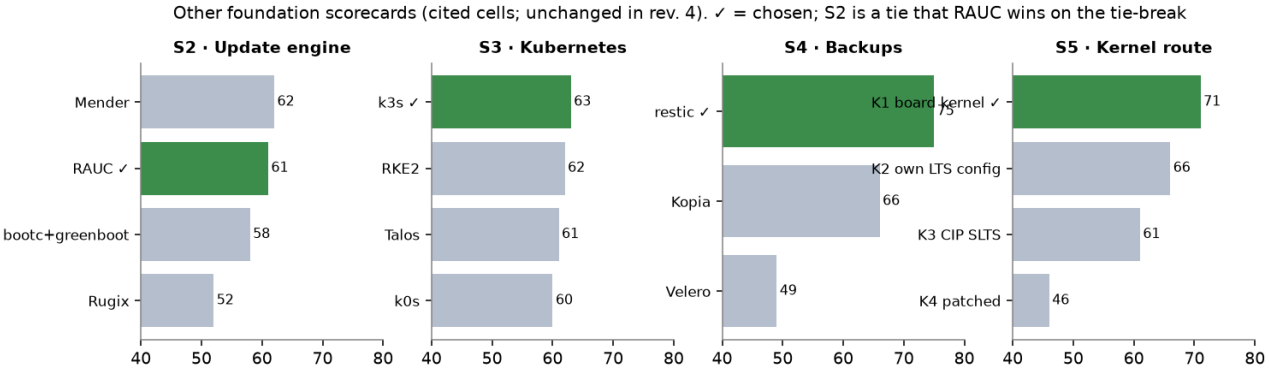


Figure 17: Update engines and the other scorecards. S2 is a tie that RAUC wins on the tie-break.

	RAUC (chosen)	Mender (runner-up)	Rugix (alternative)
Used by	Home Assistant OS, ZimaOS [C-TD-casaos-zimaos-012]	umbrelOS earlier; still a legacy variant [C-W3-040]	umbrelOS 2.0
Bus factor	Core mostly two engineers at one company [C-REF-W3-longevity-001]	Company-backed	One person wrote 303 of 304 commits in a year [C-W3-001]
Lab on our layout	Real k3s, Secure Boot, 1.15.2, native arm64: pass [C-LAB-rauc-1152-011, C-LAB-aarch64-gate-004]	Pass, with a one-try limit [C-LAB-mender-013]	Pass (x86 and arm64) [C-LAB-debian-rugix-022]
Deltas, phased rollout	Free	Paid [C-STEEL-updates-mender-015]	—
Debian/Ubuntu precedent	**None field-proven** [C-DEC-013 (UNVERIFIED: negative search)]; IronFlockOS pre-1.0 [C-CIT4-001]	umbrelOS	umbrelOS

RAUC wins the recount 5–3 on Secure Boot, revocation, free deltas, the Pi 5 path and three boot tries. **The disk layout stays engine-neutral**, so switching engines later is an update, not a reflash (Z-03).

7.4 The disk layout, decided now

#	Partition	Size	Content
1	`onemind-esp`	1 GiB	Recovery UKI (efi route) or first-stage GRUB
2–3	`onemind-boot-a/b`	512 MiB each, FAT32 ESP type	Kernel + initrd now; signed UKIs later
4–5	`onemind-system-a/b`	10 GiB each, ≥ 10% verity headroom	Read-only root
6	`onemind-recovery`	1 GiB	Reserved
7	`onemind-data`	Rest	**LUKS2 from first boot** [C-W6-047]

Changing a layout after boxes ship means a reflash, so it is fixed on day 1 even though Secure Boot enforcement comes later. Under emulated Secure Boot this exact layout booted signed kernels per slot, updated, rolled back, refused tampered and unsigned kernels and unlocked the data by TPM after every update [C-LAB-secureboot-rauc-020, C-LAB-secureboot-rauc-013].

7.5 The backup flow

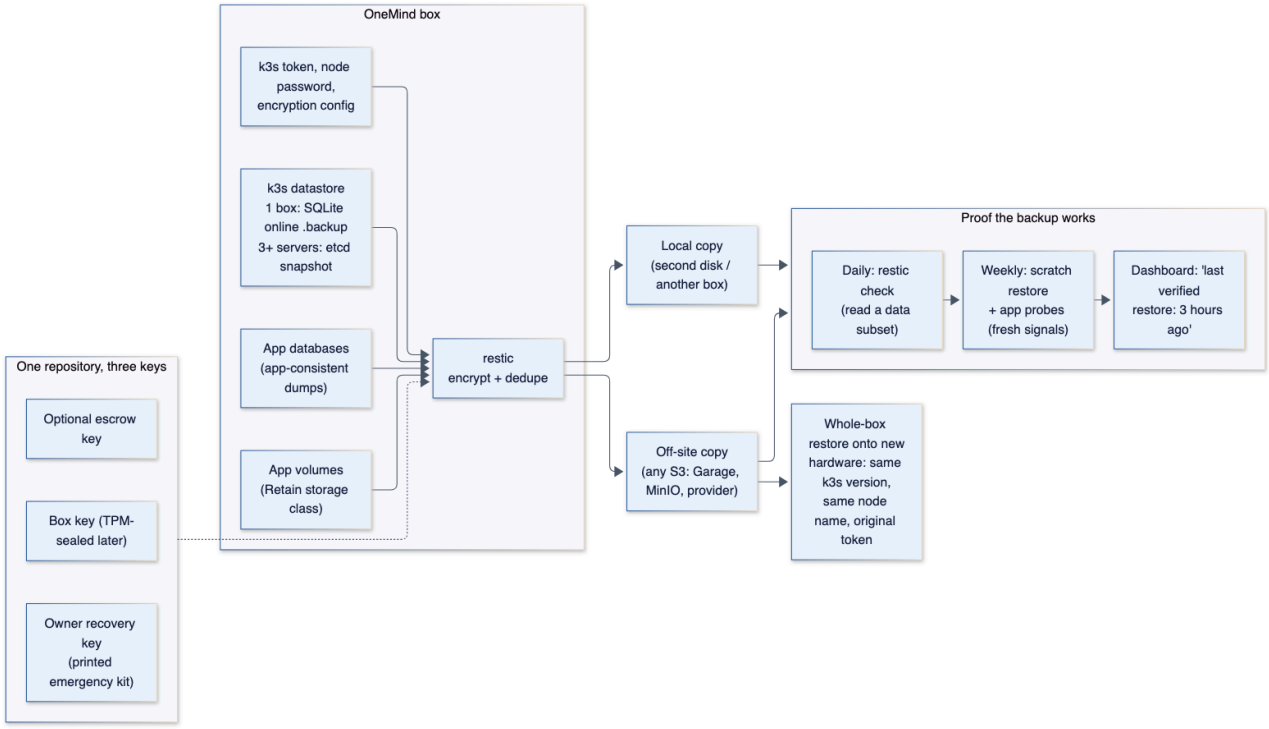


Figure 18: What is backed up, where it goes, which keys open it, and how the box proves it can restore.

- **ADR-001's `k3s etcd-snapshot` fails on a single box**, which uses SQLite, not etcd [C-W5-001]. The fix: an online SQLite backup plus the token, inside restic. In our lab a whole box restored onto a fresh VM with all 32 pods and identical data [C-LAB-k3s-footprint-restore-011].
- **Kubernetes snapshots are not encrypted**, so they always go inside restic [C-W5-013].
- **restic over Kopia:** one repository can hold several keys, so the owner gets a printed recovery key beside the box's own [C-W5-023]. restic scores 75, Kopia 66 (`01` §3.4).
- **The default storage class deletes app data when a volume claim is deleted** [C-W5-019]; OneMind uses a Retain class.
- **After a restore, Kubernetes briefly reports the old saved status** [C-W5-010], so restore checks use fresh signals too.
- **A backup counts as "verified" only with datastore-health checks** (revision 4): the k3s database inside a size budget, no compaction lag, and a k3s datastore library (kine) at or above OneMind's pinned floor (`01` §4.6). kine itself is mostly one maintainer's work [C-REF4-DC-longevity-016].
- **Nobody among the competitors proves a backup can be restored.** Your cloud cluster already does for its databases [C-W5-056]. The dashboard line "last verified restore: 3 hours ago" is the feature to lead with.

Competitor	What it backs up	Constraint fit
Home Assistant	Everything, encrypted, with an "emergency kit" key	High: copy the kit
umbrelOS	Whole box every hour with Kopia	High
Olares	restic, but restores one app at a time	Medium

7.6 The chain of trust

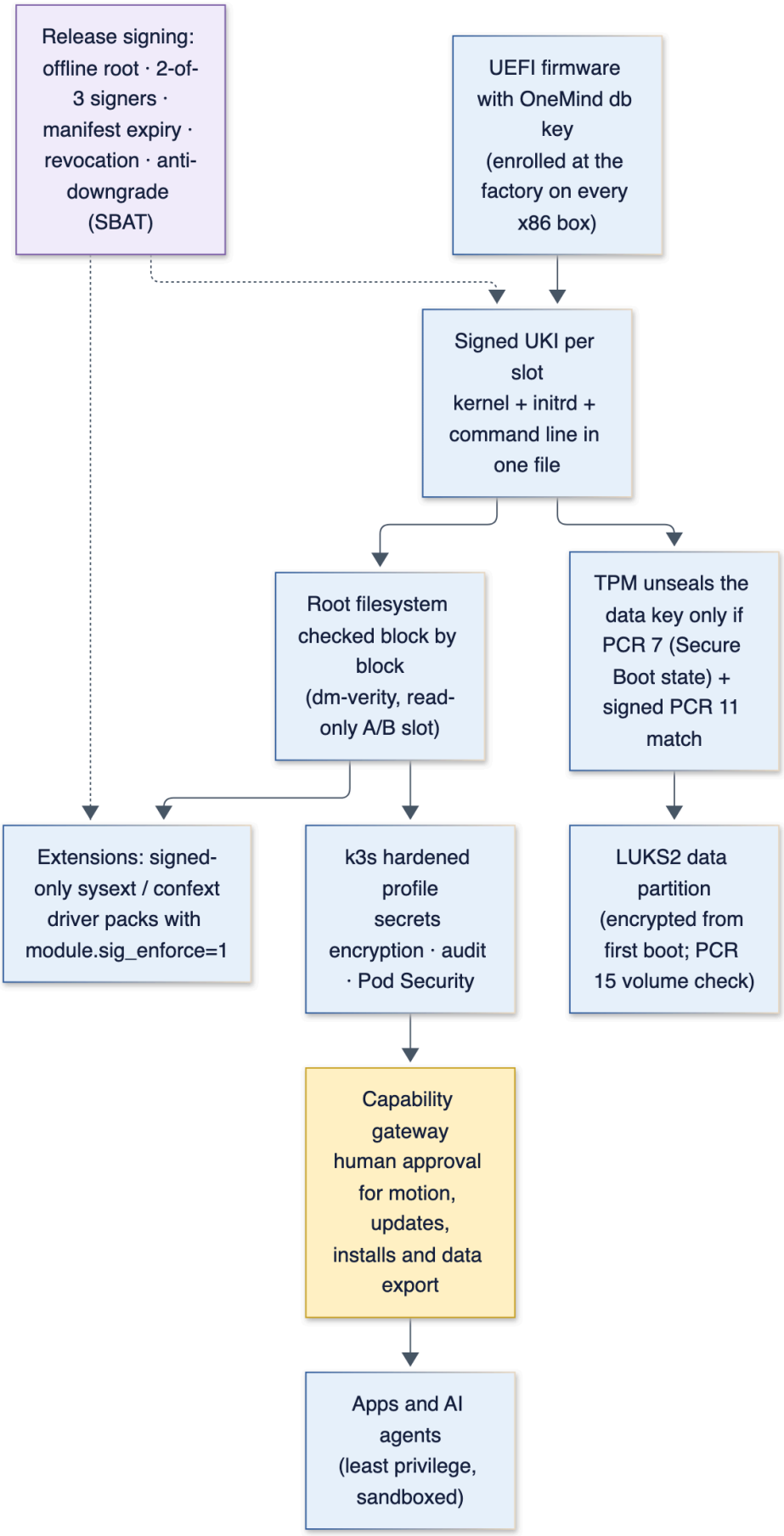


Figure 19: Each stage checks the next; the gateway is the last line between AI and the physical world.

- **Secure Boot route on x86:** RAUC's `efi` backend with one signed unified kernel image (UKI) per slot, proven in emulation with only OneMind's keys [C-LAB-secureboot-rauc-013]. **Anti-downgrade:** an old signed kernel booted against a newer root and was committed [C-LAB-secureboot-rauc-018], so the gate checks that kernel and root versions match, and security releases bump SBAT or rotate keys.
- **Data key:** bound to PCR 7 plus a signed PCR 11 policy; a literal PCR 7 binding alone turns every key change into a recovery prompt [C-REF-DC-technical-008]. The initrd also checks the unlocked volume's identity, against an attack that swaps in a fake encrypted volume [C-REF3-DC-security-009].
- **Factory key enrolment on every x86 box from the first shipment**, because the 2026 Microsoft CA rotation left some machines unbootable or needing physical access [C-REF3-DC-longevity-008].
- **Burning keys into a Pi 5 or a Jetson can never be undone**, so fusing is only for OneMind-built hardware (Z-15).
- **AI safety:** human approval enforced in the gateway for actuation, OS updates, installs and data export; prompt injection has no reliable prevention [C-W6-068].
- **Real time and robots:** an RT kernel exists on every flavor, but on the Pi 5 memory contention gave spikes over 1 ms under stress [C-REF-DC-technical-001], so control loops tighter than about 1 ms default to a microcontroller.
- **Public safety:** FIPS 140-3 modules, not RKE2's old 140-2 certificate [C-W6-026]. k3s crashes in Go's FIPS-only mode today [C-REF-DC-longevity-010], so the regulated tier runs RKE2 until that is fixed (Z-25).

7.7 The law: the Cyber Resilience Act and friends

- **The EU CRA applies from 2027-12-11** [C-W6-001]. An operating system is an "important, class I" product [C-W6-011]. Annex I asks for security updates installed automatically by default, with an opt-out, "where applicable" [C-W6-008]; that is Z-02.
- **The image and the box are classified differently:** a downloadable OS image is Class I; a box is probably classified by its own core function [C-REF-W6-technical-008].
- **Art. 13(10)** lets OneMind patch only the latest version, but only where the customer can actually upgrade on hardware they own [C-REF-W6-technical-001]; a Jetson frozen on an old JetPack must be patched on its old version for the whole declared period.
- **Route:** self-assessment as free and open-source with public documentation stays primary, with a parallel notified-body pre-assessment from 2027-Q1 (`01` §4.7). The EU Machinery Regulation (from 2027-01-14) requires robots to log safety-related decisions for a year [C-W7-046]; the AI Act's high-risk duties start 2027-12-02 and 2028-08-02 [C-W6-042]. Not legal advice: counsel items are in `07`–… §13.

Requirements served: R-1, R-2, R-3, M-2, M-4, M-8, U-4, UX-6, AI-4, AI-5, AI-7, P-7, H-4.

Chapter 8
The hardware line and box designs

Explain it like I'm the founder. The foundation already decides most of the hardware: a box that runs k3s needs at least 4 GB of RAM and an SSD, smaller gadgets join over NATS, Jetsons run NVIDIA's Ubuntu, and everything else runs the same core image (Ubuntu 26.04 under the current decision, or Debian 13 if the decision map picks it; see Chapter 4). So the boxes are not six products; they are **six sizes of one OS**. Each machine gets a "superpower" label automatically from what is inside it. And the money advice is blunt: **don't sell your own boxes in year 1.** Memory and Jetson prices jumped in 2026, so publish a certified hardware list and flashable images first, and have a factory partner build your boxes in year 2.

8.1 Six sizes of one OS

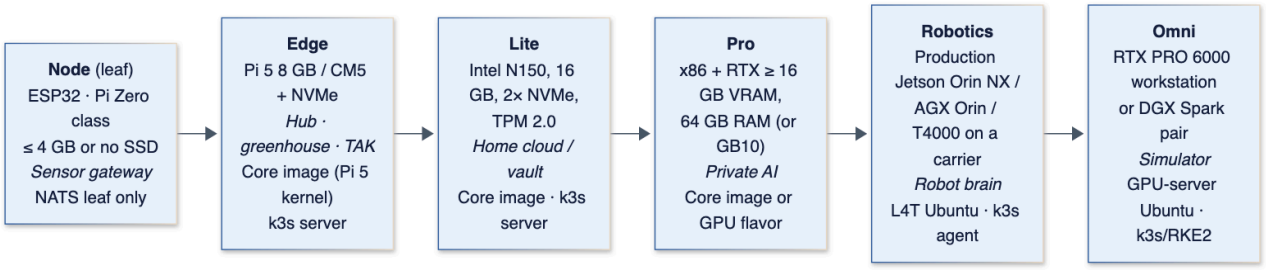


Figure 20: The line, from a sensor leaf to a simulation workstation.

Size	Superpower	Reference spec	Image · k3s role	Sourced price anchor
Node (leaf)	Sensor gateway	ESP32, Pi Zero class, ≤ 4 GB or no SSD	Vendor firmware or Debian minimal · NATS leaf only	HA Green $199 [C-W10-021]
Edge	Hub / greenhouse / TAK node	Pi 5 **8 GB** or CM5 + **NVMe**; Zigbee/Thread + LoRa; optional Hailo-10H HAT (40 TOPS, $200 [C-W10-036])	Core image (Pi 5 kernel) · k3s server	Pi 5 16 GB $385 [C-W10-034]; CM5 from $67.50 [C-W10-035]
Lite	Home cloud / vault	Intel **N150**, **16 GB** (the chip's maximum), 2+ NVMe, 2.5 GbE, TPM 2.0, custom Secure Boot keys	Core image · k3s server	Umbrel Home $499 / Pro $699 [C-W10-018]
Pro	Private AI	x86 + NVIDIA RTX ≥ 16 GB VRAM, 64 GB+ RAM; or a GB10 system	Core image + device plugin, or GPU flavor · k3s server	DGX Spark $4,699 [C-W10-032]; Olares One $4,999 [C-W10-019]
Robotics	Robot brain	**Production** Jetson Orin NX 16 GB / AGX Orin / T4000 on a partner carrier with CAN, GMSL/CSI, PoE	L4T Ubuntu 24.04 · k3s agent (or server)	Orin NX 16 GB $999, T4000 $2,999 (1k units) [C-W10-026]
Omni	Simulator	RTX PRO 6000 Blackwell workstation, 64 GB+ RAM, 16+ cores (Isaac Sim "ideal" [C-W10-016]), or a DGX Spark pair	GPU-server Ubuntu · k3s/RKE2	Not priced (no consumer competitor)

The rest of each bill of materials (enclosure, power supply, NVMe, carrier, radios, certification, freight, duty, warranty reserve) is **not priced**, because no primary or quoted source exists yet (11–… §7.3). **Why N150, not N300, for Lite:** both cap at 16 GB, but only the N150 has embedded long-availability options [C-W10-039, C-W10-040]; the 16–32 GB "Lite" in `docs/HARDWARE.md` is impossible on these chips. **Why a production Jetson module:** NVIDIA says its developer kits "are not intended for production use" [C-W10-027].

8.2 Machine superpowers and "can my box run this?" (P-6, A-8, N-3)

The OS reads what is inside each box (GPU, radios, disks, TPM, CAN bus) using Kubernetes **Node Feature Discovery** and NVIDIA's **GPU Feature Discovery**, the same parts Olares uses [C-W10-009]; NVIDIA's version understands Jetson [C-W10-014]. Rules over those labels give each machine a superpower: *Sensor gateway* (a radio dongle by USB ID), *Always-on hub* (low power, NVMe), *Vault* (two or more disks), *Private AI* (≥ 16 GB VRAM or ≥ 32 GB unified memory), *Robot brain*, *Simulator*.

Each app declares what it needs, and the dashboard answers **runs / runs with caveats / runs on another of your machines / needs new hardware**, for example: "Ollama will run on *garage-pro* (RTX, 24 GB); *greenhouse-pi* can't run it." Olares checks resources before every install [C-W10-006, C-W10-007]; Umbrel's app manifest has no hardware field at all [C-W10-004].

8.3 Bundles and the distributor model (A-7, H-1)

Bundles are data: a hardware rule, a profile and pinned apps. The same **Greenhouse** bundle can be applied to bring-your-own hardware that passes the rule, so a self-builder gets it without buying a box (P-7). **Distribution in phases:** year 1, a "Works with OneMind" certified list and flashable images, selling other makers' hardware through their normal channels; year 2, own Lite and Edge boxes from an ODM, sold direct; years 3–5, partners and distributors (11–… §7, 17–…).

8.4 2026 prices: why no own boxes in year 1

- DRAM contract prices were forecast to rise 90–95% in a single quarter, 1Q26 (SECONDARY) [C-W10-037].
- NVIDIA's Jetson Orin Nano Super kit launched at $249 [C-W10-025] and lists at $399 today [C-W10-026].
- Raspberry Pi raised the 16 GB Pi 5 by $80 to $385 on 2026-11-02, citing scarce DRAM [C-W10-034].
- Export controls, tariffs and a possible Taiwan disruption could cut SKUs (Chapter 12). Mitigations: memory-indexed pricing, second-source SKUs (RK3588-class), and riscv64 in CI.

8.5 What competitors ship (constraint fit)

Box	Price	Constraint fit	Lesson
Home Assistant Green	$199 [C-W10-021]	High	A cheap, fixed, low-support entry box sells
Umbrel Home / Pro	$499 / $699 [C-W10-018]	High	Consumer x86 boxes at the Lite price point
ZimaCube 2 Creator	$2,499 [C-W10-020]	Medium	NAS + apps hybrid
Olares One	$4,999 [C-W10-019]	High for Pro	A GPU "private AI" box is a real category
NVIDIA DGX Spark	$4,699 [C-W10-032]	Medium	Omni-class compute in a small box
ClawBox	€749 [C-W10-024]	Low	Built on a dev kit, which NVIDIA says is not for production

8.6 Your hardware: what each machine can and cannot prove

Machine	Can validate	Cannot validate
Mac (Apple silicon)	All VM labs, kernel and image builds in Lima [C-W18-066]	Real firmware, power loss, GPUs

Machine	Can validate	Cannot validate
Dell Latitude 5540	Secure Boot with your own keys: its BIOS allows custom PK/KEK/db [C-W10-043]; real power pulls; x86 install	Pi, Jetson, GPU
Beelink U55	A “worst-case old box”: i3-5005U, 16 GB max, no TPM firmware listed [C-W10-041]; power pulls	Secure Boot assurance; new CPU features
Android phone	OneMind client, ATAK	—
DigitalOcean cluster	Read-only conversion checks (22–…)	Anything physical

Buy this week (by 2026-11-16): a **Pi 5 8 GB** with an M.2 NVMe HAT and SSD, and a **Jetson Orin Nano Super** kit, so ARM and Jetson are on your desk for the week-4 to week-8 bake-off. Then a Zigbee/Thread dongle and two LoRa/Meshtastic radios by week 3 (11–… §10.1).

Requirements served: H-1, H-2, H-3, H-4, P-5, P-6, A-7, A-8, N-3, D-004.

Chapter 9
The plugin ecosystem and the Marketplace

Explain it like I'm the founder. A plugin system is a promise: *anyone can add to OneMind without asking us to change the OS.* Most plugins (apps, widgets, AI tools, automations) live inside k3s and never touch the OS. The hard ones are **drivers and OS profiles**, because the OS image is read-only. The fix is a standard Linux feature, **system extensions**, that lays a signed add-on over the read-only OS and peels it off cleanly; we tested it in a Debian 13 VM [C-W19-006]. So OneMind builds **ten sockets** into release 1. The lesson from other ecosystems is about trust: Omarchy, Obsidian, HACS and Umbrel's community stores install unsigned code from whatever is on GitHub today [C-W19-011, C-W19-021], and npm and Open VSX were hit by self-spreading worms in 2025 [C-W19-025]. OneMind controls robots, so it takes the strict path: **pinned, signed, permission-declared plugins, with a remote "pull this now" list** from day one.

9.1 Seven plugin types

#	Type	Examples	Runs where	Touches the host?
1	**Market app**	Jellyfin, Frigate, a robot fleet UI	k3s pods	Only via a declared grant
2	**Dashboard widget / UI extension**	Greenhouse gauge, robot teleop panel	Sandboxed iframe on its own origin (MCP Apps) [C-W19-035]	No
3	**Desktop/shell plugin**	Bar widget on the Omarchy edition	User's home directory, as Omarchy does [C-W19-011]	No
4	**Device / driver / protocol adapter**	ROS 2 arm driver, OPC UA bridge, MAVLink bridge, MHS device profile	k3s pod + optional udev rule or kernel module	Sometimes, through E3/E4 only
5	**AI skill / agent / tool**	Agent Skill bundle, MCP server	Agent sandbox	No
6	**Automation / workflow**	"If soil < 20 %, open valve 2"; Node-RED or n8n flow	Automation engine pod	No; every action through gateway scopes
7	**OS / image profile**	Greenhouse, TAK node, industrial gateway, kiosk	Host + Market	Yes, through E3/E4 only, never by editing the image

9.2 The ten extension points (DC-16) and how plugins plug in

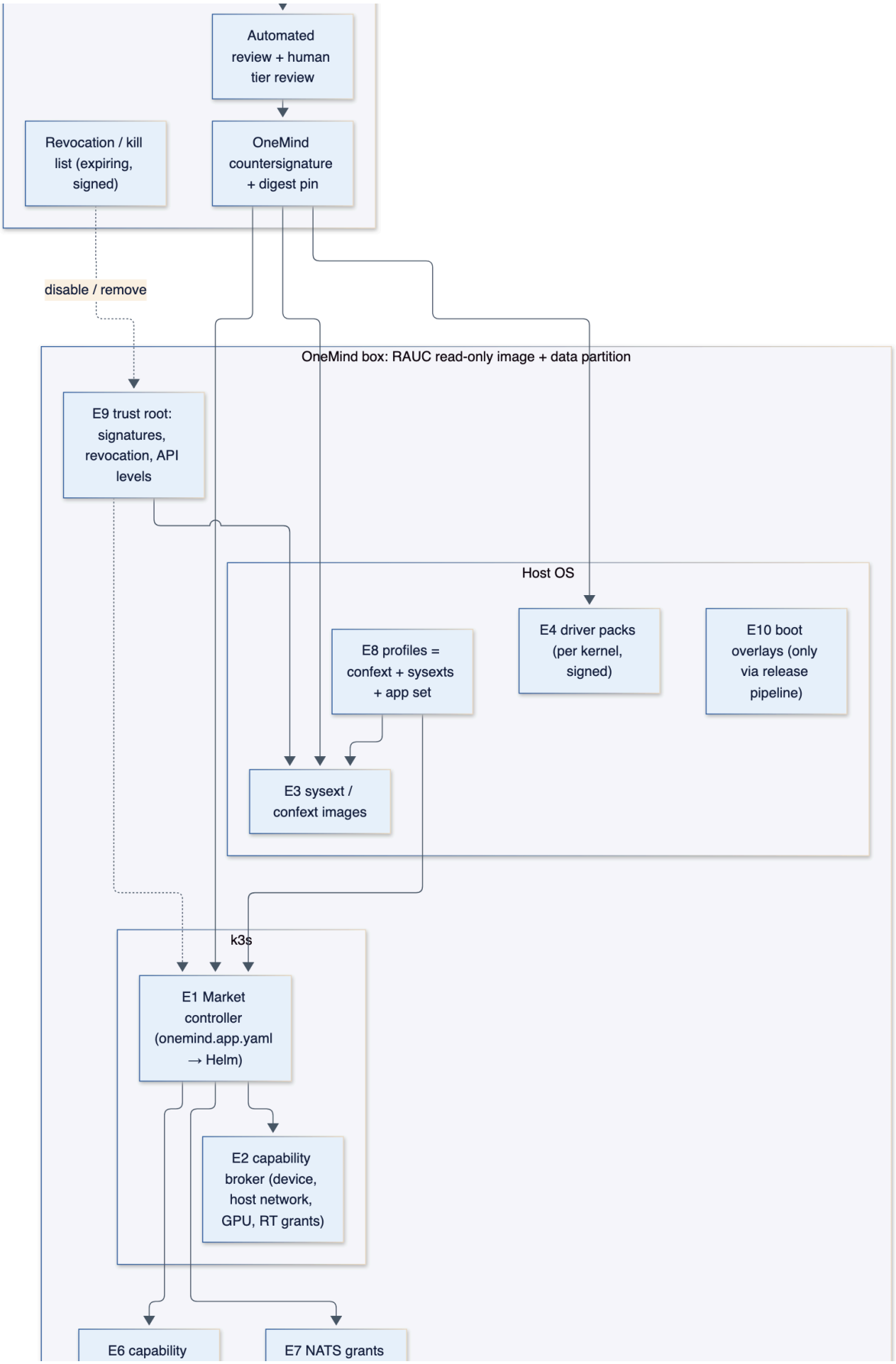

Automated review + human tier review
Revocation / kill list (expiring, signed)
OneMind countersignature + digest pin
disable / remove
OneMind box: RAUC read-only image + data partition
E9 trust root: signatures, revocation, API levels
Host OS
E4 driver packs (per kernel, signed)
E10 boot overlays (only via release pipeline)
E8 profiles = confext + sysexts + app set
E3 sysext / confext images
k3s
E1 Market controller (onemind.app.yaml → Helm)
E2 capability broker (device, host network, GPU, RT grants)
E6 capability
E7 NATS grants

Figure 21: How a third-party plugin travels from a publisher to a box, and which socket it uses.

ID	Socket	Release-1 status	Key enforcement
E1	Market controller reading `onemind.app.yaml`	Frozen v1	Helm is a compile target; Compose is imported
E2	Capability broker for host grants (devices, host network, GPU, real time)	Frozen v1	Any host grant makes a plugin effectively host root [C-REF-W19-security-003]; only counter-signed plugins get them
E3	Signed system/config extensions (sysext/confext)	Frozen v1	systemd accepts unsigned images by default [C-REF-W19-security-004]: signed-only, no directories; one bad image blocks every merge, so stage first [C-REF-W19-cost-005]
E4	Driver packs: signed kernel modules built per kernel by OneMind	Frozen v1	Per-vendor packs, as Flatcar fixed it [C-REF-W19-cost-010]; `module.sig_enforce=1` everywhere
E5	Dashboard UI slot (sandboxed iframes, MCP Apps)	v1-beta	Contract may change until real plugins use it
E6	Capability registry + approval gateway	v1-beta	Approval rules themselves are not beta
E7	NATS grants inside the locked subject grammar	Frozen v1	—
E8	Profiles (confext + sysexts + app set + defaults)	v1-beta	—
E9	Trust root: signatures, countersignature, revocation, API levels	Frozen v1	Expiring, rollback-protected revocation list [C-REF-W19-security-008]; signing proves *who*, not *what* [C-REF-W19-security-009]
E10	Boot-stage overlays (device trees, early drivers)	Frozen v1	Only through the release pipeline, because sysext cannot reach the boot partition. **Rev. 4:** the Pi EEPROM and Jetson QSPI firmware versions are pinned, gated and recorded per release, kernel × firmware combinations are tested on real boards, and the trial gate checks each profile's overlay device nodes. On a Pi that is not fused, this is a configuration control, not a security control [C-REF4-DC-security-007, C-REF4-DC-security-008]

A third-party driver binary cannot drop into any box: kernel drivers must be built by OneMind's pipeline for each OneMind kernel, which is how Flatcar does it [C-W19-005]. **The one condition on the whole design:** a test must show an attacker cannot make an unsigned extension merge after a reboot or slot switch (T-W19-01). If that fails, host extensions ship only inside the signed image and third-party host plugins wait (Z-22).

9.3 How a third party builds and publishes a plugin



Figure 22: From `onemind plugin init` to an owner's box.

Command	Does
`onemind plugin init --kind adapter`	Scaffold manifest, chart, tests and a CI workflow
`onemind plugin validate`	Same validator code as the store and the box (the Omarchy idea [C-W19-010])
`onemind plugin dev --box <name>`	Live-install onto a dev box or a VM on the Mac
`onemind plugin test`	Conformance tests per type, including a simulated robot
`onemind plugin pack`	OCI artifacts, SBOM, provenance; optional offline `.ompkg`

Command	Does
`onemind plugin publish`	From CI only, with short-lived OIDC credentials (trusted publishing [C-W19-026])

Quality ladder: Bronze is fully machine-checked, as 60% of Home Assistant's own integrations are unrated [C-REF3-W19-technical-007]; Silver and Gold add tests and support duties, modelled on Home Assistant's quality scale [C-W19-018]. **Review is a pipeline:** in a sample of Omarchy's store, one listing in five had moved past its validated commit [C-REF3-W19-technical-002], so boxes pin digests and re-validate each new one automatically, with human review only when permissions change.

9.4 "Connect a robot → use OneMind" (F-3)

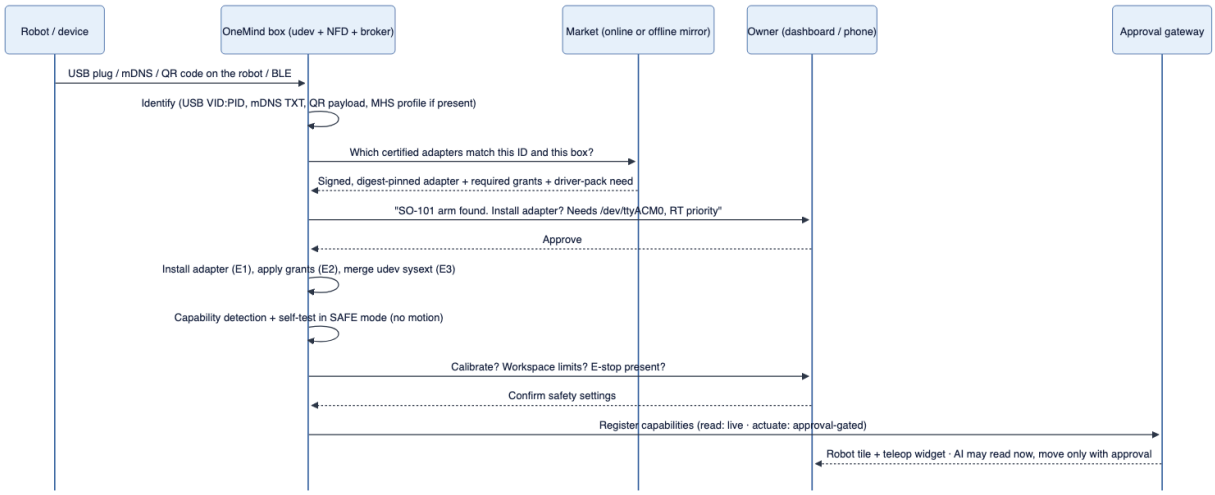


Figure 23: Robot plug-and-play: identify, match a signed adapter, approve, self-test in safe mode, register capabilities.

Non-negotiable safety gates: new robots start in *observe-only*; motion needs a declared workspace, speed limits and a hardware e-stop check; every AI-initiated motion is approval-gated; revoking an adapter means a safe stop. **First-class at launch:** robots that already have ROS 2 drivers and USB-serial or Ethernet discovery, such as LeRobot SO-100/101 arms (Apache-2.0 [C-W7-051]; Hugging Face is an MHS early adopter [C-W7-047]), TurtleBot 4, PX4 drones over MAVLink, and Jetson kits. **OneMind wraps ROS rather than competing with it**: ROS's index already holds 859 repositories for Jazzy, and its new Lyrical release targets Debian trixie [C-W19-038].

9.5 What proven ecosystems do (constraint fit)

Ecosystem	Signing / pinning	Kill switch	Constraint fit	OneMind copies
Omarchy	Clones mutable HEAD [C-W19-012]	Delisting only	High for manifest + validator; low for trust	Manifest shape, validator CLI
Home Assistant + HACS	None for custom code	HACS "critical" list auto-uninstalls every 6 h [C-W19-015]	High	Quality scale, version blocklist, kill switch
HA add-ons	Optional	—	High	Declared-privilege security rating [C-TD-home-assistant-os-025]
Works with Home Assistant	—	Badge withdrawal	High	Certification template: lab test, CHF 500/year [C-W19-020]

Ecosystem	Signing / pinning	Kill switch	Constraint fit	OneMind copies
umbrelOS community stores	Unsigned, re-cloned every 5 minutes [C-W19-033]	None	Medium	Multi-store model, ID prefixes
Olares OAC	Linter; privileged check off by default [C-W19-034]	—	High for format	k3s + Helm + manifest
VS Code Marketplace	Signs every extension [C-W19-023]	Block list + auto-uninstall	High for supply chain	Signing, malware scan, verified publishers
Grafana	Signature levels [C-W19-030]	—	High	Tiers that map to paid plugins
WordPress	Directory updates	Directory replaced a plugin by auto-update [C-W19-028]	Warning case	Kill switch needs governance
ROS	Tagged release repos [C-W19-038]	EOL distros sealed	High for robots	Wrap ROS packages
Viam	Typed `api` + `model` modules [C-W19-036]	—	High	Typed robot adapters

9.6 Hybrid self-hosted + cloud, certification and money (F-5, F-6)

- **Every plugin must work local-only.** Cloud features (relay, off-site backup, burst GPU, fleet view, store mirror) are optional plugins; the store, revocation list and fleet manager are **self-hostable**, the openBalena lesson [C-W19-044].
- **"Works with OneMind"**: a public compatibility definition per device class, an affordable annual fee, a lab test on reference boxes (Pi 5 Lite, x86 Pro, Jetson Robotics), and a badge withdrawn if the adapter is revoked. Android made compatibility a written, testable contract before granting access to its services [C-W19-039].
- **Revenue share:** free plugins list free; paid plugins keep **100% of the first USD 1M lifetime, then 85%**, the Shopify shape [C-W19-042], the most startup-friendly reference found (Apple and Google keep 15% up to $1M a year [C-W19-040, C-W19-041]).

Requirements served: A-1, A-2, A-5, A-6, AI-1, AI-3, AI-5, P-4, P-8, F-3, F-4, F-5, F-6, W-8, N-4, N-10.

Chapter 10
What the competitors actually do

Explain it like I'm the founder. Your rule was: *"Use the proven competitors for real-life documentation and use cases. They already figured it out."* We took apart 30 products, reading their build scripts, update code and app manifests at pinned commits, not their marketing. Four things stand out. **Nobody checks that an update really worked before keeping it. Signing is the industry's weak spot. Nobody does robots.** And **small teams ship these OSes**, so a solo founder can, but the key-person risk is real. Each pattern is copied only when the competitor's situation (hardware, team, Kubernetes, robots) matches OneMind's; that is the "constraint fit" column.

10.1 Four findings across every teardown

1. **No health-gated commit anywhere.** Home Assistant OS marks an update good when its Supervisor starts, even if it flags itself unhealthy [C-TD-home-assistant-os-018]; umbrelOS commits at daemon start with no watchdog found [C-TD-umbrelos-019]; ZimaOS marks good after services *start* [C-TD-casaos-zimaos-013]; Talos trusts a possibly stale "Ready" [C-TD-talos-015]. OneMind's fresh-signal gate is stricter than all of them.
2. **Signing is weak.** umbrelOS's stable release runs a server-supplied script with the bundle check switched off, and signing exists only in its beta [C-TD-umbrelos-010, C-TD-umbrelos-013]; Olares checks an MD5 from the same server [C-TD-olares-004]; ZimaOS's signing CA never expires and is named "Development" [C-TD-casaos-zimaos-049]; Synology removed package signing in DSM 7 [C-TD-tier-c-nas-036]. StartOS signs both OS updates and packages [C-TD-startos-014].
3. **No robots.** Across 11 code bases, zero files mention ROS 2 or MAVLink [C-MATRIX-002]. Home Assistant is the IoT champion but stops at vacuums and mowers.
4. **Small teams.** umbrelOS's code came from five people in a year, three of whom wrote 99% [C-MATRIX-005]; Home Assistant OS's top committer wrote 69% of a year's commits [C-MATRIX-006]; Project N.O.M.A.D. has about two maintainers [C-TD-project-nomad-004].

10.2 The comparison

Product	Base	Updates and rollback	Health gate	Apps / Kubernetes	Licence	Constraint fit
Home Assistant OS	Buildroot per board [C-TD-home-assistant-os-003]	RAUC A/B verity, 3 tries [C-TD-home-assistant-os-009]	No [C-TD-home-assistant-os-018]	Docker Supervisor	Apache-2.0	**High** (update engine, signing, UX)
umbrelOS 2.0	Debian 13 [C-TD-umbrelos-003]	Rugix A/B [C-TD-umbrelos-010]	No [C-TD-umbrelos-019]	Compose	PolyForm-NC [C-TD-umbrelos-045]	**High** (base, Pi 5, team size)
Olares	Host Ubuntu/Debian [C-TD-olares-007]	No A/B; retries forward, never back [C-TD-olares-014]	Partial [C-TD-olares-012]	k3s + Helm (OAC) [C-TD-olares-021]	AGPL-3.0 [C-TD-olares-047]	**High** (app platform); none (OS safety)

Product	Base	Updates and rollback	Health gate	Apps / Kubernetes	Licence	Constraint fit
VillageOS	None (a .NET server app) [C-TD-villageos-009]	Replace and restart [C-TD-villageos-014]	n/a	Plain processes	AGPL client, proprietary server [C-TD-villageos-007]	Low (see below)
balenaOS	Own Yocto [C-TD-balenaos-010]	Container OS into spare slot [C-TD-balenaos-012]	Partial [C-TD-balenaos-017]	Docker fleet	Apache-2.0 [C-TD-balenaos-003]	Medium (fleet; meta-tegra proof)
CasaOS	Host Debian/Ubuntu [C-TD-casaos-zimaos-020]	Downloaded script as root, no rollback [C-TD-casaos-zimaos-023]	None	Docker	Apache-2.0	Low (stalled [C-MATRIX-008])
ZimaOS	Own Buildroot [C-TD-casaos-zimaos-004]	RAUC verity, 3 tries [C-TD-casaos-zimaos-012]	No [C-TD-casaos-zimaos-013]	Docker	No licence file [C-TD-casaos-zimaos-002]	High for the engine
Kairos	Meta-distro over others [C-TD-kairos-008]	Active/passive/recovery [C-TD-kairos-019]	No [C-TD-kairos-022]	k3s	Apache-2.0 [C-TD-kairos-002]	Medium (no Pi 5)
Project N.O.M.A.D.	Host Ubuntu/Debian [C-TD-project-nomad-012]	No OS updates [C-TD-project-nomad-015]	Apps only	Docker	Apache-2.0 [C-TD-project-nomad-002]	Medium (offline library)
Proxmox VE	Debian 13 [C-TD-proxmox-ve-003]	In-place apt [C-TD-proxmox-ve-013]	None	VMs + LXC	AGPL-3.0 [C-TD-proxmox-ve-004]	Medium
StartOS	Debian 13 [C-TD-startos-005]	Image + pointer flip, no OS rollback [C-TD-startos-017]	None	Own packages	MIT [C-TD-startos-002]	Medium (signing model)
Talos	Built from source [C-TD-talos-002]	A/B with pending flag [C-TD-talos-013]	Partial [C-TD-talos-015]	Kubernetes only	MPL-2.0 [C-TD-talos-035]	Medium (no Orin)
TrueNAS SCALE	Debian 12 [C-TD-truenas-scale-003]	ZFS boot environments, manual [C-TD-truenas-scale-011]	None	Moved k3s → Docker [C-DEC-007]	GPL/LGPL [C-TD-truenas-scale-025]	Medium (portability lesson)

The full four-slice matrix (base, updates and trust, platform, market and fit) covers 22 products plus Tier C robotics fleet vendors (Formant, Foxglove, Viam, balena for robots) in `02-competitors/MATRIX.md`; Tier C products (Unraid, HexOS, Synology, Cosmos, YunoHost, Runtipi, Cloudron, Puter, Sandstorm, FreedomBox, Coolify/Dokploy, ClawBox and other AI boxes) each have a short teardown in `02-competitors/`.

10.3 What OneMind copies, and what it does better

From	Copy	Do better
Home Assistant OS	RAUC, verity bundles, 3 boot tries, CA rotation by cross-signing [C-W3-024], the encrypted backup "emergency kit", the add-on security rating, "Works with" certification	Commit only on fresh signals; Kubernetes app platform; robots

From	Copy	Do better
umbrelOS	Debian image built like a container, Pi 5 kernel pipeline, over-the-air major rebase [C-W3-039], household accounts, the beta's offline-root signing design	Signed stable releases; health gate; licence that allows OEM partners
Olares	Five-layer model, k3s + Helm app format, manifest linter, owner-signed dangerous operations, AI router and CLI skills	Own image with rollback; default-deny egress; no linter self-exemption [C-TD-olares-030]
StartOS	Signing both OS updates and packages	OS rollback
TrueNAS	Keep apps portable across runtimes	—

10.4 VillageOS: host and partner, don't compete (F-13)

You shared villageos.ai and said *"this is also why I'm going operating systems."* **VillageOS is not an operating system.** It boots nothing and has no image, no update system and no hardware; it is a server application that keeps the history of every value in a model of a village (homes, tanks, solar, farms), judges values against targets, and runs simulation services; the company calls it "a temporal graph platform" [C-TD-villageos-007]. The website's "EdgeAI without centralized dependencies" [C-TD-villageos-002] does not exist in public code: there is one central server, with readings arriving over web requests [C-TD-villageos-016, C-TD-villageos-023]. The "Stanford spin-off" claim stays unconfirmed; the code comes from ReGen Villages.

Option	Verdict
Compete on land design and finance	**No**: its simulation services are design arithmetic for pre-construction planning [C-TD-villageos-024], which no OneMind requirement asks for
Host VillageOS-style apps on OneMind	**Yes, as an optional Market app**, AGPL client parts only; the server is proprietary [C-TD-villageos-007]
Partner: OneMind as the edge OS under their "EdgeAI" stage	**Yes, low priority**: a bridge into their HTTP observation API is cheap [C-TD-villageos-022]
Own the use case (food, water, energy)	**Yes, the real answer:** a greenhouse / homestead profile with open components such as Home Assistant, OpenEMS and farmOS [C-TD-villageos-037, C-TD-villageos-040]

VillageOS is a planning brain without a body. OneMind is meant to be the body. It confirms your instinct: community resilience needs an OS on the box in the greenhouse, not another web service.

10.5 The founder-flagged repos (REA, Craft Anything, BLA)

These are AI-assisted *tools*, not operating systems: **REA** lets a coding agent take a program apart, **Craft Anything** rebuilds an app by studying the original, and **BLA** poisons your own release files so such agents waste their effort (`02-competitors/flagged-repos.md`). The decision:

1. **Don't use them for competitor teardowns.** Home Assistant OS, umbrelOS and Olares publish their source, and reading it at pinned commits is faster and legally cleaner than decompiling images.
2. **Use REA for one thing: attacking OneMind's own releases**, in a throwaway VM, before hardware ships.
3. **Don't put BLA's decoys in OneMind images.** They contradict the promise that everything on the box is readable by AI (AI-1) and would confuse OneMind's own agents.
4. **Copy the honest part of BLA** as release checks (no source maps, debug ports or secrets in shipped files; everything signed), and Craft Anything's licence allowlist as a guard for the Apache-2.0 code (D-001).

Requirements served: RS-2, RS-5, M-3, M-6, F-13, P-3, D-001.

Chapter 11
The first 180 days and the ten-year roadmap

Explain it like I'm the founder. The research decided *what* to build on. Two of those choices won only on a tie-break (the base OS, now Ubuntu 26.04 if Canonical allows it, and the update engine), and almost everything was proven in virtual machines, never on a real Pi, Jetson or PC. So the first 180 days do two jobs at once: **build the product in thin slices**, in the Blueprint's order, and **prove the foundation on real metal early**, so a wrong choice shows up in week 4–8, when switching costs days. There are six phases and eight **gates**: dates where you look at test results and follow a rule written *in advance*. After that, the ten-year roadmap has four phases, each with a different job: prove it, become a product, become a platform, become the default.

11.1 The 180 days on one page

Day 1 is Monday **2026-11-16**; Day 180 is Friday **2027-05-14**. The plan assumes about 45 hours a week from you, with AI coding agents (the second build session) writing most of the code under your review; these hours are planning assumptions, not evidence (`20-PLAN-180-DAYS.md` §2).

Phase	Weeks · dates	Goal	Blueprint milestone	Gate
P1 Workbench + update engine on metal	1–4 · 11-16 → 12-13	Mac workbench, CI, core by script on a Debian VM, first RAUC image in both base legs (Ubuntu 26.04 and Debian 13), x86 update tests on real hardware, **bootc on a real Pi 5 (W2-HW-3)**	M0 Reconcile	G0 (wk 1), G1 x86 update engine + base-OS reading (wk 4)
P2 Secure vertical slice	5–8 · 12-14 → 01-10	Identity, agent identity, gateway, action ledger, one read and one bounded action, clean restore; Pi 5, Secure Boot, Jetson proofs	M1	G2 (wk 6), G3 (wk 8)
P3 Extensions and the Market	9–13 · 01-11 → 02-14	Signed packages, the ten sockets (v1-beta), four reference adapters, GPU and kernel-route decisions, robot runtime	M2	G4 (wk 13)
P4 Disconnected field node + profiles	14–17 · 02-15 → 03-14	Offline capture and sync, NATS leaf buffering, `tak-node` and `greenhouse` on Pi 5, one cluster across Pi/x86/Orin	M3	G5 (wk 17)
P5 Independent owner alpha	18–21 · 03-15 → 04-11	Installer, onboarding, QR join, whole-box restore on metal, capability detection, first Mar-	M4	G6 (wk 21)

Phase	Weeks · dates	Goal	Blueprint milestone	Gate
		ket seed set; Jetson base lab; **base lock**		
P6 Managed appliance beta	22–26 · 04-12 → 05-14	Signed release pipeline with offline root, air-gapped rebuild, every release-gate test re-run, a second operator succeeds unaided, “connect a robot” demo	M5	**G7 beta go/no-go**

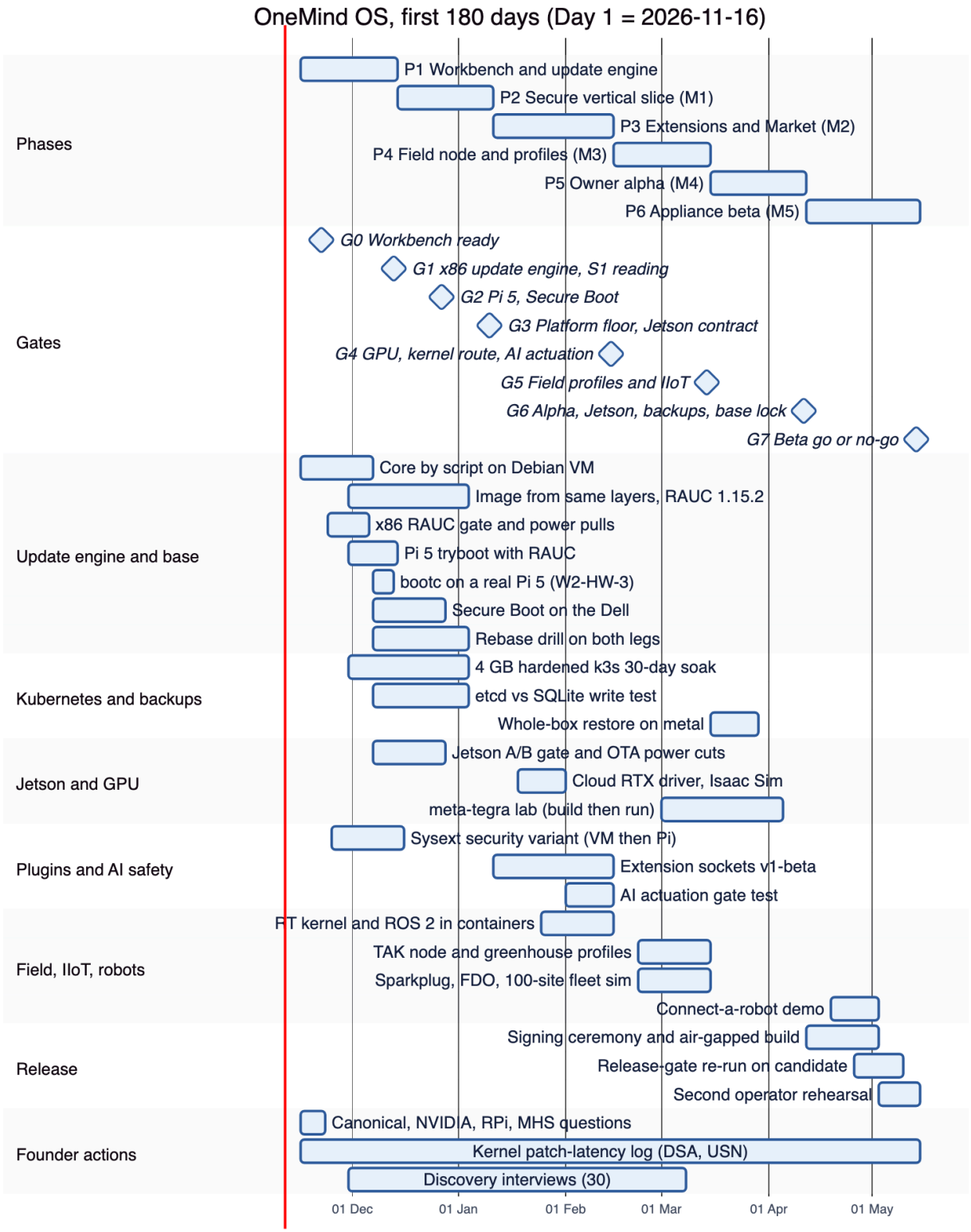


Figure 24: The 180-day Gantt: phases, gates, and every lab test scheduled against them (from 20 §1).

11.2 The gates: what passes, what fails, and what happens then

Gate	When	GO →	NO-GO → (an alternative already evaluated)
G0 Workbench ready	2026-11-22	Plan as written	No KVM on the Mac → VM labs run in CI; Dell lacks custom Secure Boot

Gate	When	GO →	NO-GO → (an alternative already evaluated)
			keys → Secure Boot tests move to the first ODM candidate
G1 x86 update engine, base-OS reading	2026-12-13	RAUC confirmed for x86 (Z-03); bootc passes on a real Pi 5 but its known bugs are still open → no change; Ubuntu's Secure Boot test (U-SB, LT-VM-20, run from day 1) passes twice unchanged → the base stays **Ubuntu 26.04**, both legs keep building	RAUC fails on metal → Mender test in week 5; if Mender passes, **the engine flips to Mender** on the engine-neutral layout. **bootc fails on a real Pi 5 (W2-HW-3)** → the base becomes **Debian 13** (6–5), and the Debian leg becomes the default. **U-SB fails**, or passes only after a partition or bootloader change → **bootc**; with W2-HW-3 also failed → **Debian**. U-SB still blocked or not run → no map event, and the record says so
G2 Pi 5, Secure Boot	2026-12-27	Pi 5 tier proven; x86 route = `efi` + signed UKIs	Pi tryboot fails → Rugix's proven Pi integration [C-W3-029]; Secure Boot fails → ship on the GRUB route with the same layout
G3 Platform floor, Jetson, rebase cost	2027-01-10	4 GB Pi 5 is a platform node; hub servers start on embedded etcd	Soak fails → floor rises to **8 GB** and 4 GB boards become NATS leaves; etcd writes too much → SQLite everywhere
G4 GPU, kernel route, AI actuation	2027-02-14	Pro GPU runs on the core leg the base decision selects; kernel stays K1	Stock kernel refuses OneMind-signed modules (expected [C-DEC-008]) → try the OneMind shim route first, K2-hybrid only if it fails. If the base is Debian: GPU fails on Debian, passes on Ubuntu → GPU tier gets the Ubuntu flavor (after Z-01); Isaac Sim fails on Debian (expected [C-REF-DC-robotics-013]) → simulation boxes use Ubuntu
G5 Field profiles + IIoT	2027-03-14	`tak-node` and `greenhouse` enter the alpha	FDO blocked on the core legs but works on EL10 bootc → the bootc Industrial profile (AlmaLinux 10 / RHEL 10 image mode) moves into year 1
G6 Alpha, Jetson base, backup engine, **base lock**	2027-04-11	restic stays; Jetson decision recorded (Z-16); **the shipping base is locked** by the decision map: Canonical agrees → Ubuntu; W2-HW-3 failed → Debian; U-SB failed → bootc; or your logged choice (Z-38). The lock record includes the LT-VM-20 result or its BLOCKED record	Canonical refuses and you reject the debranded rebuild → **bootc**, a different build, so the alpha slips by weeks; refused **and** W2-HW-3 failed → **you decide**; no Canonical answer by G6 → you decide (the alpha runs the Debian leg, labelled "base not locked"). Restore fails or takes > 4 h → no beta until fixed; restic runs out of memory where Kopia doesn't → backups flip to Kopia
G7 Beta go/no-go	2027-05-14	Public beta: x86 Lite image + Pi 5 Edge profile on a certified-hardware	Any release-gate test fails, or no base is locked → **no beta**. "Do not

Gate	When	GO →	NO-GO → (an alternative already evaluated)
		list; Robotics beta on Orin only if G3 and G6 passed	weaken safety to hit a date" (Blueprint)

Weeks 1–2, day by day (20–… §7): D1 order the hardware and apply to NVIDIA Inception; D2 inventory the Dell and Beelink (TPM, Secure Boot custom keys, CPU level) and run read-only cluster checks; D3 reconcile every Blueprint contract onto the layer map; D4 the monorepo and a two-legged root-filesystem build (`ubuntu:26.04` and `debian:trixie`); D3 also sends the Canonical and counsel questions (Z-01), asking for a written price; D5 OneMind's own RAUC from the v1.15.2 tag [C-PLAN180-001]. Week 2 brings the first real-hardware update test and ten power pulls on x86, the sysext security test, and the first CI-built image passing the gate in QEMU.

The cut list, if you fall behind: first the non-gate validation tests, then the desktop client and XR, then fleet scale at 1k, runner-up tests already settled, the industrial pull-forward, real time, and only then the "connect a robot" demo. **Never cut** the update, power-cut, Secure Boot, restore, sysext-security and AI-actuation tests: each protects users from a bricked or unsafe device.

11.3 The ten-year roadmap

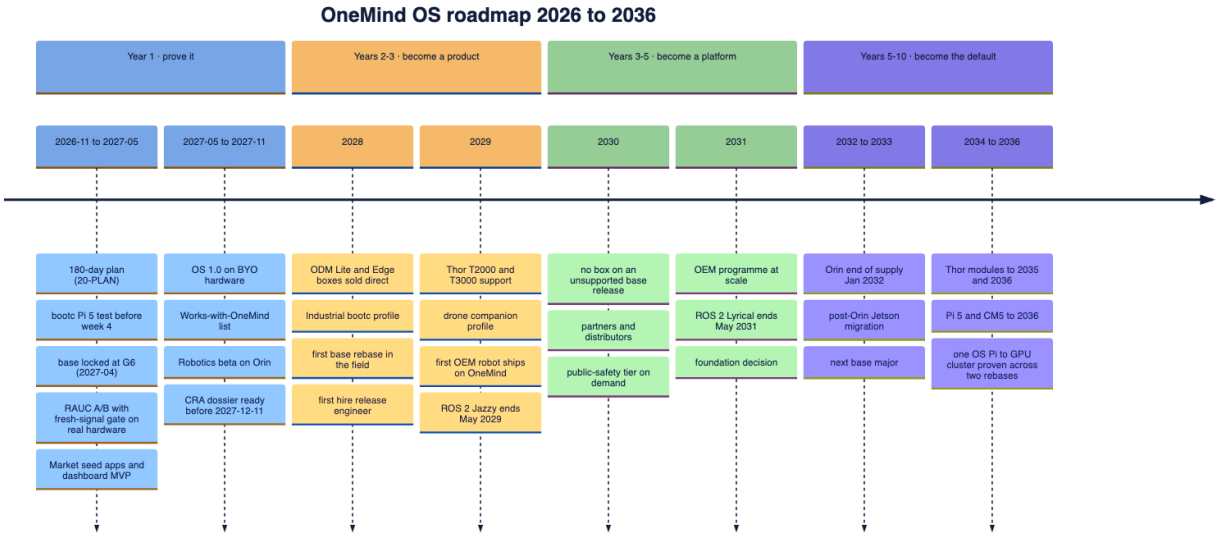


Figure 25: The roadmap from 21 §1.1.

Phase	Job	Product	Hardware	Team	Revenue	One hard exit criterion
Year 1 (2026-11 → 2027-11)	Prove the foundation	OS beta → 1.0 on bring-your-own hardware; Market; dashboard; greenhouse and TAK profiles	No own boxes; certified list [C-W10-037]	Solo + contracted counsel	Support, pre-orders, education, grants	First image passing every release gate, run by ≥ 25 outside users before 2027-12-11
Years 2–3	Become a product	Lite, Edge, Pro, Robotics profiles; Industrial bootc profile; mirror; WebXR	ODM Lite and Edge boxes, direct	1–3 hires, release engineer first	Boxes, support plans, certification fees	First major rebase of the locked base (or, on Ubuntu, a priced Pro window) done in

Phase	Job	Product	Hardware	Team	Revenue	One hard exit criterion
						the field on ≥ 95% of boxes, no reflash
Years 3–5	Become a platform	OEM programme; Cluster; public-safety tier on demand	Partners and distributors	5–12 people (proposal)	OEM fees, fleet console, partner share	≥ 10 third-party products shipping on OneMind; every SKU inside its declared support period
Years 5–10	Become the default	One OS from Pi to GPU cluster across two more base majors and the post-Orin Jetson	Second-source silicon (RK3588-class; riscv64 if credible)	Company or foundation that outlives the founder's attention	Platform economics	No robot or box stranded on an unsupported branch; a third party can rebuild and sign a release

Year-1 exit criteria (21–… §3.6) include: release gates 100% on Dell, Pi 5 and Orin; **zero** bad commits across at least 200 forced-bad updates (the lab already showed the naive gate fails [C-LAB-rauc-k3s-gate-008]); whole-box restore onto new hardware in under 60 minutes, green monthly for six months [C-LAB-k3s-footprint-restore-011]; a box offline 90 days accepts its next signed update from USB; CRA-ready before 2027-12-11 [C-W6-001]; three third-party robots onboarded by QR by a non-developer in under 10 minutes each.

11.4 The calendar OneMind does not control

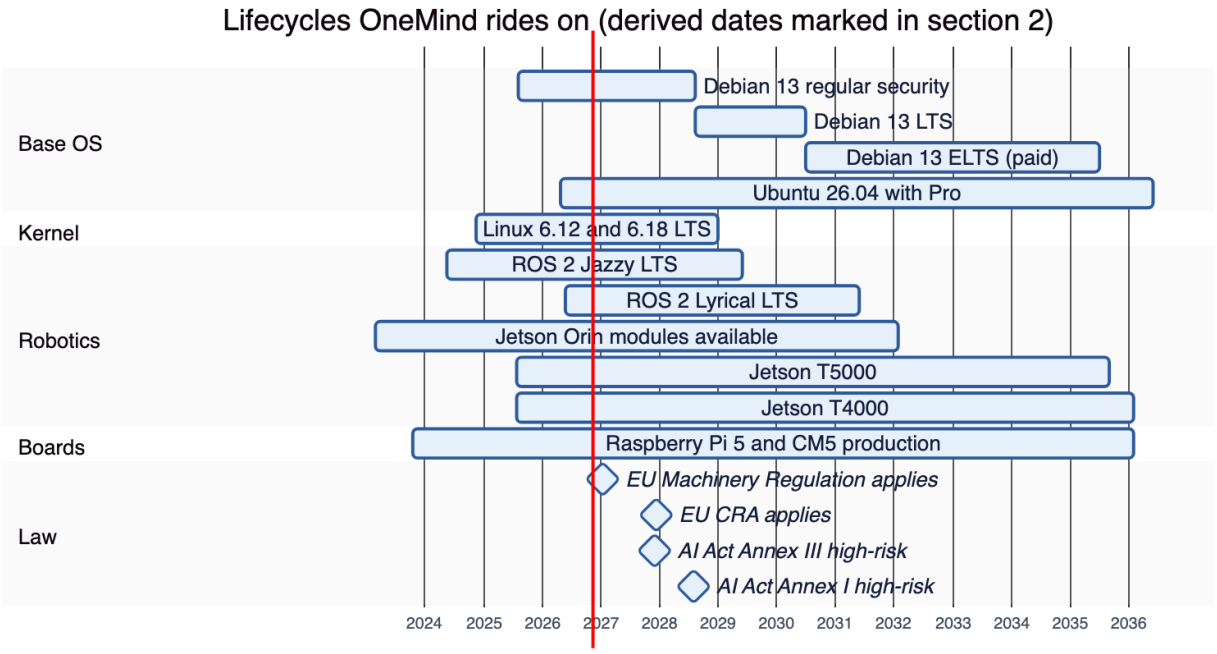


Figure 26: Lifecycles and legal dates the roadmap must fit (21 §1.2; bar starts approximate).

Date	Event	What must be done by then
2026-10-30	JetPack 5 ended [C-REF-W7L-001]	Jetson flavor on JetPack 7.2+
2027-01-14	EU Machinery Regulation applies [C-W7-045]	Robot decision log in the audit layer
2027-12-11	**EU CRA applies** [C-W6-001]	First sold image CRA-ready

Date	Event	What must be done by then
2028-08-09	Debian 13 regular security ends [C-W2-001]	If the base is Debian: most boxes on Debian 14
Dec 2028	Linux 6.12 and 6.18 LTS projected end [C-W18-001]	Kernel hop shipped as an image update
May 2029	ROS 2 Jazzy ends [C-W7-029]	Robot containers on Lyrical or newer
2030-06-30	**Debian 13 LTS ends** [C-W2-001]	If the base is Debian: zero supported boxes on Debian 13
Jan 2032	Jetson Orin end of supply [C-W7-001]	Post-Orin robot SKU available
Jan 2036	Pi 5 / CM5 production commitment ends [C-W2-029]	Edge successor board chosen
May 2036	Ubuntu 26.04 support with Ubuntu Pro ends [C-W2-005]	If the base is Ubuntu: boxes rebased to a later LTS, or retired

11.5 The second build session (F-9)

`30-NEXT-ULTRACODE-BUILD-BRIEF.md` is the one-shot brief for the AI session that builds release v0. It builds **your own read-only image** from one recipe in two legs, **Ubuntu 26.04** (the decision of record) and **Debian 13**, kept at parity until you lock the base, with nothing Ubuntu-based published before Canonical answers; A/B slots under **RAUC**, **k3s** on top, and the **OneMind Core**: sign-in, the capability gateway, the action ledger, backups, the Market and the dashboard. It works in VMs on the Mac and in CI; for your Pi, Jetson, Dell and Beelink, your cluster, your signing keys and any vendor, it writes a **kit** of exact commands and pass/fail rules for you to run. Success looks like: a signed image boots in QEMU for x86 and arm64 (Pi 5 as a kit); a broken update rolls back by itself; an AI agent asks to switch on a test light and **cannot until you approve on your phone**; a simulated robot arm is discovered, starts observe-only and stops safely when its plugin is revoked; a backup restores onto a fresh VM.

11.6 What competitors did at the same stage (constraint fit)

Product	Early path	Constraint fit	Lesson
Home Assistant	Software first, boxes years later, foundation-funded [C-W16-012]	High	Revenue from support and partners before hardware
umbrelOS	Script installs, then an image, dropped scripts at 1.0 [C-REF-DC-longevity-005]	High	Image is the product
Olares	Crowdfunded its box (Kickstarter) [C-W16-018]	Medium	Pre-orders test price before inventory
Sandstorm	Enterprise sales too early [C-W16-050]	Medium	Don’t lead with enterprise

Requirements served: RS-3, RS-7, M-6, M-7, F-9, F-10, D-002, D-004, R-1, R-3, H-1.

Chapter 12
Physical AI, NVIDIA and the next ten years

Explain it like I'm the founder. You want to keep pace with NVIDIA and become the OS that robotics companies build on. Two truths sit side by side. **NVIDIA moves fast and ships its robot software as containers and open repos**, so OneMind can list new models and tools the day they ship, if its Jetson image tracks NVIDIA's current software line. But **NVIDIA's Jetson software has ended before its hardware every time so far**, and NVIDIA lock-in is the opposite of sovereignty. So OneMind keeps pace by design and keeps an exit by design: the same app manifest must run on non-NVIDIA chips, and a Jetson image that does not depend on Ubuntu is tested in the first 180 days. Around all this, the world itself is moving (chip blocs, sovereignty rules, shutdowns, walled AI gardens), and the foundation holds in all four futures we wrote.

12.1 Does the base OS hold for robots for ten years?

Yes, with three conditions (`08-robotics-drones-physical-ai.md`):

1. **Real time** is part of normal Linux since kernel 6.12 [C-W18-005]; Debian, Raspberry Pi and NVIDIA all ship real-time kernels, NVIDIA's still as a preview. On the Pi 5, memory-bus contention gave spikes over 1 ms under stress [C-REF-DC-technical-001], and two identical Orin modules differed by about 10 μs versus 90+ μs [C-REF-DC-technical-002], so each unit is acceptance-tested and loops tighter than about 1 ms go to a microcontroller.
2. **ROS 2 runs in containers on every flavor**, because ROS 2 does not officially support Debian, and its newest release does not support Jetson's Ubuntu 24.04 [C-W7-026, C-W7-027].
3. **On a Jetson, NVIDIA owns the kernel.** Orin modules are sold until January 2032 and Thor until 2035–2036 [C-W7-001, C-W7-002], but NVIDIA publishes no software support period. Branch lives have ranged from about a month to 5.7 years, and security fixes land only at the head of a branch [C-REF-W7-T-004, C-REF-W7-T-005]. So every Jetson is flashed with A/B slots at the factory and must be able to move to a new NVIDIA software line in the field.

12.2 What NVIDIA has shipped (dated, primary)

Date	Release	Why it matters
2025-08	Jetson Linux 38.2 (JetPack 7.0), Thor only [C-W7-006]	New silicon gets the new stack first
2026-04	GR00T N1.7, code Apache-2.0 [C-W7-051]	An open robot foundation model that runs on Jetson
2026-06	Cosmos 3 world models; check each model's licence [C-W7-051]	World models reach the edge
2026-06	JetPack 7.2: Orin joins JetPack 7; official Yocto via OE4T; RT kernel [C-W7-007, C-W7-008]	One stack across Orin and Thor
2026-07	Jetson T3000/T2000 announced, shipping Q1 2027; IGX T3000 with functional safety [C-W7-048]	Mainstream Thor price points
2026-10	Isaac Lab 3.0; Newton physics 1.7 (Apache-2.0) [C-W7-051]	The simulation stack moves fast
2026-11	KubeCon NA: NVKE/NVKR preview [C-W4-039]; Agent Substrate to CNCF Sandbox [C-W4-040]	Data-centre tools, not edge tools (re-check by 2026-12-10)

“Keep pace on day one” (F-2), realistically: list NVIDIA’s apps (Isaac ROS, GR00T, Cosmos) in the Market the day they ship, as long as the Jetson flavor already runs the JetPack major they target; keep the Jetson flavor on the **current JetPack major within 60 days** of release (30 is the stretch goal); boot new modules such as T3000/T2000 on the same r39 line with a device-tree addition, not a new flavor, and test this when the dev kits ship.

12.3 Physical AI 2026–2036: what to watch

These are **proposals**: scenario reasoning on the evidence above, not sourced forecasts (`08` §12.3).

Horizon	Expected shift	Signal	What OneMind does
2027–2028	Vision-language-action models run on robots with Thor-class modules	GR00T and Cosmos list T2000/ T3000 as targets	Robot flavor on r39+; per-device model engines
2027–2029	ROS 2 Lyrical becomes the default; Jazzy ends May 2029 [C-W7-029]	Isaac ROS and Open-RMF support Lyrical	Ship Lyrical containers
2028–2031	Rust robot runtimes (dora-rs, copper-rs) gain share [C-W7-053]	Vendors ship their nodes	They are just new Market apps
2028–2032	EU certification of AI-driven machines becomes normal	Harmonised standards under the Machinery Regulation	Decision log and MCU safety split already in place [C-W7-046]
2030–2036	Orin end of supply (2032); a Jetson successor	NVIDIA’s lifecycle page; a new L4T major	One planned in-field branch upgrade per robot lifetime
Any time	NVIDIA changes Jetson’s base OS	Release notes	meta-tegra keeps OneMind independent of the root filesystem [C-STEEL-jb-002]

Humanoids and robot hardware. The research did not find primary sources that would let it forecast humanoid volumes or prices; it treats humanoids as one more robot class on the same three-brain architecture (Chapter 6) and watches NVIDIA’s GR00T targets and the EU Machinery Regulation as the signals.

12.4 How robot makers ship and update their computers (constraint fit)

Product	Robot OS and updates	Constraint fit with OneMind
Boston Dynamics Spot CORE I/O	JetPack 5.1.2; apps as Docker “Extensions” [C-W7-040]	High for app delivery (containers on a vendor BSP); low for longevity, two JetPack majors behind
Clearpath (ROS 2 Jazzy)	Lightly customised Ubuntu 24.04 ISO plus a setup script [C-W7-039]	Low: no image updates or rollback
Auterion Skynode	OS plus flight firmware in one signed file; remote, app or USB updates [C-W7-042]	High for the update model
ModalAI VOXL 2	Ubuntu 18.04, kernel 4.19, on a Qualcomm BSP [C-W7-041]	Cautionary: what vendor-BSP stagnation looks like
Home Assistant OS, umbrelOS, Olares	No robotics support or Jetson image [C-MATRIX-002]	None for robotics: robot vendors are the proof instead

12.5 Non-NVIDIA silicon: the lock-in guard

Option	2026 evidence	Ten-year fit
Rockchip RK3588 NPU	Mainline driver `drivers/accel/rocket` since Linux 6.18 [C-W7-049]	Good and improving: no vendor-BSP lock
Hailo (Pi AI HAT+ class)	HailoRT 5.5.0, Oct 2026 [C-W7-053]; Hailo-10H HAT, 40 TOPS [C-W10-036]	Good for Pi-tier vision
Google Coral	Repository archived since 2021 [C-W7-050]	Avoid for new SKUs

Option	2026 evidence	Ten-year fit
Qualcomm robotics boards	VOXL 2 still on Ubuntu 18.04 / kernel 4.19 [C-W7-041]	Risky; vendor containers only

Apps declare accelerator *capabilities*; models ship as ONNX plus per-backend engines; no app may hard-require CUDA unless marked NVIDIA-only (W7-R13). That is how "the same manifests run on AMD, Intel, Hailo and RK3588" (F-2, F-6).

12.6 Robot safety and drone rules

- **EU Machinery Regulation 2023/1230 applies from 14 January 2027**; machines with self-evolving behaviour must stay within their task space and **record safety-related decision data for one year** [C-W7-045, C-W7-046]. OneMind builds that log into the audit layer in phase 1.
- **ISO 10218 (2025)** adds cybersecurity requirements where they affect robot safety (SECONDARY, catalogue only) [C-W7-055].
- **Drones:** PX4 is BSD-3 and ArduPilot GPL-3.0 [C-LIC-047]; selling locked hardware favours PX4. FAA Part 108 (BVLOS) was not final in the latest sources (SECONDARY) [C-W7-054]. Drones are a year-2+ line; in year 1 they are integrated third-party devices. The update model to copy is a single signed artifact carrying the companion OS **and** the flight firmware [C-W7-042].
- **MHS:** design adapters in its shape (capabilities, read/write primitives, declared safety limits enforced by the adapter), but don't depend on it until its spec is public [C-W7-047].

12.7 The world around OneMind, 2030–2036

Four scenarios from `16-world-context-scenarios.md`. They are not predictions; real life will be a mix. Each has warning lights (triggers) wired into the roadmap.

Scenario	Already visible	What OneMind must be able to do
A · Fortress blocs	Taiwan makes about two-thirds of the most advanced chips [C-W15-023]; China paused, not cancelled, rare-earth export controls, and the pause ran out on 2026-11-10 (SECONDARY) [C-W15-004]; a 16 GB Pi 5 costs more than three times its launch price [C-REF3-DC-cost-001]	Run on more than one kind of chip; second-source SKUs
B · Sovereignty mandate	The EU Commission bought cloud with a sovereignty score as an entry gate [C-W15-015]; CRA, Data Act and right-to-repair in force or coming [C-W6-001]	Sell legally into Europe; publish a sovereignty dossier
C · Grey grid	313 government internet shutdowns in 2025, including 14 of satellite internet [C-W15-001, C-W15-002]; all of mainland Spain and Portugal went dark in April 2025 [C-W15-003]	Build and update itself with **no internet**; store-and-forward everywhere
D · Walled AI gardens	Google will require registered developers for apps installed outside its store, globally from 2027 [C-W15-010]	Keep phones working with boxes: PWA first, an APK channel outside Play

The foundation holds in all four, because an open image, k3s, NATS and Apache-2.0 belong to no single company or country. The one exception is the base: an Ubuntu 26.04 core depends on one company's permission (Canonical, Z-01), which is why the community-run Debian 13 leg is built alongside it until the base is locked, and why a Canonical refusal moves the base by a rule written in advance (Chapter 4). The work is at the edges: twelve no-regret moves (`16-…` §6), most of which cost days, not months, if done now.

Requirements served: F-1, F-2, F-6, M-7, N-1, N-2, N-7, N-8, N-10, AI-2, AI-5, R-3.

Chapter 13
Risks

Explain it like I'm the founder. The research raised about 120 risk statements across 30-odd files; many were the same risk seen from different angles, so `23-RISK-REGISTER.md` merges them into **62 risks**. Each has a likelihood, an impact, what we do now, an owner, and a **tripwire**: the specific event that says the risk is turning real. None of them says "change the foundation". Most say "keep the exit cheap and run the test before you depend on it".

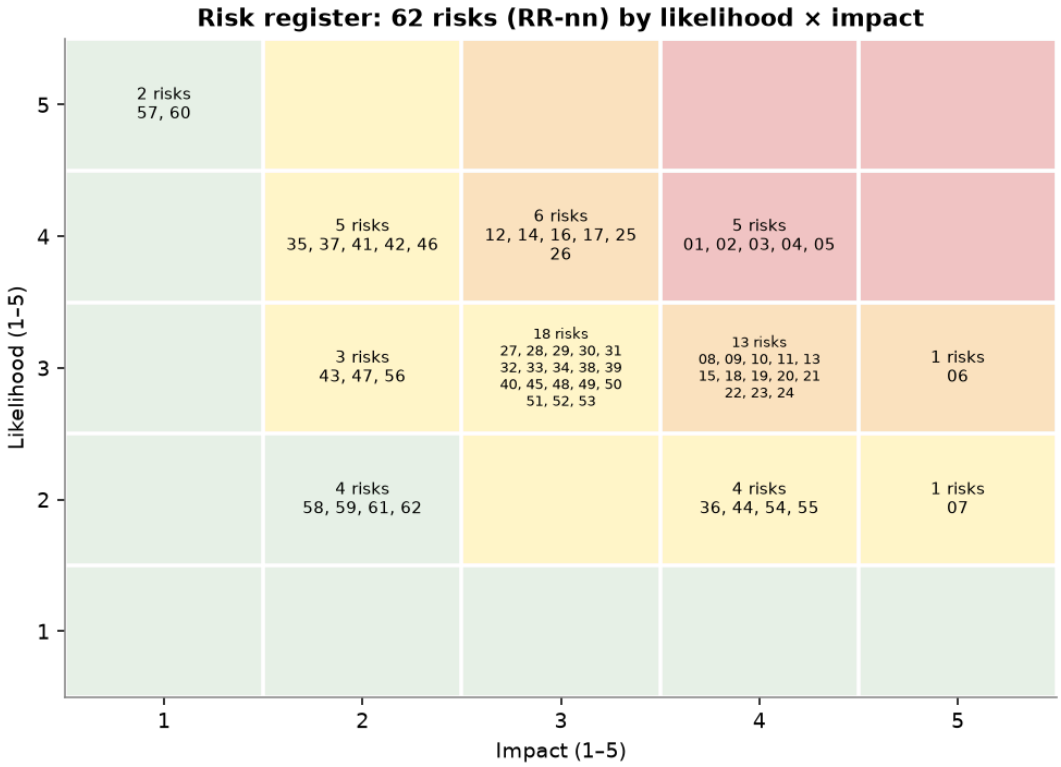


Figure 27: All 62 risks by likelihood and impact. Red cells are critical (score ≥ 16), orange high (12–15).

13.1 The five critical risks

ID	Risk	Mitigation now	Tripwire
RR-01	**Jetson software ends before the robot does.** JetPack 4 ended 26–32 months before Nano/TX2 NX stopped shipping [C-REF-W7-T-001]; JetPack 5 ended 2026-10-30 [C-REF-W7L-001]; fixes land only at branch head [C-REF-W7-T-005]	Per-module support periods; A/B from the factory; track every JetPack major and point release; meta-tegra LTS route as the exit	A JetPack branch ends with OneMind robots still on it
RR-02	**Kubernetes upgrade treadmill on offline boxes.** About 8–10 minors to 2036, none skippable [C-W4-012]	Snapshot-gated stepwise upgrades inside the A/B release; support window in the CRA declaration	A box returns after a year offline, 2–3 minors behind

ID	Risk	Mitigation now	Tripwire
RR-03	**Your own time.** Labour, not licences, dominates the 10-year cost on every base (about 1,680 base + 1,480 CRA hours on Debian, computed in `evidence/lifecycle-cost-2036.md`)	One image pipeline; three plugin contracts in beta; costed review plan; release engineer as first hire	A gate slips more than a week, or < 80% of a phase is done
RR-04	**Your cloud cluster still serves public traffic through ingress-nginx**, retired with no security fixes since March 2026 [C-W4-016] (reported state)	Phase-0 move to Gateway API + Traefik before any other conversion work	Any new ingress-nginx CVE, or public exposure found by the read-only check
RR-05	**CRA route or classification wrong** for the image or the box [C-REF-W6-technical-008]	Counsel opinion (Z-05); decide by 2027-Q2; notified-body pre-assessment in parallel from 2027-Q1	Counsel disagrees with the FOSS route, or a harmonised OS standard is cited

13.2 The high risks, in one line each

ID	Risk	Key mitigation
RR-06	Prompt injection drives a physical or destructive action [C-W6-068]	Gateway-enforced human approval; tainted-data rules
RR-07	Signing-key loss or theft, including the module key and the root's own lifetime	Offline root, HSM, 2-of-3, rotation drill, vendor-exit clause
RR-08	k3s on small, capturable devices exposes every Secret [C-REF-W4-security-008]	Agent-only robots; hardened re-measure; 8 GB floor if needed
RR-09	Canonical refuses or overprices modified-Ubuntu redistribution. Under revision 4 this is the condition on the **base-OS decision itself** (Ubuntu 26.04 core), not only on Jetson [C-REF-DC-cost-007]; the mitigation has no cost bound until a quote exists [C-DEC4-010]. Canonical sells "Trademark use" inside Ubuntu Pro for Devices, per device and unpriced; a provisional bracket is $0–$250 per box over 10 years, an assumption, not a quote [C-FIX4-001, C-FIX4-004]	Ask in week 1 with a request for a written price (Z-01); both core legs built until the G6 lock; nothing Ubuntu-based ships before an answer; the pre-registered map decides a refusal (bootc, or you if W2-HW-3 also failed); meta-tegra lab for Jetson
RR-10	Secure Boot downgrade [C-LAB-secureboot-rauc-018]	Version-match gate rule; SBAT and key rotation
RR-11	A third-party plugin persists through an unsigned extension [C-REF-W19-security-004]	Signed-only policy; T-W19-01 as the ADR condition
RR-12	Secure Boot enforcement forces a OneMind-built kernel on a solo founder [C-DEC-008]	Plan K2; measure bump time; seek a console-free route
RR-13	The locked base's major rebase costs more than budgeted or strands boxes (Debian 13 → 14, Pi paced by Raspberry Pi; on Ubuntu, the alternative is an unpriced Pro window to 2036 [C-W2-005, C-W2-006]); umbrelOS's Debian rebase stranded or reflashed some units [C-REF4-DC-longevity-006]	Timed rebase drill on both legs with founder-hours, then N → N+2 updates from the old slot; pinned Pi archive
RR-14	Hardware price and supply volatility [C-W10-037, C-W10-034]	No own boxes in year 1; second-source SKUs

The remaining 16 high, 30 medium and 6 low risks are in `23-RISK-REGISTER.md` §4, with a deduplication map in §6 showing where each original risk statement went. Also tracked: Pi 5 real time unreliable under load [C-REF-DC-technical-001] and k3s crashing in Go's FIPS-only mode [C-REF-DC-longevity-010].

13.3 What competitors do about these risks (constraint fit)

Risk	Competitor practice	Constraint fit
Bad update bricks a box	Home Assistant OS: RAUC A/B, 3 tries [C-TD-home-assistant-os-009], but commits while unhealthy [C-TD-home-assistant-os-018]	High; OneMind adds the fresh gate
Key compromise	umbrelOS beta: offline root, M-of-N signers, two-source revocation [C-TD-umbrelos-013]	High; copy the design for release 1
Key person leaves	Every small competitor carries it [C-MATRIX-005]	High; document, automate, hire a release engineer first
Licence change backlash	HashiCorp, Elastic, Redis [C-W16-007, C-W16-008, C-W16-009]	High; stay Apache-2.0, protect the trademark

Chapter 14
Decisions for Zeus

Explain it like I'm the founder. The research could decide most things from evidence. These are the ones it couldn't: they depend on your money, your risk appetite, a lawyer, or access we didn't have. Each one has a recommendation, so you can say "yes" and move on, or pick another option. **The first ten matter before any code ships or any base is locked**; the rest can wait weeks. Full options, reasoning and what each blocks are in `26-OPEN-QUESTIONS-FOR-ZEUS.md`. Items marked *(legal)* need counsel.

14.1 Decide before code ships

ID	Question	Recommendation	Why
Z-01 *(legal)*	Ask Canonical and counsel about redistributing a modified Ubuntu?	**Yes, in week 1**: counsel's reading plus Canonical's terms and a **written** Pro for Devices price at 1k/10k units	Under revision 4 this gates the **base-OS decision of record** (Ubuntu 26.04), not only Jetson [C-W2-050, C-REF-DC-cost-007]. Agreed with a price → Ubuntu outright; refused and you reject a debranded rebuild → bootc. **Assumption meanwhile:** the Ubuntu 26.04 result with both legs built, nothing Ubuntu-based shipped
Z-02	Security updates on by default?	**Yes for the security channel**, with opt-out, postpone, a window, health gate and auto-rollback; feature updates stay manual; robots get a "mission-safe reboot" rule	The CRA expects it [C-W6-008]; rollback makes it safe; R-1's intent (you stay in control) is kept
Z-03	Update engine	**RAUC**, with the layout engine-neutral (Mender runner-up, Rugix alternative)	Tie on score; RAUC wins on Secure Boot, revocation, free deltas, Pi 5, 3 tries [C-LAB-rauc-1152-011]
Z-04	Signing-key custody	**Offline root in a hardware key/HSM you hold, 2-of-3 release signers, a written succession/escrow plan**, plus a rotation drill before the first box	Home Assistant's release key lives in CI secrets [C-TD-home-assistant-os-013]; umbrelOS's beta shows the better design [C-TD-umbrelos-013]
Z-05 *(legal)*	Open-core boundary vs the CRA route	**Keep every OneMind-authored part of the base image Apache-2.0; paid features as separate apps**; counsel on classification and firmware blobs; route decided by 2027-Q2	Keeps the FOSS self-assessment route open and D-001's promise
Z-06	Support period per SKU	**At least 5 years from end of sale**, capped by each board's kernel window (Jetson shortest), met by in-place rebases	CRA support-period duty [C-W6-002]; ELTS is quote-only and own-use-only [C-W2-004]
Z-38	Base OS: accept the scorecard's result (Ubuntu 26.04 + RAUC), or override it as your own decision?	**(a) Accept it as the default**, with its conditions: Z-01 first, both legs built, base locked at G6 by the deci-	The locked rules produce only this answer, and it already holds the hedge: the Debian leg is built anyway, and the map sends the base to

ID	Question	Recommendation	Why
		sion map, which now includes U-SB (Z-41)	Debian if bootc fails on a real Pi 5 [C-DEC4-006]. Choosing Debian now is allowed, but it is then logged as **your** decision in the scorecard's change log, with the ranking under both rules (Chapter 4)
Z-39	Does an unmet condition that depends on someone else (like Canonical's permission) disqualify a candidate?	**No (the literal rule)**: a conditional candidate stays ranked; close the missing cost when Canonical answers	The locked rules remove a candidate only on a FAIL [C-PROTO4-002]. Reading an uncosted condition as a FAIL would remove Ubuntu, and through the Jetson flavor also Debian and bootc [C-DEC4-007]. Any other reading is a rule change and must be logged before Canonical answers
Z-40	Write four scoring rules down before the next tests?	**Yes, before gate G1 (week 4)**: full-table row count with a draw going to you; "experimental" Pi 5 support scored 3 until W2-HW-3 measures it; "proven in production" scored on the base layer; never a third finalist	Each of the four decides the base on its own (`01` §3.1.6 f–g). Chosen after the Pi 5 result is known, any of them would look like fitting the rules to the result
Z-41	Run the Ubuntu Secure Boot test (U-SB, lab test LT-VM-20) yourself, or approve an Ubuntu mirror for the lab?	**Run it on your Mac on day 1** (2026-11-16): a disposable VM, about 1.5–2 hours unattended, twice. Also say whether the Ubuntu block in the research environment is deliberate	It is the cheapest test that can change the base: a fail moves it to bootc. The kit and its pass/fail rule were frozen before any build [C-FIX5-002]; the research environment could not reach Ubuntu's servers and did not work around that [C-FIX5-001]

14.2 Platform and product decisions

ID	Question	Recommendation
Z-07	k3s floor for small boxes (REQUIREMENTS open question 2)	k3s server on ≥ 8 GB; 4 GB + SSD as a limited profile; SD-only or smaller = NATS leaf; confirm with the Pi 5 soak
Z-08	When to add RKE2	One distribution (k3s) until a buyer needs FIPS/STIG or an AI-conformant GPU cluster [C-W4-038]; k0s in the bake-off
Z-09	GPU-server flavor on Ubuntu + RKE2	Allow it once Z-01 is answered
Z-10	Fund the bootc Industrial profile; run W2-HW-3 when?	**Yes, year 2**, on AlmaLinux 10 or RHEL 10 image mode, with CentOS Stream and Fedora IoT as CI canaries now: your Fedora IoT answer. **Approve the bootc Pi 5 test W2-HW-3 before gate G1 (week 4)**: if it fails, the base becomes Debian [C-DEC4-007], and a box shipped on Ubuntu or Debian + RAUC cannot move to bootc without a reflash [C-STEEL4-bootc-025]
Z-11	Industrial certification route	Partner with an already-certified platform for the first industrial SKU [C-W8-033, C-W2-042]
Z-12	Default TAK server; where TAK lives	TAK Server container on ≥ 8 GB boxes [C-W8-018], OpenTAKServer on 4 GB profiles; cloud TAK as relay until the home bus is the authority
Z-13	What "mirror" means	Warm standby (continuous restore, one-click promote) by default; live HA only on 3+ nodes

ID	Question	Recommendation
Z-14	Platform first or image first	Platform first inside an image-first plan; precedent is mixed (Home Assistant platform first [C-W18-062], Umbrel image first [C-W18-063])
Z-15	Fusing Pi 5 / Jetson	Only OneMind-built hardware, after a soak on test-fused units; fusing is irreversible
Z-16	Jetson base	L4T Ubuntu for the beta **and** the meta-tegra lab within 180 days decides the default [C-STEEL-jb-002]
Z-17	Simulation GPU boxes	Ubuntu GPU flavor after Z-01, unless Isaac Sim passes on Debian [C-REF-DC-robotics-013]
Z-18	x86 boot route	`efi` + signed UKIs from the first image if the combined power-cut spike passes, else GRUB
Z-19	Secure Boot key model	Factory-enrolled db key; OneMind-built LTS kernel for the enforcing release [C-DEC-008]; decide whether owners may take over their box's keys ("sovereign mode")
Z-20	Fleet server for updates	OneMind's own release service on NATS for RAUC; the 10k fleet test picks a fleet plane in year 1
Z-21	Plugin contracts, review staffing, money	Freeze E1–E4, E7, E9, E10; E5, E6, E8 as beta; Community tier only for sandboxed types; costed review plan first [C-REF-W19-cost-004]
Z-22	Third-party host-level plugins in release 1	Allow only if the sysext security test passes; otherwise One-Mind-built only
Z-23	k3s on robots and field kits	Agent only [C-REF-W4-security-008]
Z-24	Sunset the script install	Developer loop only after image GA [C-REF-DC-longevity-004]
Z-25	FIPS/STIG as a P0 property	No, unless a named public-safety buyer is in year-1 scope; k3s fails a FIPS-only gate today [C-REF-DC-longevity-010]
Z-26	Desktop edition plugins	Accept Omarchy's manifest shape, Verified publishers only, off by default
Z-27	Pull two industrial tests forward	Yes: FDO on the core legs and AlmaLinux bootc (FDO is experimental; device-certificate onboarding plus a manual claim stays the fallback), plus a Flight Control fleet simulation, within 180 days

14.3 The REQUIREMENTS open questions and the recurring Blueprint questions

Question	Recommendation
Kubernetes everywhere, or a hybrid? (REQ Q2)	Kubernetes on every box sold; NATS leaf on devices it only connects (Z-07)
Licence model for OneMind's own code (REQ Q3)	Settled by D-001; write the pro/open ADR before the first paid feature, consistent with Z-05
First hardware tier and segment (REQ Q4)	**Lite x86 mini PC for homesteaders and self-hosters first** (most lab-proven path); Pi 5 Edge greenhouse/TAK profile second, in the same release
First internal app (REQ Q5)	**The Blueprint's first vertical slice**: sign-in, site status, Legacy summary, an approved Home Assistant light toggle, TAK observation, offline sync, restore. It exercises the whole foundation
Is Google/Pixel hardware acceptable?	Yes for field EUDs: Pixel 8 or newer with GrapheneOS [C-W14-002], after the GrapheneOS + ATAK test
What state is the Dell 5540 in?	Run the inventory on day 2: Secure Boot custom-key mode is supported in its BIOS [C-W10-043]; record TPM and CPU level
What is the first vertical slice?	As REQ Q5

Question	Recommendation
Which task system?	Still open: no evidence was gathered; pick what you already use and expose it through the capability layer

14.4 Access the research needed but could not get

The live DOKS cluster (read-only commands are in `22-HARDWARE-LAB-TEST-PLAN.md`); founder hardware (Pi 5, Jetson Orin Nano Super, an RTX box) for the bake-off; real power pulls on the Beelink and Dell; vendor quotes (Canonical Pro for Devices, Freexian ELTS for arm64 and redistribution, Red Hat, NVIDIA), with drafts written but **not sent**, because the brief forbids contacting vendors; counsel; the Blueprint originals; paywalled standard texts (EN 18031, IEC 62443); a bot-walled CIP wiki; and new Wayback captures. Each is listed with what you can do in `26-…` §D.

Chapter 15
Glossary

Term	Meaning
A/B update	Two copies (slots) of the OS. Updates go into the copy that isn't running; if the new copy fails, the box boots the old one.
Agent principal	An AI agent with its own identity and permissions, acting as "a named user intersected with a grant", never as the owner.
aiPolicy	A per-integration flag saying whether its data may go to AI: allowed, consent-required, official-MCP-only, display-only or prohibited.
ATAK / iTAK / CloudTAK	Team Awareness Kit map apps for Android, iPhone and the browser; TAK Server links them. CoT (Cursor on Target) is their message format.
batman-adv	A Linux kernel mesh-networking protocol; with WireGuard and NATS leaf links it forms OneMind's native mesh.
Base lock (G6)	The gate (April 2027) after which one base (Ubuntu, Debian or bootc) is the shipping base; changing it later means reflashing boxes.
bootc / image mode	Shipping the whole OS as a container image; Red Hat's long-life lineage (Fedora → CentOS Stream → RHEL) uses it.
BSP	Board support package: the vendor's kernel, drivers and firmware for a board (for example NVIDIA's L4T).
Decision map	The base-OS result for each outcome of the three pending results (Canonical's reply, the bootc Pi 5 test, Ubuntu's Secure Boot test U-SB), written on 2026-11-11 before any is known.
Decision of record	What the research assumes until you rule otherwise: for the base, the result the frozen rules give (Ubuntu 26.04 + RAUC, conditional).
Capability gateway	OneMind's single enforcement point: it turns app capabilities into MCP tools, REST, CLI and Agent Skills, checks permissions, asks for approvals and logs everything.
CRA	EU Cyber Resilience Act (Regulation (EU) 2024/2847); main obligations from 2027-12-11.
DC-nn	A decision-critical question: one of 24 questions whose answer could flip ADR-001.
DKMS	A system that rebuilds a kernel driver (such as NVIDIA's) on the device after each kernel update; OneMind builds and signs drivers in CI instead.
dm-verity / verity bundle	A kernel feature that checks every block of a read-only filesystem against a hash; RAUC's verity bundles use it.
Driver pack	A signed system extension carrying kernel modules built for one exact OneMind kernel.
E1–E10	OneMind's ten plugin extension points (Chapter 9).
FDO	FIDO Device Onboard: zero-touch onboarding of a new device to its owner's management system.
Flavor / profile	A flavor is a variant image for a hardware family (Jetson, GPU server); a profile is a configuration of the same image for a use (greenhouse, TAK node, Industrial).
Fresh signal	Health evidence produced *after this boot*, as opposed to status saved from an earlier boot.
greenboot	Fedora/RHEL health checks that trigger an automatic rollback.
Health-gated commit	The new slot becomes the default only after checks pass.
IEC 62443	The industrial cybersecurity standard family: 4-1 for the development process, 4-2 for components, 3-3 for systems.
JetPack / L4T	NVIDIA's Jetson software stack; Jetson Linux (L4T) is its Ubuntu-based OS and kernel.
k3s / RKE2 / k0s	Small Kubernetes distributions; k3s and RKE2 come from SUSE, k0s from Mirantis.
Kernel contract	The list of kernel features OneMind needs, checked in CI on every kernel bump.

Term	Meaning
LUKS2	Linux disk encryption; OneMind encrypts the data partition from first boot.
MCP	Model Context Protocol: the standard way AI clients call tools; current version 2026-07-28.
meta-tegra / OE4T	The Yocto layer that builds Jetson images; official in NVIDIA's documentation since JetPack 7.2.
MHS	Anthropic's Model Hardware Standard, a research preview since 2026-08-27 for AI control of devices.
NATS / JetStream / leaf node	The message bus; JetStream adds storage; a leaf node is a small NATS server that forwards to a hub and buffers during outages.
NFD / GFD	Kubernetes Node Feature Discovery and NVIDIA GPU Feature Discovery, which label a machine's hardware.
PCR 7 / 11 / 15	TPM registers recording Secure Boot state (7), the booted kernel image (11) and the unlocked volume's identity (15).
PREEMPT_RT	Linux's real-time mode, in mainline since 6.12.
RAUC / Mender / Rugix	Open-source A/B update engines (Home Assistant OS uses RAUC; umbrelOS uses Rugix; umbrelOS once used Mender).
Rebase	Moving a fielded box to a new major OS release (Debian 13 → 14) without wiping it.
Refuter / steelman	An agent told to disprove a conclusion / an agent told to argue for the runner-up. A **blind** refuter sees only the claim, not the research's reasoning.
restic	An encrypting, de-duplicating backup program that allows several keys per repository.
ROS 2	Robot Operating System 2, the standard robotics framework; Jazzy and Lyrical are long-term releases.
SBAT	A generation number inside signed boot files that lets firmware refuse older signed files.
Secure Boot / UKI	Firmware checks each boot file's signature; a unified kernel image (UKI) packs kernel, initrd and command line into one signed file.
sysext / confext	systemd system and configuration extensions: signed images laid over a read-only `/usr` or `/etc`.
SQLite / etcd	The two k3s databases: SQLite for one server, etcd for several.
Tie / tie-break	When the **top two** candidates on the frozen scorecard are less than 3 points apart, they are tied; the tie goes to the decision-critical evidence between those two, counted row by row, never to changed weights or to a third candidate.
Tryboot	The Raspberry Pi firmware's one-shot "try this boot partition next" mechanism.
U-SB / LT-VM-20	Ubuntu's own Secure Boot test: the Debian Secure Boot lab re-run on Ubuntu 26.04 in a virtual machine. A fail moves the base to bootc. Attempted on 2026-11-11 and blocked; LT-VM-20 is the founder-run version.
Z-nn	A founder decision item in `26-OPEN-QUESTIONS-FOR-ZEUS.md`.

Appendix B · How this was researched, and its limits

Method (`RESEARCH-MISSION.md` §2): every factual claim cites a primary source the research actually opened (official docs, release notes, legal text, or source code at a pinned commit), or is labelled SECONDARY or UNVERIFIED. Scorecard weights were frozen before scoring. Every P0 conclusion went to at least three independent refuters with different lenses (technical, longevity, security, cost, robotics), and each runner-up got a written steelman. Thirty products were torn down from code.

The VM lab (disposable Lima/QEMU VMs on a Mac; pass/fail criteria written before each run):

Spike	Result
Debian 13 + Rugix (x86 and arm64)	Pass; found the stale-status trap [C-LAB-debian-rugix-023]
Debian 13 + RAUC + real k3s gate	Pass twice; naive gate committed a broken kubelet [C-LAB-rauc-k3s-gate-008]
RAUC under enforced Secure Boot	Pass twice; downgrade gap found [C-LAB-secureboot-rauc-018]
RAUC 1.15.2 under enforced Secure Boot (rev. 4)	Pass; the old-signed-kernel gap persists [C-LAB-secure-boot-1152-005, C-LAB-secureboot-1152-009]
RAUC 1.15.2 + k3s on native arm64 (rev. 4)	Pass with the unchanged gate [C-LAB-aarch64-gate-004]
RAUC 1.15.2 + real k3s gate	Pass [C-LAB-rauc-1152-011]
Ubuntu 26.04 leg	Pass with RAUC 1.15.1; RAUC 1.13 fails on Linux 7.0 [C-LAB-ubuntu-2604-leg-010]
Ubuntu 26.04 under enforced Secure Boot (U-SB, rev. 4.1)	**Blocked, not run**: rule frozen before the build; Ubuntu's archive does not resolve from the research environment. Founder runs it as LT-VM-20 [C-FIX5-001, C-FIX5-002]
Mender 5.2.0 + same gate	Pass, with the one-try limit [C-LAB-mender-013]
Fedora bootc + greenboot	Mostly pass; power cut in finalize silently downgraded a healthy box [C-LAB-fedora-bootc-011]
Fedora IoT bootc + k3s gate	Automation pass; 32-minute rollback fails the time target [C-W2-054]
k3s footprint + restore	Pass; whole-box restore onto a fresh VM [C-LAB-k3s-footprint-re-store-011]
NATS leaf buffering	600/600 delivered [C-W8-050]
Minimal image, Debian vs Buildroot	4–6× smaller, 7–18% fewer CVE IDs [C-LAB-minimal-image-dc20-011]
Mac kernel build	arm64 kernel in 5.5 minutes [C-W18-066]
systemd sysext/confext	Functional pass; signing and slot switches not yet tested [C-W19-006]

What the lab cannot show: real power loss, real PC firmware, Pi 5, Jetson, GPU drivers or real-time latency on metal. These are the founder tests in `22-HARDWARE-LAB-TEST-PLAN.md` and the gates in Chapter 11.

Honest limits of this report. It summarises; the reasoning and the full evidence are in the numbered research files. Confidence is **low** on the base OS (a fragile result, conditional on Canonical) and **low-medium** on the update engine (a tie-break win), **high** on the health gate and backups. Vendor prices (Ubuntu Pro for Devices, Freexian ELTS for arm64, Red Hat) are unpublished, so cost figures depend on stated assumptions. KubeCon NA 2026 was checked mid-event (re-check by 2026-12-10). Humanoid and post-Thor NVIDIA forecasts are proposals, not sourced. Legal readings (CRA, licences, trademarks, AB 1043) are for counsel. The research

could not contact vendors, use the live cluster or touch hardware. The completeness critic's later passes and any remaining ledger gaps are tracked in `evidence/critic/`.

Appendix A · Topic checklist: every topic from the founding session

Built from `context/00-CONVERSATION-RECORD.md` (sections 1–14, numbered in the first column) plus the deep dives the brief names (marked –). Each row links to where the topic is answered. The build's verification step (`verify_pdf.py`) checks that every row's key terms appear in this PDF's text; the terms are listed in `report/checklist.tsv`.

§	Topic	Answered in
1	Samsung S24 Ultra, locked bootloader, Knox e-fuse	Ch. 2: Phones and Android sovereignty (where it started)
1	No-root customisation (Shizuku, UAD, Obtainium, Shelter, Tasker, DeX)	Ch. 2: Phones and Android sovereignty (where it started)
1	GrapheneOS on Pixel as a hardened ATAK device	Ch. 2: Phones and Android sovereignty (where it started)
1	The reported ATAK crash on a GrapheneOS Pixel Tablet	Ch. 2: Phones and Android sovereignty (where it started)
1	Linux phones (postmarketOS, Ubuntu Touch, Droidian, Mobian, Sailfish, Librem 5, PinePhone, FuriLabs)	Ch. 2: Phones and Android sovereignty (where it started)
1	Termux / Android Linux Terminal (AVF)	Ch. 2: Phones and Android sovereignty (where it started)
1	ATAK via Waydroid vs CloudTAK in a browser	Ch. 2: Phones and Android sovereignty (where it started)
1	Android forks (GrapheneOS, LineageOS, /e/OS, CalyxOS, iodéOS), Fairphone, Volla, KaiOS	Ch. 2: The mobile OS landscape, HarmonyOS and the dead OSes
1	HarmonyOS NEXT and platform sovereignty	Ch. 2: The mobile OS landscape, HarmonyOS and the dead OSes
1	Dead OSes and Fuchsia	Ch. 2: The mobile OS landscape, HarmonyOS and the dead OSes
1	Four "build your own OS" levels; a OneMind EUD; custom ROM deferred	Ch. 2: "Build your own OS" levels and the OneMind EUD
2	Earlier planning documents; the Blueprint is canonical; Build Handoff outdated	Ch. 2: The earlier planning documents (the Blueprint)
2	Recurring Blueprint questions (Pixel hardware, Dell state, first slice, task system)	Ch. 14: The REQUIREMENTS open questions and the recurring Blueprint questions
3	OneMind as a platform first, then its own image	Ch. 2: The strategy conversation
3	The moat: homestead + AI agents + TAK + self-hosting + robots	Ch. 2: The strategy conversation
3	Market view and first customers; AI as the turning point	Ch. 2: The strategy conversation
3	Business model: open core + boxes + support + education	Ch. 3: Business, licensing and go-to-market
3	NVIDIA at KubeCon NA 2026 (KAI, Dynamo, AICR, DRA, Agent Substrate)	Ch. 2: The strategy conversation
3	NVIDIA lock-in vs hardware-agnostic	Ch. 12: Non-NVIDIA silicon: the lock-in guard
3	Fedora CoreOS / Fedora IoT as a candidate core	Ch. 4: Your Fedora IoT question (F-12)
4	Reverse-engineer competitors	Ch. 10: What the competitors actually do
4	Research all Linux distributions	Ch. 4: Which Linux, and why Ubuntu wins narrowly
4	Sell hardware and become a distributor	Ch. 8: Bundles and the distributor model (A-7, H-1)
4	OS profiles (situational awareness / TAK node, greenhouse)	Ch. 3: IoT, Industrial IoT, mesh, off-grid and TAK
4	Robotics, drones and physical AI as the moat	Ch. 12: Physical AI, NVIDIA and the next ten years

§	Topic	Answered in
4	Everything is a node; native mesh networking	Ch. 6: The NATS bus: one nervous system from sensor to cloud
4	Every app readable by AI: MCP, CLI, API, SSH, MHS	Ch. 3: The AI capability layer: everything readable and actionable by AI
4	Native alert system	Ch. 3: Dashboard, clients and alerts
4	Cloud apps (Strava) with API keys and AI context	Ch. 3: The AI capability layer: everything readable and actionable by AI
4	Internal apps: finance, wellness, robotics, infrastructure, PM, mesh, IoT, family, OSINT	Ch. 3: OneMind's own apps: build the thin parts, integrate the heavy ones
4	Multi-user family accounts and parental controls	Ch. 3: Family accounts, parental controls and a child's device
4	Kids' education and offline knowledge	Ch. 3: Education and offline knowledge
4	Hardware capability auto-detection	Ch. 8: Machine superpowers and "can my box run this?" (P-6, A-8, N-3)
4	QR onboarding for robots and IoT	Ch. 9: "Connect a robot → use OneMind" (F-3)
4	Multiple machines in one UI, mirror, infrastructure map	Ch. 3: Dashboard, clients and alerts
4	Lite / Pro / Omni editions	Ch. 8: Six sizes of one OS
4	NVIDIA as a partner (Inception)	Ch. 2: Founder decisions D-001 to D-006
4	Website: blog, Circle.so, docs, OS download, store/cart, affiliate, Zoho Forms, support desk, headless CMS	Ch. 3: Website, store and community
4	Omarchy-style plugin ecosystem	Ch. 3: The Omarchy desktop edition
4	WebXR and Android XR	Ch. 3: XR (WebXR, Android XR, visionOS)
4	Updates that users control; strong support	Ch. 14: Decide before code ships
4	Become the industry standard for sovereign operations	Ch. 11: The ten-year roadmap
4	The Olares Market catalog and Olares One images	Ch. 2: The big brief (82, now 97, requirement IDs)
5	Phase-1 verified facts (licences, Kairos, N.O.M.A.D., MHS, Strava, MCP)	Ch. 2: What phase-1 research reported, re-checked
6	Founder decisions D-001 to D-006	Ch. 2: Founder decisions D-001 to D-006
7	Learning track; Omarchy lesson	Ch. 2: The learning track
8	ADR-001 questions: Ubuntu vs Debian, own image, k3s vs k8s, dashboard	Ch. 2: ADR-001 and your questions about it
8	Dashboard design: Simple vs Pro, Control Hub	Ch. 3: Dashboard, clients and alerts
8	Backups, rollback, security and updates as top concerns	Ch. 7: Updates, rollback, backups and security, as flows
8	EU Cyber Resilience Act dates	Ch. 7: The law: the Cyber Resilience Act and friends
8	DOKS converts "with three changes" (corrected)	Ch. 6: Converting your DigitalOcean cluster
9	Existing cluster as the proven concept (Backstage etc.); why an OS now	Ch. 2: Your existing cloud cluster: the proven concept, and why an OS now
9	Cover all hardware	Ch. 8: The hardware line and box designs
9	5–10-year plan; world events	Ch. 11: The ten-year roadmap
10	Physical AI over 5–10 years; keep pace with NVIDIA	Ch. 12: Physical AI 2026–2036: what to watch
10	Ease of use; supersede other OSes	Ch. 1: "Pi greenhouse → enterprise robotics simulation": one OS at every scale (F-14)
10	Marketplace startups integrate into; self-hosted + cloud	Ch. 9: Hybrid self-hosted + cloud, certification and money (F-5, F-6)
10	The test sentence: "connect a robot → use OneMind OS"	Ch. 9: "Connect a robot → use OneMind" (F-3)
10	Own kernel or own image	Ch. 5: What "our own OS" realistically means
10	Development tools, especially from the Mac	Ch. 5: The Mac development setup and CI/CD (F-8)

§	Topic	Answered in
10	A second ultracode build session	Ch. 11: The second build session (F-9)
10	This PDF: visuals, distro comparisons, everything from the chat	Ch. 1: The executive answer
11	IoT and IIoT without limitations	Ch. 3: IoT, Industrial IoT, mesh, off-grid and TAK
11	Fedora IoT is “super important”	Ch. 4: Your Fedora IoT question (F-12)
11	VillageOS (“this is also why I’m going operating systems”)	Ch. 10: VillageOS: host and partner, don’t compete (F-13)
11	Mesh and off-grid (Meshtastic, LoRaWAN, Reticulum)	Ch. 3: IoT, Industrial IoT, mesh, off-grid and TAK
11	TAK Server, CloudTAK, ATAK, iTAK	Ch. 3: IoT, Industrial IoT, mesh, off-grid and TAK
12	As simple as a Pi, up to full enterprise (greenhouse → robotics simulation)	Ch. 1: “Pi greenhouse → enterprise robotics simulation”: one OS at every scale (F-14)
13	Kernel families: Darwin/XNU, Mach, BSD, Linux	Ch. 5: Kernel families: Darwin, BSD, Linux and microkernels (F-15)
14	The plugin ecosystem (community can edit and add plugins; website distribution)	Ch. 9: The plugin ecosystem and the Marketplace
-	Wellness deep dive (Strava, Oura, HealthKit, Health Connect)	Ch. 3: OneMind’s own apps: build the thin parts, integrate the heavy ones
-	Finance deep dive (SimpleFIN, Plaid, Actual)	Ch. 3: OneMind’s own apps: build the thin parts, integrate the heavy ones
-	OSINT legal and ethical limits	Ch. 3: OneMind’s own apps: build the thin parts, integrate the heavy ones
-	Competitors: Umbrel, Olares, CasaOS/ZimaOS, Home Assistant OS, Project N.O.M.A.D.	Ch. 10: The comparison
-	Flagged repos (REA, Craft Anything, BLA)	Ch. 10: The founder-flagged repos (REA, Craft Anything, BLA)

Appendix C · Sources for every claim ID in this report

This report cites **484 claim IDs**. Each row gives the claim's source as recorded in the research ledger (`research-output/evidence/claims/*.csv`), with its source type and status. The ledger also holds the version, release date, support end, pinned commit or tag, re-check date and the files that use each claim; saved page text with SHA-256 hashes is in `research-output/evidence/sources/`. The full bibliography is `research-output/SOURCES.md`. PRIMARY = official docs, legal text or code at a pinned commit; PRIMARY-METADATA = catalogue or metadata page; SECONDARY = press or third party; lab rows are this run's own VM experiments.

Claim ID	Type · status	Accessed	Source
C-CIT4-001	PRIMARY · verified	2026-11-11	https://github.com/IronFlock/ironflockos/blob/bb33a82dfa959366bb90896fc69b601604131185/README.md#L18-L77
C-CIT4-003	PRIMARY · verified	2026-11-11	https://docs.nvidia.com/cuda/cuda-installation-guide-linux/index.html
C-COST-002	PRIMARY · verified	2026-11-11	https://ubuntu.com/pricing/devices
C-COST-007	PRIMARY · verified	2026-11-11	https://www.debian.org/releases/
C-COST-009	PRIMARY · verified	2026-11-11	https://www.redhat.com/en/blog/red-hat-device-edge-decision-framework
C-DEC-005	PRIMARY · verified	2026-11-10	https://github.com/rauc/rauc/blob/4fb7c798d6ae412344fb8f8d310d773046af3441/docs/reference.rst#L104-L110
C-DEC-006	PRIMARY · verified	2026-11-10	https://bugs.launchpad.net/ubuntu/+source/linux-raspi/+bug/2170238
C-DEC-007	PRIMARY · verified	2026-11-10	https://www.truenas.com/docs/scale/24.10/gettingstarted/scalereleasenotes/
C-DEC-008	PRIMARY · verified	2026-11-11	https://github.com/torvalds/linux/blob/adc218676eef25575469234709c2d87185ca223a/kernel/module/signing.c#L64-L65
C-DEC-010	PRIMARY · verified	2026-11-11	https://github.com/torvalds/linux/blob/adc218676eef25575469234709c2d87185ca223a/security/integrity/Kconfig#L65-L75
C-DEC-013	UNVERIFIED · partial	2026-11-11	https://rauc.io/ ; https://github.com/rauc/rauc/blob/4fb7c798d6ae412344fb8f8d310d773046af3441/docs/integration.rst#L1801-L1833
C-DEC4-001	PRIMARY · verified	2026-11-11	research-output/01-FOUNDATION-DECISION.md §3.1.6 (d) and §3.6 script; research-output/evidence/scorecard-rev3.md §3.2
C-DEC4-002	PRIMARY · verified	2026-11-11	research-output/01-FOUNDATION-DECISION.md L239-L247
C-DEC4-006	PRIMARY · verified	2026-11-11	research-output/01-FOUNDATION-DECISION.md §3.1.6 (e)-(f)

Claim ID	Type · status	Accessed	Source
C-DEC4-007	PRIMARY · verified	2026-11-11	research-output/01-FOUNDATION-DECISION.md §3.1.6 (g)-(h)
C-DEC4-010	PRIMARY · verified	2026-11-11	research-output/23-RISK-REGISTER.md L125; research-output/evidence/lifecycle-cost-2036.md L53
C-FIX4-001	PRIMARY · verified	2026-11-11	https://github.com/canonical/ubuntu.com/blob/ec15716b1c4b95f1860672d4274dda730ee9b4fe/templates/pricing/devices.html
C-FIX4-004	PRIMARY · verified	2026-11-11	https://github.com/canonical/ubuntu.com/blob/ec15716b1c4b95f1860672d4274dda730ee9b4fe/templates/pricing/pro.html
C-FIX4-010	PRIMARY · verified	2026-11-11	research-output/evidence/lab/ubuntu-2604-leg/versions.md L20-L23
C-FIX4-014	PRIMARY · verified	2026-11-11	research-output/evidence/scorecard-weights.md L106, L159
C-FIX5-001	PRIMARY · verified	2026-11-11	research-output/evidence/lab/ubuntu-secure-boot/logs/env-probe.log
C-FIX5-002	PRIMARY · verified	2026-11-11	research-output/evidence/lab/ubuntu-secure-boot/README.md
C-FIX5-003	PRIMARY · verified	2026-11-11	research-output/evidence/lab/ubuntu-secure-boot/scripts/
C-G4U-001	PRIMARY · verified	2026-11-11	https://github.com/canonical/real-time-ubuntu-docs/blob/f8b1bb14746140cbd8442ac3e56543aca4b2180a/docs/reference/releases.rst#L26-L41
C-G4U-002	PRIMARY · verified	2026-11-11	https://github.com/canonical/real-time-ubuntu-docs/blob/f8b1bb14746140cbd8442ac3e56543aca4b2180a/docs/how-to/enable-real-time-ubuntu.md#L16-L36
C-G4U-003	PRIMARY · verified	2026-11-11	https://github.com/eclipse-mosquitto/mosquitto/blob/fda14f2c196d9d2fcdd07d6c5d31a673524f7368/ChangeLog.txt#L1685-L1688
C-G4U-004	PRIMARY · verified	2026-11-11	https://github.com/fido-device-onboard/go-fdo-client/tree/ce60de283b3ce9ffeed18320308ed30e1cb64bdd/build/package
C-G4U-005	PRIMARY · verified	2026-11-11	https://github.com/fido-device-onboard/go-fdo-server/tree/dbe2b32366d0a72a00aeba13ccc34d1c0a6257a8/build/package
C-G4U-008	PRIMARY · verified	2026-11-11	https://github.com/canonical/landscape-documentation/blob/466a33a01f980f9e6f9cba7e9a6fdfddb59a1831/docs/reference/api/rest-api-endpoints/package-ch…
C-G4U-010	PRIMARY · verified	2026-11-11	https://github.com/rauc/rauc/blob/2d17b27041db7fa8b13a61c5e168f49310a06566/docs/advanced.rst#L2043-L2110

Claim ID	Type · status	Accessed	Source
C-LAB-aarch64-gate-004	PRIMARY · verified	2026-11-11	evidence/lab/aarch64-gate/logs/driver-run4.log ; evidence/lab/aarch64-gate/logs/driver-run5.log ; evidence/lab/aarch64-gate/logs/serial-run4-trimme…
C-LAB-debian-rugix-022	PRIMARY · verified	2026-11-11	research-output/evidence/lab/debian-rugix/logs/run4-arm64/events.log
C-LAB-debian-rugix-023	PRIMARY · verified	2026-11-11	research-output/evidence/lab/debian-rugix/logs/run3-arm64/events.log
C-LAB-fedora-bootc-009	PRIMARY · verified	2026-11-10	research-output/evidence/lab/fedora-bootc/logs/03b-after-rollback.log
C-LAB-fedora-bootc-011	PRIMARY · verified	2026-11-10	research-output/evidence/lab/fedora-bootc/logs/04e-finalize-root-cause.log
C-LAB-fedora-bootc-015	PRIMARY · verified	2026-11-10	research-output/evidence/lab/fedora-bootc/logs/05-rebase-and-data.log
C-LAB-k3s-footprint-restore-007	PRIMARY · verified	2026-11-10	evidence/lab/k3s-footprint-restore/logs/run-sqlite.log
C-LAB-k3s-footprint-restore-008	PRIMARY · verified	2026-11-10	evidence/lab/k3s-footprint-restore/logs/run-sqlite.log
C-LAB-k3s-footprint-restore-011	PRIMARY · verified	2026-11-10	evidence/lab/k3s-footprint-restore/logs/restore-sqlite.log
C-LAB-mender-013	PRIMARY · verified	2026-11-11	research-output/evidence/lab/mender/logs/serial-run1-trimmed.log ; research-output/evidence/lab/mender/logs/serial-run2-trimmed.log
C-LAB-minimal-image-dc20-011	PRIMARY · verified	2026-11-11	evidence/lab/minimal-image-dc20/results/sbom-cve.md
C-LAB-rauc-010	PRIMARY · verified	2026-11-10	https://github.com/home-assistant/operating-system/blob/ab7d0f1a0174760353780a21be2befecbc80a672/buildroot-external/board/pc/grub.cfg#L10-L61
C-LAB-rauc-1152-011	PRIMARY · verified	2026-11-11	evidence/lab/rauc-1152/logs/driver-run1.log ; evidence/lab/rauc-1152/logs/driver-run2.log ; evidence/lab/rauc-1152/logs/serial-run2-trimmed.log
C-LAB-rauc-1152-012	PRIMARY · verified	2026-11-11	https://github.com/raspberrypi/documentation/blob/21d772e7ee022e36a8f43c3ee75e20ca359fb832/documentation/asciidoc/computers/raspberry-pi/bootflow-e…
C-LAB-rauc-k3s-gate-001	PRIMARY · verified	2026-11-11	https://github.com/rauc/rauc/blob/2d17b27041db7fa8b13a61c5e168f49310a06566/src/install.c#L1444-L1661 ; https://github.com/rauc/rauc/blob/1183a396a2…
C-LAB-rauc-k3s-gate-004	PRIMARY · verified	2026-11-11	https://github.com/k3s-io/k3s/blob/f6973cb850e9b245585363d15bde80904ac862fd/pkg/rootless/mounts.go#L34-L40 ; https://github.com/k3s-io/k3s/blob/f69…
C-LAB-rauc-k3s-gate-008	PRIMARY · verified	2026-11-11	research-output/evidence/lab/rauc-k3s-gate/logs/serial-run1-trimmed.log and serial-run2-trimmed.log

Claim ID	Type · status	Accessed	Source
C-LAB-secureboot-1152-005	PRIMARY · verified	2026-11-11	evidence/lab/secureboot-1152/logs/driver-run1.log
C-LAB-secureboot-1152-009	PRIMARY · verified	2026-11-11	evidence/lab/secureboot-1152/logs/driver-e1.log
C-LAB-secureboot-1152-010	PRIMARY · partial	2026-11-11	https://github.com/rauc/rauc/blob/4fb7c798d6ae412344fb8f8d310d773046af3441/src/mark.c#L235-L243
C-LAB-secureboot-rauc-003	PRIMARY · verified	2026-11-11	https://github.com/tianocore/edk2/blob/fbe0805b2091393406952e84724188f8c1941837/MdeModulePkg/Universal/BdsDxe/BdsEntry.c#L1052-L1093
C-LAB-secureboot-rauc-012	PRIMARY · verified	2026-11-11	evidence/lab/secureboot-rauc/logs/serial-run1-trimmed.log
C-LAB-secureboot-rauc-013	PRIMARY · verified	2026-11-11	evidence/lab/secureboot-rauc/logs/serial-run1-trimmed.log
C-LAB-secureboot-rauc-018	PRIMARY · verified	2026-11-11	evidence/lab/secureboot-rauc/logs/serial-e1-trimmed.log
C-LAB-secureboot-rauc-020	PRIMARY · verified	2026-11-11	evidence/lab/secureboot-rauc/logs/driver-run1.log
C-LAB-ubuntu-2604-leg-010	PRIMARY · verified	2026-11-11	research-output/evidence/lab/ubuntu-2604-leg/logs/serial-u1-run1-trimmed.log
C-LAB-ubuntu-2604-leg-011	PRIMARY · verified	2026-11-11	research-output/evidence/lab/ubuntu-2604-leg/logs/driver-u2-run3.log ; research-output/evidence/lab/ubuntu-2604-leg/logs/driver-u2-run4.log
C-LIC-047	PRIMARY · verified	2026-11-11	https://github.com/meshtastic/firmware/blob/54e0d8d0ab2ff56b3a9ce967e53f79e49af560fb/LICENSE
C-MATRIX-002	PRIMARY · verified	2026-11-11	research-output/evidence/sources/MATRIX/protocol-grep-2026-11-11.txt
C-MATRIX-005	PRIMARY-METADATA · verified	2026-11-11	https://api.github.com/repos/getumbrel/umbrel/commits ; research-output/evidence/sources/MATRIX/commit-author-concentration-2026-11-11.txt
C-MATRIX-006	PRIMARY-METADATA · verified	2026-11-11	https://api.github.com/repos/home-assistant/operating-system/commits ; research-output/evidence/sources/MATRIX/commit-author-concentration-2026-11-…
C-MATRIX-008	PRIMARY-METADATA · verified	2026-11-11	https://api.github.com/repos/IceWhaleTech/CasaOS/commits ; research-output/evidence/sources/MATRIX/commit-author-concentration-2026-11-11.txt
C-PLAN180-001	PRIMARY · verified	2026-11-11	https://github.com/rauc/rauc (git ls-remote –tags)
C-PLAN180-003	PRIMARY · verified	2026-11-11	https://github.com/coreos/bootupd/pull/935
C-PROTO4-002	PRIMARY · verified	2026-11-11	<repo>@414d60a2528a2e0172ff35180c78f6801ec4b1af:research-output/evidence/scorecard-weights.md#L35-L41

Claim ID	Type · status	Accessed	Source
C-PROTO4-018	PRIMARY · verified	2026-11-11	<repo>@7836316f8e63128fbd3c9095a74770b33cc63597:research-output/evidence/scorecard-weights.md#L67;L196
C-REF-DC-cost-007	SECONDARY · verified	2026-11-10	https://forums.developer.nvidia.com/t/l4t-redistribution-license-issue-for-manually-generated-root-file-system/306285
C-REF-DC-cost-009	PRIMARY · verified	2026-11-10	https://github.com/NVIDIA/open-gpu-kernel-modules/issues/1258
C-REF-DC-longevity-004	PRIMARY · verified	2026-11-10	https://www.home-assistant.io/blog/2025/05/22/deprecating-core-and-supervised-installation-methods-and-32-bit-systems/
C-REF-DC-longevity-005	PRIMARY · verified	2026-11-10	https://github.com/getumbrel/umbrel/issues/1826
C-REF-DC-longevity-008	PRIMARY · verified	2026-11-10	https://github.com/kairos-io/kairos/issues/2010
C-REF-DC-longevity-010	PRIMARY · verified	2026-11-10	https://github.com/k3s-io/k3s/issues/13651
C-REF-DC-robotics-013	PRIMARY · verified	2026-11-10	https://docs.isaacsim.omniverse.nvidia.com/latest/installation/requirements.html
C-REF-DC-technical-001	SECONDARY · verified	2026-11-10	https://forums.raspberrypi.com/viewtopic.php?t=382231
C-REF-DC-technical-002	SECONDARY · verified	2026-11-10	https://forums.developer.nvidia.com/t/jetpack-6-2-agx-orin-64gb-preempt-rt-large-cyclictest-latency-variation-between-different-modules-up-to-90-us…
C-REF-DC-technical-008	PRIMARY · verified	2026-11-10	https://github.com/systemd/systemd/issues/32946
C-REF-W19-cost-004	PRIMARY · verified	2026-11-10	https://eclipse-foundation.blog/2025/09/23/businesses-built-on-open-infrastructure-have-a-responsibility-to-sustain-it/
C-REF-W19-cost-005	PRIMARY · verified	2026-11-10	https://github.com/systemd/systemd/issues/32762
C-REF-W19-cost-010	PRIMARY · verified	2026-11-10	https://github.com/flatcar/scripts/pull/2976
C-REF-W19-security-003	PRIMARY · verified	2026-11-11	https://kubernetes.io/docs/concepts/security/pod-security-standards/
C-REF-W19-security-004	PRIMARY · verified	2026-11-11	https://www.freedesktop.org/software/systemd/man/latest/systemd-sysext.html
C-REF-W19-security-008	PRIMARY · verified	2026-11-11	https://theupdateframework.io/docs/security/
C-REF-W19-security-009	SECONDARY · verified	2026-11-11	https://www.techworm.net/2025/08/malicious-vs-code-extensions-exploit-marketplace-loophole.html
C-REF-W3-longevity-001	PRIMARY · verified	2026-11-11	https://api.github.com/repos/rauc/rauc/commits?since=2025-11-10T00:00:00Z
C-REF-W4-security-008	PRIMARY · verified	2026-11-11	https://arxiv.org/abs/2405.10131
C-REF-W6-technical-001	PRIMARY · verified	2026-11-10	https://ec.europa.eu/newsroom/dae/redirection/document/131456
C-REF-W6-technical-008	PRIMARY · partial	2026-11-10	https://ec.europa.eu/newsroom/dae/redirection/document/131456
C-REF-W7-T-001	PRIMARY · verified	2026-11-10	https://forums.developer.nvidia.com/t/announcing-end-of-life-for-nvidia-jetpack-4-with-the-release-of-jetpack-4-6-6/314300

Claim ID	Type · status	Accessed	Source
C-REF-W7-T-004	PRIMARY · verified	2026-11-10	https://developer.nvidia.com/embedded/jetson-linux-archive
C-REF-W7-T-005	PRIMARY · verified	2026-11-10	https://nvidia.custhelp.com/app/answers/detail/a_id/5797/
C-REF-W7L-001	PRIMARY · verified	2026-11-10	https://forums.developer.nvidia.com/t/announcing-end-of-life-for-nvidia-jetpack-5-with-the-release-of-jetpack-5-1-8/387449
C-REF3-DC-cost-001	SECONDARY · verified	2026-11-11	https://www.theregister.com/personal-tech/2026/11/03/raspberry-pi-hikes-prices-again-barely-a-month-after-the-last-round/5307004
C-REF3-DC-longevity-008	SECONDARY · verified	2026-11-11	https://lwn.net/Articles/1079808/
C-REF3-DC-security-008	PRIMARY · verified	2026-11-11	https://docs.k3s.io/security/secrets-encryption ; https://docs.k3s.io/cli/secrets-encrypt
C-REF3-DC-security-009	SECONDARY · verified	2026-11-11	https://oddlama.org/blog/bypassing-disk-encryption-with-tpm2-unlock/
C-REF3-DC-technical-004	PRIMARY · verified	2026-11-11	https://github.com/home-assistant/architecture/discussions/1198
C-REF3-DC-technical-005	PRIMARY · verified	2026-11-11	https://github.com/home-assistant/architecture/discussions/1197
C-REF3-W19-technical-002	PRIMARY-METADATA · verified	2026-11-11	https://github.com/omacom/omarchy-plugin-marketplace/blob/d2bf9a7a2005550b705ea271ff5cdd59aac05359/registry.json
C-REF3-W19-technical-007	PRIMARY · verified	2026-11-11	https://github.com/home-assistant/core/tree/d9115f3e0492aa94b06edb439ab1ebcd594b4738/homeassistant/components
C-REF4-DC-longevity-001	PRIMARY · verified	2026-11-11	https://www.centos.org/download/
C-REF4-DC-longevity-006	COMMUNITY · verified	2026-11-11	https://community.umbrel.com/t/umbreld-crashes-every-36h-with-gpf-in-libc-so-6-node-process-on-amd64-umbrelos-1-5-0-update-system-also-broken/24970
C-REF4-DC-longevity-016	PRIMARY · verified	2026-11-11	https://api.github.com/repos/k3s-io/kine/commits?since=2025-11-11T00:00:00Z
C-REF4-DC-security-001	PRIMARY · verified	2026-11-11	https://rauc.readthedocs.io/en/latest/reference.html
C-REF4-DC-security-007	PRIMARY · verified	2026-11-11	https://github.com/raspberrypi/usbboot/blob/master/docs/secure-boot.md
C-REF4-DC-security-008	PRIMARY · verified	2026-11-11	https://www.raspberrypi.com/documentation/security/security-architecture.html
C-REF5-S1-technical-003	PRIMARY · verified	2026-11-11	https://developer.nvidia.com/embedded/jetpack/downloads/archive-7.1.md
C-REF5-S1-technical-004	PRIMARY · verified	2026-11-11	https://docs.nvidia.com/jetson/archives/r39.2/ReleaseNotes/Jetson_Linux_Release_Notes_r39.2.pdf
C-REF5-S1-technical-006	PRIMARY · verified	2026-11-11	https://github.com/rauc/rauc/pull/1599
C-REF5-S1L-002	PRIMARY · verified	2026-11-11	https://github.com/canonical/canonical.com/blob/eebaf62d63ae3d51bd34f18fdf6e26e35a35dd96/templates/legal/intellectual-property-policy/index.md#L18

Claim ID	Type · status	Accessed	Source
C-REF5-S1L-009	PRIMARY · verified	2026-11-11	https://developer.nvidia.com/embedded/jet-pack
C-REF5S1SEC-001	PRIMARY · verified	2026-11-11	https://web.archive.org/web/20261023180559/https://ubuntu.com/legal/intellectual-property-policy
C-REF5S1SEC-002	PRIMARY · verified	2026-11-11	https://github.com/rhboot/shim-review/blob/e9b3a73d0e74fb1258a3ef68ea6e79e483134d24/README.md
C-REF5S1SEC-006	PRIMARY · verified	2026-11-11	https://web.archive.org/web/20261107214224/https://ubuntu.com/pro
C-SCORE-023	PRIMARY · verified	2026-11-11	https://canonical.com/blog/bosch-rexroth-adopts-ubuntu-core-and-snaps-for-app-based-ctrlx-automation-platform
C-SCORE-025	PRIMARY · verified	2026-11-11	https://github.com/AlmaLinux/bootc-images-rpi/blob/bb8e83e6b898db63ca2b53a69343f067dfa4f28a/README.md#L3-L44
C-SCORE-027	SECONDARY · verified	2026-11-11	https://nullr0ute.com/2026/10/raspberry-pi-5-on-fedora/
C-STEEL-jb-002	PRIMARY · verified	2026-11-11	https://github.com/OE4T/meta-tegra/blob/c5efa2ff644b6d90f8ac2ed09f57cbfb22194253/recipes-kernel/linux/linux-yocto-tegra.inc
C-STEEL-jb-010	PRIMARY · verified	2026-11-11	https://github.com/balena-os/balena-jetson-orin/blob/862a3522f4292c6bf8a00432b5087038fae00083/.gitmodules
C-STEEL-ubu-003	PRIMARY · verified	2026-11-11	https://canonical.com/blog/nvidia-gtc-2026
C-STEEL-ubu-016	PRIMARY · verified	2026-11-11	https://launchpad.net/ubuntu/resolute/+source/linux-restricted-modules
C-STEEL-updates-mender-015	PRIMARY · verified	2026-11-10	https://mender.io/pricing
C-STEEL4-bootc-013	PRIMARY · partial	2026-11-11	https://repo.download.nvidia.com/jetson/rhel-9.4/r36.3.1/nvidia-l4t.repo
C-STEEL4-bootc-014	PRIMARY · verified	2026-11-11	https://almalinux.org/p/the-almalinux-os-trademark-usage-policy/
C-STEEL4-bootc-025	PRIMARY · partial	2026-11-11	https://github.com/bootc-dev/bootc/blob/27f1e64ea93639929ed0088cdc230c220dac0fe6/docs/src/bootc-installation.7.md#L121-L131
C-TD-balenaos-003	PRIMARY · verified	2026-11-11	https://github.com/balena-os/meta-balena/blob/48184b689a43b762c5eea6efbc16b018eed70839/COPYING.Apache-2.0
C-TD-balenaos-010	PRIMARY · verified	2026-11-11	https://github.com/balena-os/meta-balena/blob/48184b689a43b762c5eea6efbc16b018eed70839/meta-balena-common/classes/image_types_balena.bbclass#L387-L402
C-TD-balenaos-012	PRIMARY · verified	2026-11-11	https://github.com/balena-os/meta-balena/blob/48184b689a43b762c5eea6efbc16b018eed70839/meta-balena-common/recipes-containers/hostapp-update/files/h…

Claim ID	Type · status	Accessed	Source
C-TD-balenaos-017	PRIMARY · verified	2026-11-11	https://github.com/balena-os/meta-balena/blob/48184b689a43b762c5eea6efbc16b018eed70839/meta-balena-common/recipes-core/balena-rollback/files/rollba…
C-TD-casaos-zimaos-002	PRIMARY · verified	2026-11-11	https://github.com/IceWhaleTech/ZimaOS/blob/8bc15999eb23085dbe5de5310cba4d7489507565/README.md#L33-L36
C-TD-casaos-zimaos-004	PRIMARY · verified	2026-11-11	https://github.com/IceWhaleTech/ZimaOS/blob/8bc15999eb23085dbe5de5310cba4d7489507565/README.md#L28 ; research-output/evidence/sources/TD-casaos-zim…
C-TD-casaos-zimaos-012	PRIMARY · verified	2026-11-11	research-output/evidence/sources/TD-casaos-zimaos/zimaos-1.8.0-bundle-inspection.txt
C-TD-casaos-zimaos-013	PRIMARY · verified	2026-11-11	research-output/evidence/sources/TD-casaos-zimaos/zimaos-1.8.0-bundle-inspection.txt
C-TD-casaos-zimaos-020	PRIMARY · verified	2026-11-11	https://github.com/IceWhaleTech/CasaOS/blob/e099dd6b4982a264663df2536a55bfea2ae5448a/README.md#L108-L142
C-TD-casaos-zimaos-023	PRIMARY · verified	2026-11-11	https://github.com/IceWhaleTech/CasaOS/blob/e099dd6b4982a264663df2536a55bfea2ae5448a/service/system.go#L373-L389 ; https://github.com/IceWhaleTech/…
C-TD-casaos-zimaos-049	PRIMARY · verified	2026-11-11	research-output/evidence/sources/TD-casaos-zimaos/zimaos-1.8.0-bundle-inspection.txt
C-TD-home-assistant-os-003	PRIMARY · verified	2026-11-10	https://github.com/home-assistant/operating-system/blob/fc301f556647b53281e853b9ce7f44915b3c7b14/Makefile#L1-L42
C-TD-home-assistant-os-009	PRIMARY · verified	2026-11-10	https://github.com/home-assistant/operating-system/blob/fc301f556647b53281e853b9ce7f44915b3c7b14/buildroot-external/ota/manifest.raucm.gtpl#L1-L10
C-TD-home-assistant-os-013	PRIMARY · verified	2026-11-10	https://github.com/home-assistant/operating-system/blob/fc301f556647b53281e853b9ce7f44915b3c7b14/.github/workflows/build.yaml#L255-L274
C-TD-home-assistant-os-018	PRIMARY · verified	2026-11-10	https://github.com/home-assistant/supervisor/blob/c1c6f0a754c0656eb1af189f11009f1cfc4b3e28/supervisor/core.py#L216-L239
C-TD-home-assistant-os-025	PRIMARY · verified	2026-11-10	https://github.com/home-assistant/supervisor/blob/c1c6f0a754c0656eb1af189f11009f1cfc4b3e28/supervisor/apps/utils.py#L26-L95

Claim ID	Type · status	Accessed	Source
C-TD-kairos-002	PRIMARY · verified	2026-11-11	https://github.com/kairos-io/kairos/blob/6f33334a6a6c88a4a38c0b7ec4df1ab0334b830d/LICENSE#L1-L3
C-TD-kairos-007	PRIMARY · verified	2026-11-11	https://github.com/kairos-io/kairos/blob/6f33334a6a6c88a4a38c0b7ec4df1ab0334b830d/images/Dockerfile#L1-L37
C-TD-kairos-008	PRIMARY · verified	2026-11-11	https://github.com/kairos-io/kairos/blob/6f33334a6a6c88a4a38c0b7ec4df1ab0334b830d/kairos-init/pkg/values/values.go#L30-L46
C-TD-kairos-019	PRIMARY · verified	2026-11-11	https://github.com/kairos-io/kairos-docs/blob/b1bf8bedc43697ce552b988a74f0f4e76afaeddf/docs/architecture/immutable.md#L104-L121
C-TD-kairos-022	PRIMARY · verified	2026-11-11	https://github.com/kairos-io/kairos-docs/blob/b1bf8bedc43697ce552b988a74f0f4e76afaeddf/docs/upgrade/boot_assessment.md#L50-L75
C-TD-olares-003	PRIMARY · verified	2026-11-10	https://github.com/beclab/Olares/blob/e7fe1c9634f26530386c2033a33be7a9495704f3/build/build.sh#L30-L48
C-TD-olares-004	PRIMARY · verified	2026-11-10	https://github.com/beclab/Olares/blob/e7fe1c9634f26530386c2033a33be7a9495704f3/.github/workflows/release.yaml#L206-L243
C-TD-olares-007	PRIMARY · verified	2026-11-10	https://github.com/beclab/Olares/blob/e7fe1c9634f26530386c2033a33be7a9495704f3/docs/manual/get-started/install-linux-iso.md#L21-L70
C-TD-olares-012	PRIMARY · verified	2026-11-10	https://github.com/beclab/Olares/blob/e7fe1c9634f26530386c2033a33be7a9495704f3/cli/pkg/upgrade/base.go#L196-L214
C-TD-olares-014	PRIMARY · verified	2026-11-10	https://github.com/beclab/Olares/blob/e7fe1c9634f26530386c2033a33be7a9495704f3/daemon/pkg/cluster/clusterop/upgrade.go#L247-L251
C-TD-olares-021	PRIMARY · verified	2026-11-10	https://github.com/beclab/Olares/blob/e7fe1c9634f26530386c2033a33be7a9495704f3/cli/pkg/common/common.go#L25
C-TD-olares-022	PRIMARY · verified	2026-11-10	https://github.com/kubernetes/website/blob/4c77b36edb121506e7eb2983777e7c4b49f98ab1/data/releases/eol.yaml#L7-L9
C-TD-olares-030	PRIMARY · verified	2026-11-10	https://github.com/beclab/apps/blob/37a1fb4f44b2428feb7978b25fdc798d6d6310ed/lares/templates/deployment.yaml#L181-L188
C-TD-olares-034	PRIMARY · verified	2026-11-10	https://github.com/beclab/Olares/blob/e7fe1c9634f26530386c2033a33be7a9495704f3/framework/app-service/pkg/security/templates.go#L317-L356
C-TD-olares-047	PRIMARY · verified	2026-11-10	https://github.com/beclab/Olares/blob/e7fe1c9634f26530386c2033a33be7a9495704f3/LICENSE.md#L13
C-TD-project-nomad-002	PRIMARY · verified	2026-11-11	https://github.com/Crosstalk-Solutions/project-nomad/blob/

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			0dbbd21c2f470975e86131c497694c147ca8bc00/LICENSE
C-TD-project-nomad-004	PRIMARY-METADATA · verified	2026-11-11	https://api.github.com/repos/Crosstalk-Solutions/project-nomad/contributors
C-TD-project-nomad-012	PRIMARY · verified	2026-11-11	https://www.projectnomad.us/install
C-TD-project-nomad-015	PRIMARY · verified	2026-11-11	https://github.com/Crosstalk-Solutions/project-nomad/blob/0dbbd21c2f470975e86131c497694c147ca8bc00/install/sidecar-updater/update-watcher.sh#L31-L117
C-TD-project-nomad-022	PRIMARY · verified	2026-11-11	https://github.com/Crosstalk-Solutions/project-nomad/blob/0dbbd21c2f470975e86131c497694c147ca8bc00/admin/database/seeders/service_seeder.ts#L37-L582
C-TD-proxmox-ve-003	PRIMARY · verified	2026-11-11	https://pve.proxmox.com/wiki/FAQ
C-TD-proxmox-ve-004	PRIMARY · verified	2026-11-11	https://github.com/proxmox/pve-manager/blob/e1e1c4afbe2dbf3ab8c32dea80aded60bd26b979/debian/copyright#L1-L8
C-TD-proxmox-ve-013	PRIMARY · verified	2026-11-11	https://github.com/proxmox/pve-manager/blob/e1e1c4afbe2dbf3ab8c32dea80aded60bd26b979/bin/pveupgrade#L38-L101
C-TD-startos-002	PRIMARY · verified	2026-11-11	https://github.com/Start9Labs/start-technologies/blob/175ce3beb370443d8db2cea30c3f99ebdcd1a8f5/LICENSE#L1-L3 ; https://github.com/Start9Labs/start-…
C-TD-startos-005	PRIMARY · verified	2026-11-11	https://github.com/Start9Labs/start-technologies/blob/175ce3beb370443d8db2cea30c3f99ebdcd1a8f5/projects/start-os/build/image-recipe/Dockerfile#L1 ;…
C-TD-startos-014	PRIMARY · verified	2026-11-11	https://github.com/Start9Labs/start-technologies/blob/175ce3beb370443d8db2cea30c3f99ebdcd1a8f5/shared-libs/crates/start-core/src/update/mod.rs#L278…
C-TD-startos-017	PRIMARY · verified	2026-11-11	https://github.com/Start9Labs/start-technologies/blob/175ce3beb370443d8db2cea30c3f99ebdcd1a8f5/projects/start-os/build/lib/scripts/startos-initramf…
C-TD-talos-002	PRIMARY · verified	2026-11-11	https://github.com/siderolabs/talos/releases/tag/v1.14.5
C-TD-talos-013	PRIMARY · verified	2026-11-11	https://github.com/siderolabs/talos/blob/dad8d3edfe7ca381044d7cad259fdf3030efa700/internal/app/machined/pkg/runtime/v1alpha1/v1alpha1_sequencer.go#…
C-TD-talos-015	PRIMARY · verified	2026-11-11	https://github.com/siderolabs/talos/blob/dad8d3edfe7ca381044d7cad259fdf3030efa700/

Claim ID	Type · status	Accessed	Source
			internal/app/machined/pkg/controllers/runtime/drop_upgrade_fallba…
C-TD-talos-035	PRIMARY · verified	2026-11-11	https://github.com/siderolabs/talos/blob/dad8d3edfe7ca381044d7cad259fdf3030efa700/LICENSE#L1-L2
C-TD-tier-c-nas-036	PRIMARY · verified	2026-11-11	https://help.synology.com/developer-guide/breaking_changes.html
C-TD-truenas-scale-003	PRIMARY · verified	2026-11-11	https://github.com/truenas/scale-build/blob/453acc8fddacf20c82a509a0177730e9479d49af/conf/build.manifest#L1-L30
C-TD-truenas-scale-011	PRIMARY · verified	2026-11-11	https://github.com/truenas/middleware/blob/29ed02b8082f38494b0d2896bada8151f0d59716/src/middlewared/middlewared/plugins/boot_/environments.py#L132-…
C-TD-truenas-scale-025	PRIMARY · verified	2026-11-11	https://github.com/truenas/middleware/blob/29ed02b8082f38494b0d2896bada8151f0d59716/LICENSE ; https://github.com/truenas/apps/blob/c85431d5f6ad13eb…
C-TD-umbrelos-003	PRIMARY · verified	2026-11-10	https://github.com/getumbrel/umbrel/blob/6f357e18d0a4faccb69d9fa59e3eae9cc3380538/packages/os/umbrelos.Dockerfile#L1-L10
C-TD-umbrelos-010	PRIMARY · verified	2026-11-10	https://github.com/getumbrel/umbrel/blob/a72dac249974d1603be9423d3cf6ac1a2d8f65f9/packages/umbreld/source/modules/system/update.ts#L52-L88
C-TD-umbrelos-013	PRIMARY · verified	2026-11-10	https://github.com/getumbrel/umbrel/blob/6f357e18d0a4faccb69d9fa59e3eae9cc3380538/packages/umbreld/source/modules/update/release-signing.ts#L5-L49
C-TD-umbrelos-019	PRIMARY · verified	2026-11-10	https://github.com/getumbrel/umbrel/blob/a72dac249974d1603be9423d3cf6ac1a2d8f65f9/packages/umbreld/source/index.ts#L229-L231
C-TD-umbrelos-045	PRIMARY · verified	2026-11-10	https://github.com/getumbrel/umbrel/wiki/License-FAQ
C-TD-villageos-002	PRIMARY · verified	2026-11-10	https://www.villageos.ai/
C-TD-villageos-007	PRIMARY · verified	2026-11-10	https://github.com/ReGen-Villages/VillageOS-API/blob/0082f5bb6763aca379cb24588b8d7153558a08d9/docs/FIELD_GUIDE.md#L294-L300
C-TD-villageos-009	PRIMARY · verified	2026-11-10	https://github.com/ReGen-Villages/VillageOS-API/blob/0082f5bb6763aca379cb24588b8d7153558a08d9/docs/FIELD_GUIDE.md#L2097-L2106
C-TD-villageos-014	PRIMARY · verified	2026-11-10	https://github.com/ReGen-Villages/VillageOS-API/blob/0082f5bb6763aca379cb24588b8d7153558a08d9/docs/FIELD_GUIDE.md#L773-L775
C-TD-villageos-016	PRIMARY · verified	2026-11-10	https://github.com/ReGen-Villages/VillageOS-API/blob/

Claim ID	Type · status	Accessed	Source
			0082f5bb6763aca379cb24588b8d7153558a08d9/deploy/Caddyfile#L1-L43
C-TD-villageos-022	PRIMARY · verified	2026-11-10	https://github.com/ReGen-Villages/VillageOS-API/blob/0082f5bb6763aca379cb24588b8d7153558a08d9/docs/FIELD_GUIDE.md#L57-L71
C-TD-villageos-023	PRIMARY · verified	2026-11-10	https://github.com/ReGen-Villages/VillageOS-API/tree/0082f5bb6763aca379cb24588b8d7153558a08d9
C-TD-villageos-024	PRIMARY · verified	2026-11-10	https://github.com/ReGen-Villages/VillageOS-API/blob/0082f5bb6763aca379cb24588b8d7153558a08d9/vos.Service.EnergyBalance/Services/Energy-BalanceCalcu…
C-TD-villageos-037	PRIMARY · verified	2026-11-10	https://github.com/farmOS/farmOS/blob/a86a15f825de836f9c0da45363a88e7b89824438/LICENSE.txt#L1-L339
C-TD-villageos-040	PRIMARY · verified	2026-11-10	https://www.home-assistant.io/docs/energy/
C-W10-004	PRIMARY · verified	2026-11-11	https://github.com/getumbrel/umbrel/blob/c43beed964074d08865d648ca9ac3359826a14ff/packages/umbreld/source/modules/apps/schema.ts#L91-L136
C-W10-006	PRIMARY · verified	2026-11-11	https://github.com/beclab/Olares/blob/e7fe1c9634f26530386c2033a33be7a9495704f3/framework/app-service/pkg/apiserver/compute_preflight_types.go#L20-L34
C-W10-007	PRIMARY · verified	2026-11-11	https://github.com/beclab/Olares/blob/e7fe1c9634f26530386c2033a33be7a9495704f3/framework/app-service/pkg/compute/preflight_types.go#L8-L50
C-W10-009	PRIMARY · verified	2026-11-11	https://github.com/beclab/Olares/blob/e7fe1c9634f26530386c2033a33be7a9495704f3/infrastructure/gpu/.olares/config/gpu/intel/node-feature-rules.yaml#…
C-W10-014	PRIMARY · verified	2026-11-11	https://github.com/NVIDIA/k8s-device-plugin/blob/de8841190ea6bbaceab572f5cb5847da54dff448/cmd/gpu-feature-discovery/main.go#L100-L141
C-W10-016	PRIMARY · verified	2026-11-11	https://docs.isaacsim.omniverse.nvidia.com/latest/installation/requirements.html
C-W10-018	PRIMARY · verified	2026-11-11	https://store.umbrel.com/products.json
C-W10-019	PRIMARY · verified	2026-11-11	https://one.olares.com/products.json
C-W10-020	PRIMARY · verified	2026-11-11	https://shop.zimaspace.com/products.json
C-W10-021	PRIMARY · verified	2026-11-11	https://www.home-assistant.io/green/
C-W10-024	PRIMARY · verified	2026-11-11	https://clawbox.com/home-ai-server
C-W10-025	PRIMARY · verified	2026-11-11	https://developer.nvidia.com/blog/nvidia-jetson-orin-nano-developer-kit-gets-a-super-boost/
C-W10-026	PRIMARY · verified	2026-11-11	https://developer.nvidia.com/embedded/faq

Claim ID	Type · status	Accessed	Source
C-W10-027	PRIMARY · verified	2026-11-11	https://developer.nvidia.com/embedded/faq
C-W10-032	PRIMARY · verified	2026-11-11	https://forums.developer.nvidia.com/t/2-23-2026-price-change-announcement/361713
C-W10-034	PRIMARY · verified	2026-11-11	https://www.raspberrypi.com/news/price-rises-for-some-16gb-raspberry-pi-products/
C-W10-035	PRIMARY · verified	2026-11-11	https://www.raspberrypi.com/products/compute-module-5/
C-W10-036	PRIMARY · verified	2026-11-11	https://www.raspberrypi.com/products/ai-hat-plus-2/
C-W10-037	SECONDARY · verified	2026-11-11	https://www.trendforce.com/presscenter/news/20260202-12911.html
C-W10-039	PRIMARY · verified	2026-11-11	https://www.intel.com/content/www/us/en/products/sku/241636/intel-processor-n150-6m-cache-up-to-3-60-ghz/specifications.html
C-W10-040	PRIMARY · verified	2026-11-11	https://www.intel.com/content/www/us/en/products/sku/231806/intel-core-i3n300-processor-6m-cache-up-to-3-80-ghz/specifications.html
C-W10-041	PRIMARY · verified	2026-11-11	https://www.intel.com/content/www/us/en/products/sku/84695/intel-core-i35005u-processor-3m-cache-2-00-ghz/specifications.html
C-W10-043	PRIMARY · verified	2026-11-11	https://www.dell.com/support/manuals/en-us/latitude-15-5540-laptop/latitude-5540-owners-manual/system-setup-options?guid=guid-3b0ef8b1-6a72-41b4-90…
C-W11-003	PRIMARY · verified	2026-11-11	https://github.com/getumbrel/umbrel/blob/6f357e18d0a4faccb69d9fa59e3eae9cc3380538/packages/ui/src/utils/misc.ts#L42-L44
C-W11-004	PRIMARY · verified	2026-11-11	https://github.com/getumbrel/umbrel/blob/6f357e18d0a4faccb69d9fa59e3eae9cc3380538/packages/umbreld/source/modules/notifications/notifications.ts#L1…
C-W11-009	PRIMARY · verified	2026-11-11	https://github.com/beclab/Olares/blob/e7fe1c9634f26530386c2033a33be7a9495704f3/docs/developer/develop/package/manifest.md#L363-L372
C-W11-011	PRIMARY · verified	2026-11-11	https://github.com/beclab/Olares/blob/e7fe1c9634f26530386c2033a33be7a9495704f3/apps/packages/app/package.json#L1-L255
C-W11-015	PRIMARY · verified	2026-11-11	https://github.com/home-assistant/android
C-W11-016	PRIMARY · verified	2026-11-11	https://companion.home-assistant.io/docs/notifications/notification-details/
C-W11-019	PRIMARY · verified	2026-11-11	https://contributors.home-assistant.io/blog/2026/05/26/advanced-mode-config-flow-deprecation
C-W11-043	PRIMARY · verified	2026-11-11	https://www.w3.org/TR/webxr/
C-W11-044	PRIMARY · verified	2026-11-11	https://developer.android.com/develop/xr/web

Claim ID	Type · status	Accessed	Source
C-W11-045	PRIMARY · verified	2026-11-11	https://developer.apple.com/videos/play/wwdc2024/10066/
C-W11-047	PRIMARY · verified	2026-11-11	https://github.com/basecamp/omarchy/blob/ceb30ed19ab1926713a7832dd4de5868c90d7b94/manual/17-ai.md#L57
C-W11-048	PRIMARY-METADATA · verified	2026-11-11	https://github.com/basecamp/omarchy/releases
C-W12-001	PRIMARY · verified	2026-11-11	https://github.com/getumbrel/umbrel-apps/tree/3a087384be96ea6890a7428f62e28a2203c1ea6e
C-W12-002	PRIMARY-METADATA · verified	2026-11-11	https://api.github.com/repos/getumbrel/umbrel-apps/license
C-W12-004	PRIMARY · verified	2026-11-11	https://github.com/beclab/apps/blob/37a1fb4f44b2428feb7978b25fdc798d6d6310ed/LICENSE
C-W12-007	PRIMARY · verified	2026-11-11	https://github.com/toeverything/AFFiNE/blob/e2767cb27186a344c5d14f1d1643caf50f638988/packages/backend/server/LICENSE#L1-L8
C-W12-008	PRIMARY · verified	2026-11-11	https://github.com/getumbrel/umbrel-apps/tree/3a087384be96ea6890a7428f62e28a2203c1ea6e
C-W12-009	PRIMARY · verified	2026-11-11	https://github.com/IceWhaleTech/CasaOS-AppStore/blob/820f99007fd521d0b534d94366371debea2ed438/Apps/Jellyfin/docker-compose.yml#L98-L105
C-W12-015	PRIMARY · verified	2026-11-11	https://github.com/open-webui/open-webui/blob/b0e6098e940942281d9c83f3529ea117287a3a22/LICENSE#L20-L36
C-W12-041	PRIMARY · verified	2026-11-11	https://developer.apple.com/app-store/review/guidelines/
C-W12-042	PRIMARY · verified	2026-11-11	https://support.google.com/googleplay/android-developer/answer/12991134
C-W12-043	PRIMARY · verified	2026-11-11	https://cloud.ouraring.com/legal/api-agreement
C-W12-044	PRIMARY · verified	2026-11-11	https://www.strava.com/legal/api
C-W12-061	PRIMARY · verified	2026-11-11	https://www.simplefin.org/protocol.html
C-W12-065	PRIMARY · verified	2026-11-11	https://plaid.com/legal/#end-user-privacy-policy
C-W12-070	PRIMARY · verified	2026-11-11	https://www.law.cornell.edu/uscode/text/18/1030
C-W12-075	PRIMARY · verified	2026-11-11	https://opensky-network.org/about/terms-of-use
C-W12-076	PRIMARY · verified	2026-11-11	https://www.maltego.com/license-agreement/
C-W12-147	PRIMARY · verified	2026-11-11	https://github.com/actualbudget/actual/blob/2ee836c222104854e0dedb904f940f4a4aca4a22/LICENSE.txt

Claim ID	Type · status	Accessed	Source
C-W12-157	PRIMARY · verified	2026-11-11	https://github.com/smicallef/spiderfoot/blob/0f815a203afebf05c98b605dba5cf0475a0ee5fd/LICENSE
C-W13-001	PRIMARY · verified	2026-11-11	https://github.com/jellyfin/jellyfin/blob/8466670e95b5f4f0b55d6353cf9fb3d6123217e3/MediaBrowser.Model/Users/UserPolicy.cs#L112-L182
C-W13-012	SECONDARY · partial	2026-11-11	https://calmatters.digitaldemocracy.org/bills/ca_202520260ab1043
C-W13-014	SECONDARY · partial	2026-11-11	https://anoni.net/news/en/2026/10/california-ab1856-open-source-exemption/
C-W13-015	PRIMARY · verified	2026-11-11	https://github.com/systemd/systemd/pull/40954
C-W13-020	PRIMARY · verified	2026-11-11	https://kolibri.readthedocs.io/en/latest/install/system_requirements.html
C-W13-022	PRIMARY · verified	2026-11-11	https://download.kiwix.org/zim/
C-W13-025	PRIMARY · verified	2026-11-11	https://help.circle.so/p/administration/analytics/export-your-data
C-W13-043	PRIMARY · verified	2026-11-11	https://github.com/home-assistant/my.home-assistant.io/blob/1bb9d44c84e9c6e3ede8c44f24178294754aaa8e/src/entrypoints/my-redirect.ts#L80-L120
C-W13-051	PRIMARY · verified	2026-11-11	https://community.umbrel.com/
C-W14-002	PRIMARY · verified	2026-11-11	https://grapheneos.org/faq#recommended-devices
C-W14-010	PRIMARY · partial	2026-11-11	https://github.com/GrapheneOS/os-issue-tracker/issues/8066
C-W14-011	PRIMARY · verified	2026-11-11	https://support.google.com/googlepixeltablet/answer/13555146?hl=en
C-W14-015	PRIMARY · verified	2026-11-11	https://github.com/TAK-Product-Center/atak-civ/blob/9f6893dd657feacc35ec5de03dad721c2e44170e/atak/ATAK/app/src/main/java/com/atakmap/android/locati…
C-W14-018	PRIMARY-METADATA · partial	2026-11-11	https://github.com/GrapheneOS/os-issue-tracker/issues?q=ATAK
C-W14-021	PRIMARY · verified	2026-11-11	https://docs.samsungknox.com/admin/fundamentals/whitepaper/samsung-knox-mobile-security/system-security/hw-backed-security/
C-W14-027	PRIMARY · verified	2026-11-11	https://calyxos.org/install/
C-W14-029	PRIMARY · verified	2026-11-11	https://nura.eco/blog/
C-W14-030	PRIMARY · verified	2026-11-11	https://furilabs.com/shop/flx1s/
C-W14-031	PRIMARY · verified	2026-11-11	https://github.com/waydroid/waydroid/issues/226
C-W14-033	PRIMARY · verified	2026-11-11	https://grapheneos.org/releases#2025030900
C-W14-036	PRIMARY · verified	2026-11-11	https://github.com/openharmony/communication_dsoftbus/blob/0e0c18c4a10b8fe2410d2097c38fd9f7a6ab8132/README.md

Claim ID	Type · status	Accessed	Source
C-W14-037	SECONDARY · unverified	2026-11-11	https://tech-ish.com/2026/08/30/harmonyos-worlds-third-largest-mobile-os/
C-W14-040	PRIMARY · verified	2026-11-11	https://grapheneos.org/releases
C-W15-001	PRIMARY · verified	2026-11-11	https://www.accessnow.org/press-release/rising-and-resisting-in-the-darkness-internet-shutdowns-in-2025/
C-W15-002	PRIMARY · verified	2026-11-11	https://www.accessnow.org/press-release/rising-and-resisting-in-the-darkness-internet-shutdowns-in-2025/
C-W15-003	PRIMARY · verified	2026-11-11	https://www.entsoe.eu/news/2026/03/20/entso-e-publishes-expert-panel-final-report-on-28-april-2025-blackout-in-spain-and-portugal/
C-W15-004	SECONDARY · partial	2026-11-11	https://www.globaltimes.cn/page/202511/1347625.shtml
C-W15-010	PRIMARY · verified	2026-11-11	https://developer.android.com/developer-verification
C-W15-015	PRIMARY · verified	2026-11-11	https://commission.europa.eu/news-and-media/news/commission-advances-cloud-sovereignty-through-strategic-procurement-2026-04-17_en
C-W15-023	SECONDARY · verified	2026-11-11	https://www.tomshardware.com/tech-industry/taiwanese-research-firm-says-tsmc-will-maintain-foundry-lead-until-at-least-2032
C-W16-001	PRIMARY · verified	2026-11-10	https://www.apache.org/licenses/LICENSE-2.0.txt
C-W16-003	PRIMARY · verified	2026-11-10	https://www.home-assistant.io/developers/cla/
C-W16-006	PRIMARY · verified	2026-11-10	https://github.com/beclab/Olares/commit/534a73122710057ea2c26763028a65117918dc90
C-W16-007	PRIMARY · verified	2026-11-10	https://www.hashicorp.com/en/blog/hashicorp-adopts-business-source-license
C-W16-008	PRIMARY · verified	2026-11-10	https://www.elastic.co/blog/elasticsearch-is-open-source-again
C-W16-009	PRIMARY · verified	2026-11-10	https://redis.io/blog/agplv3/
C-W16-012	PRIMARY · verified	2026-11-10	https://www.openhomefoundation.org/structure/
C-W16-013	PRIMARY · verified	2026-11-10	https://www.openhomefoundation.org/media-room/annual-report-factsheet/
C-W16-018	PRIMARY · verified	2026-11-10	https://www.prnewswire.com/news-releases/olares-showcases-private-ai-at-intels-ai-nas–thunderbolt-technology-forum-in-japan-302909516.html
C-W16-021	PRIMARY · verified	2026-11-10	https://start9.com/
C-W16-031	PRIMARY · verified	2026-11-10	https://www.nvidia.com/en-us/startups/
C-W16-050	PRIMARY · verified	2026-11-10	https://sandstorm.io/news/2017-02-06-sandstorm-returning-to-community-roots
C-W17-001	PRIMARY · verified	2026-11-11	https://bootlin.com/docs/
C-W18-001	PRIMARY · verified	2026-11-10	https://www.kernel.org/category/releases.html

Claim ID	Type · status	Accessed	Source
C-W18-003	PRIMARY · verified	2026-11-10	https://cip-project.org/blog/2025/05/26/cip-is-now-supporting-five-slts-kernels
C-W18-005	PRIMARY · verified	2026-11-10	https://github.com/torvalds/linux/blob/adc218676eef25575469234709c2d87185ca223a/kernel/Kconfig.preempt#L70-L72
C-W18-006	PRIMARY · verified	2026-11-10	https://packages.debian.org/trixie/linux-image-rt-arm64
C-W18-008	PRIMARY · verified	2026-11-10	https://github.com/home-assistant/operating-system/blob/fc301f556647b53281e853b9ce7f44915b3c7b14/buildroot-external/configs/generic_x86_64_defconfi…
C-W18-009	PRIMARY · verified	2026-11-10	https://github.com/home-assistant/operating-system/blob/fc301f556647b53281e853b9ce7f44915b3c7b14/buildroot-external/configs/rpi5_64_defconfig#L20-L24
C-W18-010	PRIMARY-METADATA · verified	2026-11-10	https://api.github.com/repos/home-assistant/operating-system/commits?path=buildroot-external/configs/generic_x86_64_defconfig
C-W18-012	PRIMARY · verified	2026-11-10	https://github.com/getumbrel/umbrel/blob/a72dac249974d1603be9423d3cf6ac1a2d8f65f9/packages/os/umbrelos.Dockerfile#L105-L111
C-W18-013	PRIMARY · verified	2026-11-10	https://github.com/getumbrel/umbrel/blob/a72dac249974d1603be9423d3cf6ac1a2d8f65f9/packages/os/build-steps/setup-raspberrypi.sh#L9-L35
C-W18-015	PRIMARY · verified	2026-11-10	https://github.com/getumbrel/umbrel/blob/a72dac249974d1603be9423d3cf6ac1a2d8f65f9/packages/os/rugix/rugix-bakery.toml#L1-L22
C-W18-018	PRIMARY · verified	2026-11-10	https://github.com/torvalds/linux/tree/7d0a66e4bb9081d75c82ec4957c50034cb0ea449/arch/arm64/boot/dts/nvidia
C-W18-019	PRIMARY · verified	2026-11-10	https://developer.nvidia.com/embedded/jetpack
C-W18-021	PRIMARY · verified	2026-11-10	https://docs.nvidia.com/jetson/archives/r39.2/DeveloperGuide/IN/QuickStart.html
C-W18-023	PRIMARY · verified	2026-11-10	https://docs.nvidia.com/jetson/orin-nano-devkit/user-guide/latest/quick_start.html
C-W18-028	PRIMARY · verified	2026-11-10	https://www.nvidia.com/en-us/drivers/unix/
C-W18-029	PRIMARY · verified	2026-11-10	https://kubernetes.io/docs/concepts/windows/intro/
C-W18-030	PRIMARY · verified	2026-11-10	https://github.com/apple-oss-distributions/xnu/blob/f6217f891ac0bb64f3d375211650a4c1ff8ca1ea/README.md#L3-L4
C-W18-031	PRIMARY · verified	2026-11-10	https://github.com/apple/container/blob/d5416e63231783c9699d31f35381caac941d3cde/README.md#L6-L20
C-W18-032	PRIMARY · verified	2026-11-10	https://docs.freebsd.org/en/books/handbook/jails/

Claim ID	Type · status	Accessed	Source
C-W18-033	PRIMARY · verified	2026-11-10	https://www.truenas.com/blog/truenas-core-13-3-plans/
C-W18-034	PRIMARY · verified	2026-11-10	https://github.com/ros-infrastructure/rep/blob/11ca24a41f31480dfb9562ba99f2a5b93d3ebda5/rep-2000.rst#L1204-L1225
C-W18-035	PRIMARY · verified	2026-11-10	https://github.com/seL4/seL4/blob/c98d70adf15e8be7d3b99085c28d057069ba7155/LICENSE.md
C-W18-036	PRIMARY · verified	2026-11-10	https://www.usenix.org/conference/osdi24/presentation/chen-haibo
C-W18-039	SECONDARY · partial	2026-11-10	https://www.techspot.com/news/89814-google-fuchsia-os-now-available-update-first-gen.html
C-W18-040	PRIMARY · verified	2026-11-10	https://github.com/redox-os/redox/blob/674d255f5fa189275e85968c52d55e6711092891/README.md#L15
C-W18-041	PRIMARY · verified	2026-11-10	https://source.android.com/docs/core/architecture/kernel/generic-kernel-image
C-W18-043	PRIMARY · verified	2026-11-10	https://www.chromium.org/chromium-os/chromiumos-design-docs/chromium-os-kernel/
C-W18-044	PRIMARY · verified	2026-11-10	https://www.collabora.com/news-and-blog/news-and-events/steamos-3-6-how-the-steam-deck-atomic-updates-are-improving.html
C-W18-045	PRIMARY · verified	2026-11-10	https://github.com/siderolabs/pkgs/blob/17fdd3333e028525c10cbd90197c4958c03936da/Pkgfile#L114-L116
C-W18-050	PRIMARY · verified	2026-11-10	https://docs.yoctoproject.org/ref-manual/system-requirements.html
C-W18-054	PRIMARY · verified	2026-11-10	https://github.com/lima-vm/lima/blob/13424c4af7393a19cb6cad1ba96e87d100d57857/templates/default.yaml#L682-L695
C-W18-058	PRIMARY · verified	2026-11-10	https://docs.isaacsim.omniverse.nvidia.com/latest/installation/requirements.html
C-W18-062	PRIMARY-METADATA · verified	2026-11-10	https://github.com/home-assistant/operating-system/releases
C-W18-063	PRIMARY-METADATA · verified	2026-11-10	https://github.com/getumbrel/umbrel-os/releases
C-W18-065	PRIMARY · verified	2026-11-10	research-output/evidence/sources/W18.md#lab
C-W18-066	PRIMARY · verified	2026-11-10	research-output/evidence/sources/W18.md#lab
C-W18-067	PRIMARY · verified	2026-11-10	research-output/evidence/sources/W18.md#lab
C-W18-069	PRIMARY · verified	2026-11-10	https://github.com/torvalds/linux/blob/7d0a66e4bb9081d75c82ec4957c50034cb0ea449/COPYING#L1-L12
C-W19-005	PRIMARY · verified	2026-11-11	https://www.flatcar.org/docs/latest/sys-ext/
C-W19-006	PRIMARY · verified	2026-11-11	evidence/lab/w19-sysext/README.md

Claim ID	Type · status	Accessed	Source
C-W19-010	PRIMARY · verified	2026-11-11	https://github.com/omacom/omarchy/blob/ceb30ed19ab1926713a7832dd4de5868c90d7b94/bin/omarchy-plugin-validate#L41-L116
C-W19-011	PRIMARY · verified	2026-11-11	https://github.com/omacom/omarchy/blob/ceb30ed19ab1926713a7832dd4de5868c90d7b94/bin/omarchy-plugin-add#L14-L115
C-W19-012	PRIMARY · verified	2026-11-11	https://github.com/omacom/omarchy-plugin-marketplace/blob/f01e1860d89bc37c952218474fbfee5457de198d/README.md#L26-L37
C-W19-015	PRIMARY · verified	2026-11-11	https://github.com/hacs/integration/blob/8a543828351b963438a1613a6f06332138e4f4d6/custom_components/hacs/base.py#L643
C-W19-018	PRIMARY · verified	2026-11-11	https://github.com/home-assistant/core/blob/d8a0dc698c8946b44e455d904000db563cfbf1ba/script/hassfest/quality_scale.py#L37-L95
C-W19-020	PRIMARY · verified	2026-11-11	https://partner.home-assistant.io/
C-W19-021	PRIMARY · verified	2026-11-11	https://github.com/obsidianmd/obsidian-releases/blob/39ae8b23b4bf31935407948fe1ee645cacf7774d/community-plugins.json
C-W19-023	PRIMARY · verified	2026-11-11	https://code.visualstudio.com/docs/configure/extensions/extension-runtime-security
C-W19-025	PRIMARY · verified	2026-11-11	https://blogs.eclipse.org/post/mika%C3%ABl-barbero/open-vsx-security-update-october-2025
C-W19-026	PRIMARY · verified	2026-11-11	https://github.blog/security/supply-chain-security/our-plan-for-a-more-secure-npm-supply-chain/
C-W19-028	PRIMARY · verified	2026-11-11	https://wordpress.org/news/2024/10/secure-custom-fields/
C-W19-030	PRIMARY · verified	2026-11-11	https://grafana.com/legal/plugins/
C-W19-033	PRIMARY · verified	2026-11-11	https://github.com/getumbrel/umbrel/blob/6f357e18d0a4faccb69d9fa59e3eae9cc3380538/packages/umbreld/source/modules/apps/app-repository.ts#L71-L97
C-W19-034	PRIMARY · verified	2026-11-11	https://github.com/beclab/Olares/blob/e7fe1c9634f26530386c2033a33be7a9495704f3/framework/oac/README.md#L222
C-W19-035	PRIMARY · verified	2026-11-11	https://github.com/modelcontextprotocol/ext-apps/blob/ea3e9d0e624a20bc1c51310292f09d9b8ed17161/specification/2026-01-26/apps.mdx#L474-L481
C-W19-036	PRIMARY · verified	2026-11-11	https://github.com/viamrobotics/rdk/blob/046082ec9d5180319bf37559ad0df7659fcb91e0/cli/module_registry.go#L53-L122
C-W19-038	PRIMARY · verified	2026-11-11	https://github.com/ros/rosdistro/blob/06990791cebf0c0789868ed7f8202a0eac1dd47d/lyrical/distribution.yaml#L5-L13

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C-W19-039	PRIMARY · verified	2026-11-11	https://source.android.com/docs/compatibility/cdd
C-W19-040	PRIMARY · verified	2026-11-11	https://developer.apple.com/app-store/small-business-program/
C-W19-041	PRIMARY · verified	2026-11-11	https://support.google.com/googleplay/android-developer/answer/112622
C-W19-042	PRIMARY · verified	2026-11-11	https://shopify.dev/docs/apps/launch/distribution/revenue-share
C-W19-044	PRIMARY · verified	2026-11-11	https://www.balena.io/pricing
C-W2-001	PRIMARY · verified	2026-11-10	https://www.debian.org/releases/
C-W2-004	PRIMARY · verified	2026-11-10	https://www.freexian.com/lts/extended/debian-elts-subscription-form.pdf
C-W2-005	PRIMARY · verified	2026-11-10	https://ubuntu.com/about/release-cycle
C-W2-006	PRIMARY · verified	2026-11-10	https://ubuntu.com/pricing/devices
C-W2-008	PRIMARY · verified	2026-11-10	https://documentation.ubuntu.com/core/reference/support-duration/
C-W2-009	PRIMARY · verified	2026-11-10	https://documentation.ubuntu.com/core/reference/release-notes/
C-W2-013	PRIMARY · verified	2026-11-10	https://wiki.almalinux.org/release-notes/
C-W2-015	PRIMARY · verified	2026-11-10	https://bodhi.fedoraproject.org/releases/F45
C-W2-019	PRIMARY · verified	2026-11-10	https://github.com/fedora-iot/iot-distro/issues/123
C-W2-024	PRIMARY · verified	2026-11-10	https://github.com/bootc-dev/bootc/blob/27f1e64ea93639929ed0088cdc230c220dac0fe6/README.md#L22-L23
C-W2-026	PRIMARY · verified	2026-11-10	https://github.com/NVIDIA/cloud-native-docs/blob/d856ea69d8d9138ac171f24e1f9d08df53448591/gpu-operator/platform-support.rst#L296-L420
C-W2-028	PRIMARY · verified	2026-11-10	https://github.com/ros2/ros2_documentation/blob/1c5c20cb9f14cbfec658690c44b70ca3d3ec81e7/source/Releases/lyrical/supported-platforms.rst#L19-L121
C-W2-029	PRIMARY · verified	2026-11-10	https://www.raspberrypi.com/products/raspberry-pi-5/
C-W2-030	PRIMARY · verified	2026-11-10	https://developer.nvidia.com/embedded/lifecycle
C-W2-034	PRIMARY · verified	2026-11-10	https://wiki.yoctoproject.org/wiki/Releases
C-W2-035	PRIMARY · verified	2026-11-10	https://buildroot.org/lts.html
C-W2-040	PRIMARY · verified	2026-11-10	https://api.balena-cloud.com/v7/device_type
C-W2-041	PRIMARY · verified	2026-11-10	https://www.redhat.com/en/blog/red-hat-device-edge-now-available-run-nvidia-jetson-orin
C-W2-042	PRIMARY · verified	2026-11-10	https://access.redhat.com/compliance/iec-62443-4-2
C-W2-043	PRIMARY · verified	2026-11-10	https://github.com/getumbrel/umbrel/blob/a72dac249974d1603be9423d3cf6ac1a2d8f65f9/packages/os/umbrelos.Dockerfile#L1-L6

Claim ID	Type · status	Accessed	Source
C-W2-044	PRIMARY · verified	2026-11-10	https://github.com/getumbrel/umbrel/blob/a72dac249974d1603be9423d3cf6ac1a2d8f65f9/packages/os/build-steps/setup-raspberrypi.sh#L13-L35
C-W2-045	PRIMARY · verified	2026-11-10	https://github.com/getumbrel/umbrel/blob/a72dac249974d1603be9423d3cf6ac1a2d8f65f9/packages/os/umbrelos.Dockerfile#L136-L150
C-W2-047	PRIMARY · verified	2026-11-10	https://github.com/home-assistant/operating-system/blob/fc301f556647b53281e853b9ce7f44915b3c7b14/buildroot-external/configs/rpi5_64_defconfig#L20-L24
C-W2-049	PRIMARY · verified	2026-11-10	https://github.com/rauc/rauc/blob/2d17b27041db7fa8b13a61c5e168f49310a06566/COPYING#L1-L501 ; https://github.com/siderolabs/talos/blob/19df1db27e88c…
C-W2-050	PRIMARY · verified	2026-11-10	https://canonical.com/legal/intellectual-property-policy
C-W2-054	PRIMARY · verified	2026-11-10	research-output/evidence/lab/w2-fedora-bootc-greenboot/h2.log
C-W2-056	PRIMARY · verified	2026-11-10	research-output/evidence/lab/w2-fedora-bootc-greenboot/run-build.sh
C-W2-057	PRIMARY · verified	2026-11-10	https://ubuntu.com/download/raspberry-pi
C-W3-001	PRIMARY · verified	2026-11-10	https://api.github.com/repos/rugix/rugix/commits?since=2025-11-10T00:00:00Z
C-W3-024	PRIMARY · verified	2026-11-10	https://github.com/home-assistant/operating-system/blob/fc301f556647b53281e853b9ce7f44915b3c7b14/buildroot-external/ota/rel-ca-xsign-oldroot.cert.p…
C-W3-025	PRIMARY · verified	2026-11-10	https://github.com/home-assistant/supervisor/blob/c1c6f0a754c0656eb1af189f11009f1cfc4b3e28/supervisor/core.py#L230-L239 ; https://github.com/home-a…
C-W3-029	PRIMARY · verified	2026-11-10	https://github.com/home-assistant/operating-system/blob/fc301f556647b53281e853b9ce7f44915b3c7b14/buildroot-external/board/raspberrypi/rpi5-64/rootf…
C-W3-034	PRIMARY · verified	2026-11-10	https://github.com/getumbrel/umbrel/blob/a72dac249974d1603be9423d3cf6ac1a2d8f65f9/packages/umbreld/source/modules/system/update.ts#L72-L90
C-W3-036	PRIMARY · verified	2026-11-10	https://github.com/getumbrel/umbrel/blob/a72dac249974d1603be9423d3cf6ac1a2d8f65f9/packages/umbreld/source/index.ts#L230-L231 ; https://github.com/g…
C-W3-039	PRIMARY · verified	2026-11-10	https://github.com/getumbrel/umbrel/blob/81d2e7c90c3f57452bf8534d075cb56d8a1fe7ba/packages/os/umbrelos.Dockerfile#L1 ; https://github.com/getumbrel…

Claim ID	Type · status	Accessed	Source
C-W3-040	PRIMARY · verified	2026-11-10	https://github.com/getumbrel/umbrel/blob/a72dac249974d1603be9423d3cf6ac1a2d8f65f9/packages/os/README.md#L69-L120 ; https://github.com/getumbrel/umb…
C-W3-049	PRIMARY · verified	2026-11-10	https://docs.nvidia.com/jetson/archives/r39.2/DeveloperGuide/SD/RootFileSystem.html
C-W3-050	PRIMARY · verified	2026-11-10	https://docs.nvidia.com/jetson/archives/r39.2/DeveloperGuide/SD/SoftwarePackagesAndTheUpdateMechanism.html
C-W3-052	PRIMARY · verified	2026-11-10	https://github.com/OE4T/tegra-demo-distro/blob/e151343f1dd3ff6c19c447214b75ba3150c2d416/README.md#L14-L90
C-W3-058	PRIMARY · verified	2026-11-10	https://github.com/bootc-dev/bootc/blob/27f1e64ea93639929ed0088cdc230c220dac0fe6/docs/src/bootc-bootloaders.7.md#L8-L16 ; https://github.com/bootc-…
C-W4-012	PRIMARY · verified	2026-11-10	https://www.suse.com/lifecycle/
C-W4-016	PRIMARY · verified	2026-11-10	https://kubernetes.io/blog/2025/11/11/ingress-nginx-retirement/
C-W4-022	PRIMARY · verified	2026-11-10	https://docs.nvidia.com/datacenter/cloud-native/gpu-operator/latest/platform-support.html
C-W4-026	PRIMARY · verified	2026-11-10	https://docs.nvidia.com/datacenter/cloud-native/container-toolkit/latest/install-guide.html
C-W4-029	PRIMARY · verified	2026-11-10	https://github.com/NVIDIA/k8s-device-plugin/blob/de8841190ea6bbaceab572f5cb5847da54dff448/README.md#L76
C-W4-030	PRIMARY · verified	2026-11-10	https://github.com/NVIDIA/k8s-dra-driver-gpu/blob/e775fdb63e593a121b40fc1cc4c9a9c89955e635/README.md#L8-L39
C-W4-032	PRIMARY · verified	2026-11-10	https://blogs.nvidia.com/blog/nvidia-at-kubecon-2026/
C-W4-034	PRIMARY · verified	2026-11-10	https://github.com/NVIDIA/KAI-Scheduler/blob/2e3015e96fbae1680125a5802d0596a1183321ef/README.md#L54-L169
C-W4-036	PRIMARY · verified	2026-11-10	https://project-hami.io/docs/next/userguide/device-supported
C-W4-037	PRIMARY · verified	2026-11-10	https://github.com/NVIDIA/aicr/blob/c760855ed7307f40731a11f9a1b7898b76167209/README.md#L84-L92
C-W4-038	PRIMARY · verified	2026-11-10	https://github.com/cncf/k8s-ai-conformance/tree/c64e5bc8085da207715df96e458f2c6585c9f53e
C-W4-039	PRIMARY · verified	2026-11-10	https://developer.nvidia.com/blog/manage-kubernetes-clusters-for-gpu-infrastructure-with-nvidia-kubernetes-engine/

Claim ID	Type · status	Accessed	Source
C-W4-040	PRIMARY · verified	2026-11-10	https://developer.nvidia.com/blog/agent-sub-strate-advancing-kubernetes-for-agents-at-scale/
C-W4-041	PRIMARY · verified	2026-11-10	https://github.com/beclab/Olares/blob/e7fe1c9634f26530386c2033a33be7a9495704f3/cli/pkg/k3s/templates/k3sService.go#L45
C-W4-042	PRIMARY · verified	2026-11-10	https://github.com/beclab/Olares/blob/e7fe1c9634f26530386c2033a33be7a9495704f3/docs/manual/get-started/reusables.md#L74-L80
C-W4-043	PRIMARY · verified	2026-11-10	https://github.com/getumbrel/umbrel/blob/a72dac249974d1603be9423d3cf6ac1a2d8f65f9/packages/umbreld/source/modules/apps/app.ts#L518
C-W4-044	PRIMARY · verified	2026-11-10	https://github.com/home-assistant/supervi-sor/tree/c1c6f0a754c0656eb1af189f11009f1cfc4b3e28/supervisor/docker
C-W4-054	PRIMARY · verified	2026-11-10	https://docs.nats.io/running-a-nats-service/configuration/leafnodes/jetstream_leafnodes
C-W4-056	PRIMARY · verified	2026-11-10	https://www.cncf.io/blog/2025/05/01/pro-tecting-nats-and-the-integrity-of-open-source-cncfs-commitment-to-the-community/
C-W4-058	PRIMARY · verified	2026-11-10	https://github.com/NousResearch/hermes-agent/blob/a55a299b6520dd30bab2871bb58dffbae2f8a915/Dockerfile#L8
C-W4-059	PRIMARY · verified	2026-11-10	https://backstage.io/docs/getting-started/
C-W4-063	PRIMARY · verified	2026-11-10	context/01-proven-infrastructure-cluster.md §2 + founder infra repo docs/portability (private, read-only 2026-11-10)
C-W4-064	PRIMARY · verified	2026-11-10	https://github.com/siderolabs/sbc-jetson/pull/23
C-W4-078	PRIMARY · verified	2026-11-11	lab W4-1: research-output/evidence/lab/w4-k3s-footprint/run-1.log (sha256 c019bc07aa57c3bdd24feff626e3e49ce6e115fcc9d8ade0533d3a0
C-W5-001	PRIMARY · verified	2026-11-10	https://github.com/k3s-io/k3s/blob/f6973cb850e9b245585363d15bde80904ac862fd/pkg/cluster/managed.go#L70-L75
C-W5-005	PRIMARY · verified	2026-11-10	https://github.com/k3s-io/k3s/blob/f6973cb850e9b245585363d15bde80904ac862fd/pkg/etcd/etcd.go#L898-L964
C-W5-010	PRIMARY · verified	2026-11-10	lab:research-output/evidence/lab/w5-restore/logs/b1.log, logs/a2.log
C-W5-013	PRIMARY · verified	2026-11-10	https://github.com/k3s-io/docs/blob/8958508d493c12778517f2a178080f221cbe1081/docs/cli/etcd-snapshot.md#L318-L330
C-W5-019	PRIMARY · verified	2026-11-10	https://github.com/k3s-io/k3s/blob/f6973cb850e9b245585363d15bde80904ac862fd/manifests/local-storage.yaml#L96-L100

Claim ID	Type · status	Accessed	Source
C-W5-023	PRIMARY · verified	2026-11-10	https://github.com/restic/restic/blob/6aa3a516ce654808a1f28f9fa21e9b7c8e6e90bf/doc/design.rst#L314-L320
C-W5-056	PRIMARY · verified	2026-11-10	context/01-proven-infrastructure-cluster.md#L389-L401
C-W6-001	PRIMARY · verified	2026-11-10	https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:32024R2847
C-W6-002	PRIMARY · verified	2026-11-10	https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:32024R2847
C-W6-008	PRIMARY · verified	2026-11-10	https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:32024R2847
C-W6-011	PRIMARY · verified	2026-11-10	https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:32024R2847
C-W6-013	PRIMARY · verified	2026-11-10	https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:32024R2847
C-W6-026	PRIMARY · verified	2026-11-10	https://csrc.nist.gov/projects/cryptographic-module-validation-program/certificate/4407
C-W6-042	PRIMARY · verified	2026-11-10	https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:32026R1744
C-W6-047	PRIMARY · verified	2026-11-10	https://www.freedesktop.org/software/systemd/man/latest/systemd-cryptenroll.html
C-W6-052	PRIMARY · verified	2026-11-10	https://github.com/getumbrel/umbrel/blob/c43beed964074d08865d648ca9ac3359826a14ff/packages/umbreld/source/modules/mcp/tools/system-management.ts#L9…
C-W6-068	PRIMARY · verified	2026-11-10	https://genai.owasp.org/llmrisk/llm01-prompt-injection/
C-W7-001	PRIMARY · verified	2026-11-10	https://developer.nvidia.com/embedded/life-cycle
C-W7-002	PRIMARY · verified	2026-11-10	https://developer.nvidia.com/embedded/life-cycle
C-W7-006	PRIMARY · verified	2026-11-10	https://developer.nvidia.com/embedded/jetson-linux-archive
C-W7-007	PRIMARY · verified	2026-11-10	https://developer.nvidia.com/embedded/jetpack
C-W7-008	PRIMARY · verified	2026-11-10	https://developer.nvidia.com/embedded/jetpack/downloads/archive-7.2
C-W7-026	PRIMARY · verified	2026-11-10	https://github.com/ros-infrastructure/rep/blob/11ca24a41f31480dfb9562ba99f2a5b93d3ebda5/rep-2000.rst#L1088-L1100
C-W7-027	PRIMARY · verified	2026-11-10	https://github.com/ros2/ros2_documentation/blob/1c5c20cb9f14cbfec658690c44b70ca3d3ec81e7/source/Releases/lyrical/supported-platforms.rst#L19-L38
C-W7-029	PRIMARY · verified	2026-11-10	https://github.com/ros2/ros2_documentation/blob/1c5c20cb9f14cbfec658690c44b70ca3d3ec81e7/source/Releases.rst#L141-L165

Claim ID	Type · status	Accessed	Source
C-W7-033	PRIMARY · verified	2026-11-10	https://fast-dds.docs.eprosima.com/en/latest/docker/shm_docker.html
C-W7-039	PRIMARY · verified	2026-11-10	https://docs.clearpathrobotics.com/docs/ros/installation/robot
C-W7-040	PRIMARY · verified	2026-11-10	https://dev.bostondynamics.com/docs/payload/docker_containers
C-W7-041	PRIMARY · verified	2026-11-10	https://docs.px4.io/main/en/flight_controller/modalai_voxl_2
C-W7-042	PRIMARY · verified	2026-11-10	https://docs.auterion.com/vehicle-operation/settings-and-maintenance/skynode-software-update
C-W7-044	PRIMARY · verified	2026-11-10	https://docs.px4.io/main/en/middleware/uxrce_dds.html
C-W7-045	PRIMARY · verified	2026-11-10	https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:32023R1230
C-W7-046	PRIMARY · verified	2026-11-10	https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:32023R1230
C-W7-047	PRIMARY · verified	2026-11-10	https://www.anthropic.com/news/model-hardware-standard-research-preview
C-W7-048	PRIMARY · verified	2026-11-10	https://blogs.nvidia.com/blog/jetson-thor-robotics-edge-ai-agent/
C-W7-049	PRIMARY · verified	2026-11-10	https://github.com/torvalds/linux/tree/7d0a66e4bb9081d75c82ec4957c50034cb0ea449/drivers/accel
C-W7-050	PRIMARY-METADATA · verified	2026-11-10	https://github.com/google-coral/edgetpu/tree/5020de9386ff370dcc1f63291a2d0f98eeb98adb
C-W7-051	PRIMARY · verified	2026-11-10	https://github.com/NVIDIA/Isaac-GR00T/blob/23ace64f17aa5015259b8609d371eb61a357c776/LICENSE#L1-L190 ; https://github.com/NVIDIA/Cosmos/blob/284be18…
C-W7-053	PRIMARY · verified	2026-11-10	https://github.com/dora-rs/dora/blob/668e59f2bbae38378f7e3d06226881ff37caabf6/LICENSE#L1-L201 ; https://github.com/copper-project/copper-rs/blob/fc…
C-W7-054	SECONDARY · partial	2026-11-10	https://avweb.com/aviation-news/aviation-law/faa-reopens-part-108-bvlos-comment-period/ ; https://dronebundle.com/blog/part-108-nprm
C-W7-055	SECONDARY · partial	2026-11-10	https://www.iso.org/standard/73933.html ; https://www.iso.org/standard/83498.html
C-W8-002	PRIMARY · verified	2026-11-10	https://sources.debian.org/data/main/l/linux/6.12.107-1/debian/config/config
C-W8-003	PRIMARY · verified	2026-11-10	https://sources.debian.org/data/main/l/linux/6.12.107-1/debian/config/config
C-W8-004	PRIMARY · verified	2026-11-10	https://packages.debian.org/trixie/libopen62541-1.4
C-W8-005	PRIMARY · verified	2026-11-10	https://sources.debian.org/api/src/mosquitto/

Claim ID	Type · status	Accessed	Source
C-W8-007	PRIMARY · verified	2026-11-10	https://github.com/nats-io/nats-server/blob/0e5b4f72ac27f7fbb3bb0fe85fd964f5d4c6b664/server/mqtt.go#L3896-L3901
C-W8-008	PRIMARY · verified	2026-11-10	https://github.com/nats-io/nats-server/blob/0e5b4f72ac27f7fbb3bb0fe85fd964f5d4c6b664/server/mqtt.go#L5288-L5300
C-W8-014	PRIMARY · verified	2026-11-10	https://github.com/markqvist/Reticulum/blob/16621a78e35e2ba5156c72af7fe492458ba092d5/LICENSE
C-W8-015	PRIMARY · verified	2026-11-10	https://github.com/meshtastic/ATAK-Plugin/blob/784530f8bc9271d59d33caf8afb40f464cab032f/README.md
C-W8-018	PRIMARY · verified	2026-11-10	https://raw.githubusercontent.com/TAK-Product-Center/Server/5187abd46d827d37cfc5708805eced197a837e49/src/docs/TAK_Server_Configuration_Guide.pdf
C-W8-019	PRIMARY · verified	2026-11-10	https://sources.debian.org/api/src/openjdk-17/
C-W8-026	PRIMARY · verified	2026-11-10	https://github.com/getumbrel/umbrel-apps/blob/51aac978c67c7c9bbe7e0b42f64ff5f3139c8763/home-assistant/docker-compose.yml#L4-L13
C-W8-029	PRIMARY · verified	2026-11-10	https://esphome.io/components/api/
C-W8-030	PRIMARY · verified	2026-11-10	https://esphome.io/components/switch/
C-W8-032	PRIMARY · verified	2026-11-10	https://www.chirpstack.io/docs/chirpstack/integrations/mqtt.html
C-W8-033	PRIMARY · verified	2026-11-10	https://www.moxa.com/en/products/industrial-computing/system-software/moxa-industrial-linux
C-W8-034	PRIMARY · verified	2026-11-10	https://www.linuxfoundation.org/press/civil-infrastructure-platform-achieves-significant-cybersecurity-milestone-with-iec-62443-compliance
C-W8-037	PRIMARY · verified	2026-11-10	https://cache.industry.siemens.com/dl/files/702/109990702/att_1333795/v1/Industrial_Edge_Own_Device_en-US.pdf
C-W8-040	PRIMARY · verified	2026-11-10	https://github.com/raspberrypi/linux/tree/e045ff670093cf3d515d121ab2678ce295f61f43/arch/arm64/configs
C-W8-041	PRIMARY · verified	2026-11-10	https://www.raspberrypi.com/documentation/security/security-architecture.html
C-W8-042	PRIMARY · verified	2026-11-10	https://fidoalliance.org/specifications/download-iot-specifications/
C-W8-044	PRIMARY · verified	2026-11-10	https://github.com/fido-device-onboard/go-fdo/blob/65061850e1c91a2cfe83c2dea748989a7e70c110/README.md
C-W8-049	PRIMARY · verified	2026-11-10	https://docs.k3s.io/installation/requirements

Claim ID	Type · status	Accessed	Source
C-W8-050	PRIMARY · verified	2026-11-10	research-output/evidence/lab/w8-nats-leaf-buffer/results.md
C-W9-001	PRIMARY · verified	2026-11-10	https://modelcontextprotocol.io/specification/versioning.md
C-W9-002	PRIMARY · verified	2026-11-10	https://modelcontextprotocol.io/specification/2026-07-28/changelog.md
C-W9-004	PRIMARY · verified	2026-11-10	https://modelcontextprotocol.io/specification/2026-07-28/basic/authorization.md
C-W9-009	PRIMARY · verified	2026-11-10	https://www.linuxfoundation.org/press/linux-foundation-announces-the-formation-of-the-agentic-ai-foundation
C-W9-032	PRIMARY · verified	2026-11-10	https://www.strava.com/legal/api_policy
C-W9-035	PRIMARY · verified	2026-11-10	https://www.ilga.gov/Legislation/ILCS/Articles?ActID=3004&ChapterID=57
C-W9-037	PRIMARY · verified	2026-11-10	https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:32016R0679
C-W9-043	PRIMARY · verified	2026-11-10	https://www.w3.org/TR/webauthn-3/